INDIAN ECONOMIC DEVELOPMENT

TEXTBOOK OF ECONOMICS FOR HIGHER SECONDARY 2ND YEAR

Assam Higher Secondary Education Council
Bamunimaidam, Guwahati-21

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Rationalisation of Content in the Textbooks

In view of the COVID-19 pandemic, it is imperative to reduce content load on students. The National Education Policy 2020, also emphasises reducing the content load and providing opportunities for experiential learning with creative mindset. In this background, the NCERT has undertaken the exercise to rationalise the textbooks across all classes. Learning Outcomes already developed by the NCERT across classes have been taken into consideration in this exercise.

Contents of the textbooks have been rationalised in view of the following:

- Overlapping with similar content included in other subject areas in the same class
- Similar content included in the lower or higher class in the same subject
- Difficulty level
- Content, which is easily accessible to students without much interventions from teachers and can be learned by children through self-learning or peer-learning
- Content, which is irrelevant in the present context

This present edition, is a reformatted version after carrying out the changes given above.

FOREWORD

The National Curriculum Framework (NCF) 2005, recommends that children's life at school must be linked to their life outside the school. This principle marks a departure from the legacy of bookish learning which continues to shape our system and causes a gap between the school, home and community. The syllabi and textbooks developed on the basis of NCF signify an attempt to implement this basic idea. They also attempt to discourage rote learning and the maintenance of sharp boundaries between different subject areas. We hope these measures will take us significantly further in the direction of a child-centred system of education outlined in the National Policy on Education (1986).

The success of this effort depends on the steps that school principals and teachers will take to encourage children to reflect on their own learning and to pursue imaginative activities and questions. We must recognise that, given space, time and freedom, children generate new knowledge by engaging with the information passed on to them by adults. Treating the prescribed textbook as the sole basis of examination is one of the key reasons why other resources and sites of learning are ignored. Inculcating creativity and initiative is possible if we perceive and treat children as participants in learning, not as receivers of a fixed body of knowledge.

These aims imply considerable change in school routines and mode of functioning. Flexibility in the daily time-table is as necessary as rigour in implementing the annual calendar so that the required number of teaching days are actually devoted to teaching. The methods used for teaching and evaluation will also determine how effective this textbook proves for making children's life at school a happy experience, rather than a source of stress or boredom. Syllabus designers have tried to address the problem of curricular burden by restructuring and reorienting knowledge at different stages with greater consideration for child psychology and the time available for teaching. The textbook attempts to enhance this endeavour by giving higher priority and space to opportunities for contemplation and wondering, discussion in small groups, and activities requiring hands-on experience.

The National Council of Educational Research and Training (NCERT) appreciates the hard work done by the textbook development committee responsible for this book. We wish to thank the Chairperson of the advisory group in Social Sciences, Professor Hari Vasudevan and the Chief Advisor for this book, Professor Tapas Majumdar for guiding the work of this committee. Several teachers contributed to the development of this textbook; we are grateful to their principals for making this possible. We are indebted to the institutions and organisations which have generously permitted us to draw upon their resources, material and personnel. We are especially grateful to the members of the National Monitoring Committee, appointed by the Department of Secondary and Higher Education, Ministry of Human Resource Development under the Chairpersonship of Professor Mrinal Miri and Professor G.P. Deshpande, for their valuable time and contribution. As an organisation committed to systemic reform and continuous improvement in the quality of its products, NCERT welcomes comments and suggestions which will enable us to undertake further revision and refinement.

Director
National Council of Educational
Research and Training

New Delhi
20 December 2005

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National Council of Educational Research and Training, New Delhi has given copyright and has granted permission to Assam Higher Secondary Education Council to print and publish this textbook for the students of H.S. 2nd year, under the jurisdiction of Assam Higher Secondary Education Council.

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Bamunimaidam
Guwahati-781021

Secretary
Assam Higher Secondary Education Council

How to use this Book

The main objective of this book, *Indian Economic Development,* is to expose you to some of the key issues facing the Indian economy. In this process, as young adults, you are expected to be sensitised about these issues, appreciate and learn to critically assess the role of the government in various economic spheres. The book also provides opportunities for knowing what economic resources are and how these resources are being utilised in different sectors. You are also exposed to numerical information relating to various aspects of Indian economy and India's economic policies. You are expected to develop analytical skills, interpret the economic events and visualise the economic future of India. Nevertheless, conscious effort has been made not to burden with concepts and data.

With regard to various economic issues and trends, this book attempts to bring out alternative views on every issue so that you can engage yourself in well-informed debates. On completion of the course, *Indian Economic Development*, you are expected to acquire skills to understand macroeconomic events which occur around you, and to critically evaluate and interpret the relevant information provided by the media.

This course comprises of many activities with each chapter. You need to carry out those activities under the guidance of teachers. In fact, the role of the teacher in enriching the understanding of the Indian economy is greater in this course. The activities include classroom discussions, collecting information from government documents such as Economic Survey, archival materials, from newspapers, television, internet and other sources. Learners should also be encouraged to read the original works of scholars on various topics.



For all this to happen, the teachers have to initiate the process by taking some steps before starting the course. At the beginning of the academic year, you can collect clippings from newspapers and magazines relating to different topics under the course — five year plans and the role of NITI Aayog in planning; allocation of funds to various sectors like agriculture, industry, services and for specific causes such as poverty and employment; key issues of rural development,

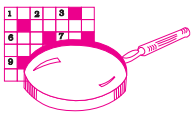
environment, various infrastructure— health, education and energy; and economic events in China and Pakistan. You may maintain these clippings, and when the teacher starts teaching a particular topic, you can cull out those news items, which you have collected from the beginning of the course, and display/use them in the classroom. It is necessary you to build this collection right from the beginning of the course so that the relevent information is at hand as and when needed; this habit will also come in handy during later stages of education.

The schools will have to buy a hard copy of the Economic Survey for the latest year. You will notice that information relating to the Indian Economy is updated in the Economic Survey. It is necessary for you to familiarise yourselves with such reports and work on relevant activities. The statistical tables available as Appendix in the Economic Survey would be immensely helpful in understanding various issues.

While discussing a particular issue, discussion of the numerical information about the issue is inevitable. For instance, when we talk of growth rates — overall growth rates and growth of different sectors — though it may be necessary for you to have a rough idea about the trends in growth rates, the process involved in reaching the level of growth and factors contributing to the trend rather than mere reproduction of tabular data of growth rate.

You will notice numbered boxes in all chapters. These boxes supplement the information given in the text. Through these boxes, an attempt has been made to lend a humane touch and, thus, bring the issue under discussion closer to real life. However, these boxes, as also 'Work These Out' activities, are not meant for examination/evaluation purposes.

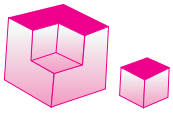
Besides the relatively conventional 'Exercises', each chapter has 'Suggested Additional Activities' at the end and 'Work This/These Out' as part of the text; the more elaborate activities of these can be treated as projects. You may explore beyond the text while attempting these activities.





Understandably, Information Technology facilities may not be available in all the schools, however, it is important to take note that various information relating to the Indian economy are available on the internet. You need to learn to use the internet and access the websites of various government departments to get the required information. For instance, the details relating to poverty are published by the NITI Aayog. You need to know that the NITI Aayog of

the Government of India has a website in which various reports relating to different aspects of India, including poverty, are available in the form of reports. Since it may not be always possible to procure such reports in hard copy, students and teachers may attempt to download such reports from websites and use them in the classroom.



The reports such as Economic Surveys of the last 10 years are available on the website: http://www.indiabudget.gov.in. Many organisations change their website addresses. In case a website given in this book is not accessible, kindly search those websites through search engines such as Google (www.google.co.in).

For the first time, each chapter has been briefly summed up in the 'Recap' to facilitate learning. Also, please take note that sources for all tables have not been given with the table as these tables were sourced from various research material which have been covered under 'References' for each chapter.

We wish to reiterate the fact that the prime objective of this course on Indian economy is to introduce the basic macro issues of the Indian economy to the student community and to initiate a well-informed debate on our economy. We also stress that collaborative learning is an important aspect of this course, hence the involvement of students, as well as, teachers in collecting information on the Indian economy from other sources is necessary and such collected information should be used as important inputs for both teaching, as well as, learning about the Indian economy.

You can send your queries and feedback relating to any part of this book to the following address.

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Constitution of India

Part IV A (Article 51 A)

Fundamental Duties

It shall be the duty of every citizen of India —

(a) to abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem;

(b) to cherish and follow the noble ideals which inspired our national struggle for freedom;

(c) to uphold and protect the sovereignty, unity and integrity of India;

(d) to defend the country and render national service when called upon to do so;

(e) to promote harmony and the spirit of common brotherhood amongst all the people of India transcending religious, linguistic and regional or sectional diversities; to renounce practices derogatory to the dignity of women;

(f) to value and preserve the rich heritage of our composite culture;

(g) to protect and improve the natural environment including forests, lakes, rivers, wildlife and to have compassion for living creatures;

(h) to develop the scientific temper, humanism and the spirit of inquiry and reform;

(i) to safeguard public property and to abjure violence;

(j) to strive towards excellence in all spheres of individual and collective activity so that the nation constantly rises to higher levels of endeavour and achievement;

*(k) who is a parent or guardian, to provide opportunities for education to his child or, as the case may be, ward between the age of six and fourteen years.

Note: The Article 51A containing Fundamental Duties was inserted by the Constitution (42nd Amendment) Act, 1976 (with effect from 3 January 1977).

*(k) was inserted by the Constitution (86th Amendment) Act, 2002 (with effect from 1 April 2010).

UNIT I

Development Policies and Experience (1947-90)

The two chapters in this unit give us an overview of the state of the Indian economy as it was at the eve of independence till after four decades of planned development, which was a path that India chose. This meant that the Government of India had to take a series of steps such as the establishment of the Planning Commission and announcement of five year plans. An overview of the goals of five year plans and a critical appraisal of the merits and limitations of planned development has been covered in this unit.

1



INDIAN ECONOMY ON THE EVE OF INDEPENDENCE

After studying this chapter, the learners will

- become familiar with the state of the Indian economy in 1947, the year of India's Independence
- understand the factors that led to the underdevelopment and stagnation of the Indian economy.

"India is the pivot of our Empire... If the Empire loses any other part of its Dominion we can survive, but if we lose India, the sun of our Empire will have set."

Victor Alexander Vruce, the Viceroy of British India in 1894

1.1 Introduction

The primary objective of this book, *Indian Economic Development*, is to familiarise you with the basic features of the Indian economy, and its development, as it is today, in the aftermath of Independence. However, it is equally important to know something about the country's economic past even as you learn about its present state and future prospects. So, let us first look at the state of India's economy prior to the country's independence and form an idea of the various considerations that shaped India's post-independence development strategy.

The structure of India's present-day economy is not just of current making; it has its roots steeped in history, particularly in the period when India was under British rule which lasted for almost two centuries before India finally won its independence on 15 August 1947. The sole purpose of the British colonial rule in India was to reduce the country to being a raw material supplier for Great Britain's own rapidly expanding modern industrial base. An understanding of the exploitative nature of this relationship is essential for any assessment of the kind and level of development which the Indian economy has been able to attain over the last seven and half decades.

1.2 Low Level of Economic Development under the Colonial Rule

India had an independent economy before the advent of the British rule. Though agriculture was the main source of livelihood for most people, yet, the country's economy was characterised by various kinds of manufacturing activities. India was particularly well known for its handicraft industries in the fields of cotton and silk textiles, metal and precious stone works etc. These products enjoyed a worldwide market based on the reputation of the fine quality of material used and the high standards of craftsmanship seen in all imports from India (See Box 1.1).

Box 1.1: Textile Industry in Bengal

Muslin is a type of cotton textile which had its origin in Bengal, particularly, places in and around Dhaka (spelled during the pre-independence period as Dacca), now the capital city of Bangladesh. 'Daccai Muslin' had gained worldwide fame as an exquisite type of cotton textile. The finest variety of muslin was called *malmal*. Sometimes, foreign travellers also used to refer to it as *malmal shahi* or *malmal khas* implying that it was worn by, or fit for, the royalty.

The economic policies pursued by the colonial government in India were concerned more with the protection and promotion of the economic interests of their home country than with the development of the Indian economy. Such policies brought about a fundamental change in the structure of the Indian economy — transforming the country into supplier of raw materials and consumer of finished industrial products from Britain.

Obviously, the colonial government never made any sincere attempt to estimate India's **national** and **per capita income**. Some individual attempts which were made to measure such incomes yielded conflicting and inconsistent results. Among the notable estimators — Dadabhai Naoroji, William Digby, Findlay Shirras, V.K.R.V. Rao and R.C. Desai — it was Rao, whose estimates during the colonial period was considered very significant. However, most studies did find that the country's growth of aggregate real output during the first half of the twentieth century was less than two per cent coupled with a meagre half per cent growth in per capita output per year.

1.3 Agricultural Sector

India's economy under the British colonial rule remained fundamentally agrarian — about 85 per cent of the country's population lived mostly in villages and derived livelihood directly or indirectly from agriculture (See Box 1.2). However, despite being the occupation of such a large population, the agricultural

Box 1.2: Agriculture During Pre-British India

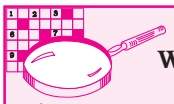
The French traveller, Bernier, described seventeenth century Bengal in the following way: "The knowledge I have acquired of Bengal in two visits inclines me to believe that it is richer than Egypt. It exports, in abundance, cottons and silks, rice, sugar and butter. It produces amply — for its own consumption — wheat, vegetables, grains, fowls, ducks and geese. It has immense herds of pigs and flocks of sheep and goats. Fish of every kind it has in profusion. From *rajmahal* to the sea is an endless number of canals, cut in bygone ages from the Ganges by immense labour for navigation and irrigation."



Fig. 1.1 *India's agricultural stagnation under the British colonial rule*

➢ Take note of the agricultural prosperity in our country in the seventeenth century. Contrast it with agricultural stagnation around the time when the British left India, around 200 years later.

sector continued to experience stagnation and, not infrequently, unusual deterioration. **Agricultural pro-ductivity** became low though, in absolute terms, the sector experienced some growth due to the expansion of the aggregate area under cultivation. This stagnation in the agricultural sector was caused mainly because of the various systems of **land settlement** that were introduced by the colonial government. Particularly, under the *zamindari* system which was implemented in the then Bengal Presidency comprising parts of India's present-day eastern states, the profit accruing out of the agriculture sector went to the *zamindars* instead of the cultivators. However, a considerable number of *zamindars,* and not just the colonial government, did nothing to improve the condition of agriculture. The main interest of the *zamindars* was only to collect rent regardless of the economic condition of the cultivators; this caused immense misery and social tension among the latter. To a very great extent, the terms of the **revenue settlement** were also responsible for the *zamindars* adopting such an attitude; dates for depositing specified sums of revenue were fixed, failing which the *zamindars* were to lose their rights. Besides this, low levels of technology, lack of irrigation facilities and negligible use of fertilisers, all added up to aggravate the plight of the farmers and contributed to the dismal level of agricultural productivity. There was, of course, some evidence of a relatively higher yield of cash crops in certain areas of the country due to **commercialisation of agriculture**.



Work These Out

- Compare the map of British India with that of independent India and find out the areas that became parts of Pakistan. Why were those parts so important to India from the economic point of view? (Refer, to your advantage, Dr Rajendra Prasad's book, *India Divided).*
- What were the various forms of revenue settlement adopted by the British in India? Where did they implement them and to what effect? How far do you think those settlements have a bearing on the current agricultural scenario in India? (In your attempt to find answers to these questions, you may refer to Ramesh Chandra Dutt's *Economic History of India,* which comes in three volumes, and B.H. Baden-Powell's *The Land Systems of British India*, also in two volumes. For better comprehension of the subject, you can also try and develop an illustrated agrarian map of British India either by hand or with the help of your school computer. Remember, nothing helps better than an illustrated map to understand the subject at hand).

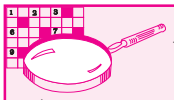
But this could hardly help farmers in improving their economic condition as, instead of producing food crops, now they were producing cash crops which were to be ultimately used by British industries back home. Despite some progress made in irrigation, India's agriculture was starved of investment in terracing, flood-control, drainage and desalinisation of soil. While a small section of farmers changed their cropping pattern from food crops to commercial crops, a large section of tenants, small farmers and sharecroppers neither had resources and technology nor had incentive to invest in agriculure.

1.4 Industrial Sector

As in the case of agriculture, so also in manufacturing, India could not develop a sound industrial base under the colonial rule. Even as the country's world famous handicraft industries declined, no corresponding modern industrial base was allowed to come up to take pride of place so long enjoyed by the former. The primary motive of the colonial government behind this policy of systematically de-industrialising India was two-fold. The intention was, first, to reduce India to the status of a mere exporter of important raw materials for the upcoming modern industries in Britain and, second, to turn India into a sprawling market for the finished products of those industries so that their continued expansion could be ensured to the maximum advantage of their home country — Britain. In the unfolding economic scenario, the decline of the indigenous handicraft industries created not only massive unemployment in India but also a new demand in the Indian consumer market, which was now deprived of the supply of locally made goods. This demand was profitably met by the increasing imports of cheap manufactured goods from Britain.

During the second half of the nineteenth century, modern industry began to take root in India but its progress remained very slow. Initially, this development was confined to the setting up of cotton and jute textile mills. The cotton textile mills, mainly dominated by Indians, were located in the western parts of the country, namely, Maharashtra and Gujarat, while the jute mills dominated by the foreigners were mainly concentrated in Bengal. Subsequently, the iron and steel industries began coming up in the beginning of the twentieth century. The Tata Iron and Steel Company (TISCO) was incorporated in 1907. A few other industries in the fields of sugar, cement, paper etc. came up after the Second World War.

However, there was hardly any **capital goods industry** to help promote further industrialisation in India. Capital goods industry means industries which can produce machine tools which are, in turn, used for producing articles for current consumption. The establishment of a few manufacturing units here and



Work These Out

- Prepare a list showing where and when other modern industries of India were first set up. Can you also find out what the basic requirements are for setting up any modern industry? What, for example, might have been the reasons for the setting up of the Tata Iron and Steel Company at Jamshedpur, which is now in the state of Jharkhand?
- How many iron and steel factories are there in India at present? Are these iron and steel factories among the best in the world or do you think that these factories need restructuring and upgradation? If yes, how can this be done? There is an argument that industries which are not strategic in nature should not continue to be in the public sector. What is your view?
- On a map of India, mark the cotton textiles, jute mills and textile mills that existed at the time of independence.

there was no substitute to the near wholesale displacement of the country's traditional handicraft industries. Furthermore, the growth rate of the new industrial sector and its contribution to the **Gross Domestic Product (GDP)** or **Gross Value Added** remained very small. Another significant drawback of the new industrial sector was the very limited area of operation of the public sector. This sector remained confined only to the railways, power generation, communications, ports and some other departmental undertakings.

1.5 Foreign Trade

India has been an important trading nation since ancient times. But the restrictive policies of commodity production, trade and tariff pursued by the colonial government adversely affected the structure, composition and volume of India's foreign trade. Consequently, India became an exporter of primary products such as raw silk, cotton, wool, sugar, indigo, jute etc. and an importer of finished consumer goods like cotton, silk and woollen clothes and capital goods like light machinery produced in the factories of Britain. For all practical purposes, Britain maintained a monopoly control over India's exports and imports. As a result, more than half of India's foreign trade was restricted to Britain while the rest was allowed with a few other countries like China, Ceylon (Sri Lanka) and Persia (Iran). The opening of the Suez Canal further intensified British control over India's foreign trade (see Box 1.3).

The most important characteristic of India's foreign trade throughout the colonial period was the generation of a large export surplus. But this surplus came at a huge cost to the country's economy. Several essential commodities—food grains, clothes,

Work These Out

- Prepare a list of items that were exported from and imported into India during the British rule.
- Collect information from the *Economic Survey* for various years published by the Ministry of Finance, Government of India, on various items of export from India and its imports. Compare these with imports and exports from the pre-independence era. Also find out the names of prominent ports which now handle the bulk of India's foreign trade.

kerosene etc. — were scarcely available in the domestic market. Furthermore, this export surplus did not result in any flow of gold or silver into India. Rather, this was used to make payments for the expenses incurred by an office set up by the colonial government in Britain, expenses on war, again fought by the British government, and the import of invisible items, all of which led to the drain of Indian wealth.

1.6 Demographic Condition

Various details about the population of British India were first collected through a census in 1881. Though suffering from certain limitations, it revealed the unevenness in India's population growth. Subsequently,



Fig.1.2 *Suez Canal: Used as highway between India and Britain*

Box 1.3: Trade Through the Suez Canal

Suez Canal is an artificial waterway running from north to south across the Isthmus of Suez in north-eastern Egypt. It connects Port Said on the Mediterranean Sea with the Gulf of Suez, an arm of the Red Sea. The canal provides a direct trade route for ships operating between European or American ports and ports located in South Asia, East Africa and Oceania by doing away with the need to sail around Africa. Strategically and economically, it is one of the most important waterways in the world. Its opening in 1869 reduced the cost of transportation and made access to the Indian market easier.

every ten years such census operations were carried out. Before 1921, India was in the first stage of **demographic transition**. The second stage of transition began after 1921. However, neither the total population of India nor the rate of population growth at this stage was very high.

The various social development indicators were also not quite encouraging. The overall literacy level was less than 16 per cent. Out of this, the female literacy level was at a negligible low of about seven per cent. Public health facilities were either unavailable to large chunks of population or, when available, were highly inadequate. Consequently, water and air-borne diseases were rampant and took a huge toll on life. No wonder, the overall **mortality rate** was very high and in that, particularly, the **infant mortality rate** was quite alarming—about 218 per thousand in contrast to the present infant mortality rate of 33 per thousand. **Life expectancy** was also very low—32 years in contrast to the present 69 years. In the absence of reliable data, it is difficult to specify the extent of poverty at that time but there is no doubt that extensive poverty prevailed in India during the colonial period which contributed to the worsening profile of India's population of the time.

1.7 Occupational Structure

During the colonial period, the occupational structure of India, i.e., distribution of working persons across different industries and sectors, showed little sign of change. The agricultural sector accounted for

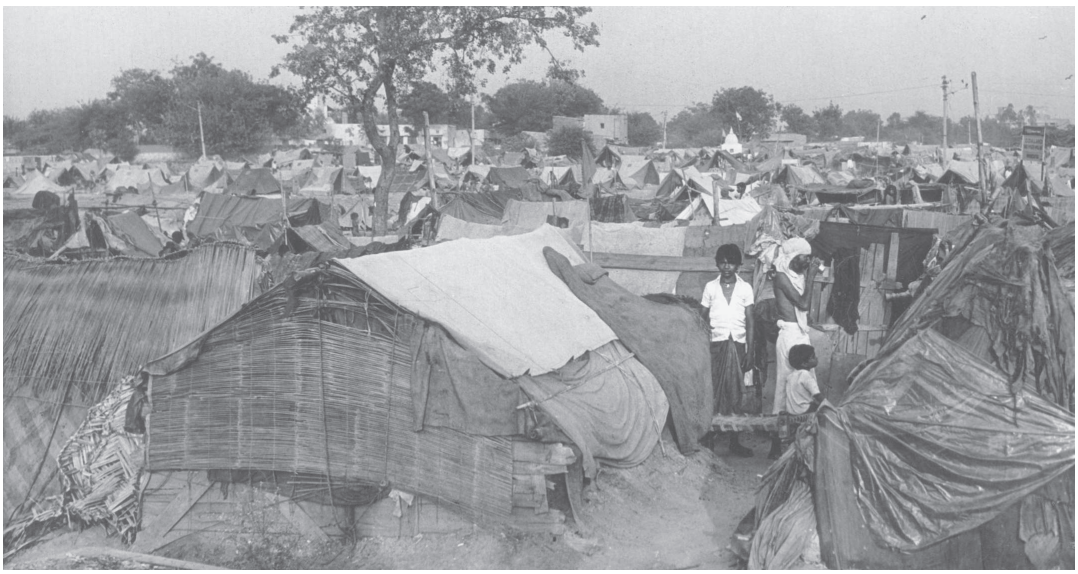
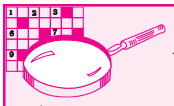


Fig. 1.3 *A large section of India's population did not have basic needs such as housing*



Work These Out

- Can you find out the reasons behind frequent occurrence of famines in India before independence? You may read from Nobel Laureate Amartya Sen's book, *Poverty and Famines.*
- Prepare a pie chart for the occupational structure in India at the time of independence.

the largest share of workforce, which usually remained at a high of 70-75 per cent while the manufacturing and the services sectors accounted for only 10 and 15-20 per cent respectively. Another striking aspect was the growing regional variation. Parts of the then Madras Presidency (comprising areas of the present-day states of Tamil Nadu, Andhra Pradesh, Kerala and Karnataka), Bombay and Bengal witnessed a decline in the dependence of the workforce on the agricultural sector with a commensurate increase in the manufacturing and the services sectors. However, there had been an increase in the share of workforce in agriculture during the same time in states such as Orissa, Rajasthan and Punjab.

1.8 Infrastructure

Under the colonial regime, basic infrastructure such as railways, ports, water transport, posts and telegraphs did develop. However, the real motive behind this development was not to provide basic amenities to the people but to subserve various colonial interests. Roads constructed in India prior to the advent of the British rule were not fit for modern transport. The roads that were built primarily served the purposes of mobilising the army within India and drawing out raw materials from the countryside to the nearest railway station or the port to send these to far away England or other lucrative foreign destinations. There always remained an acute shortage of all-weather roads to reach out to the rural areas during the rainy season. Naturally, therefore, people mostly living in these areas suffered grievously during natural calamities and famines.

The British introduced the railways in India in 1850 and it is considered as one of their most important contributions. The railways affected the structure of the Indian economy in two important ways. On the one hand it enabled people to undertake long distance travel and thereby break geographical and cultural barriers while, on the other hand, it fostered commercialisation of Indian agriculture which adversely affected the self-sufficiency of the village economies in India. The volume of India's exports undoubtedly expanded but its benefits rarely accrued to the Indian people. The social benefits, which the

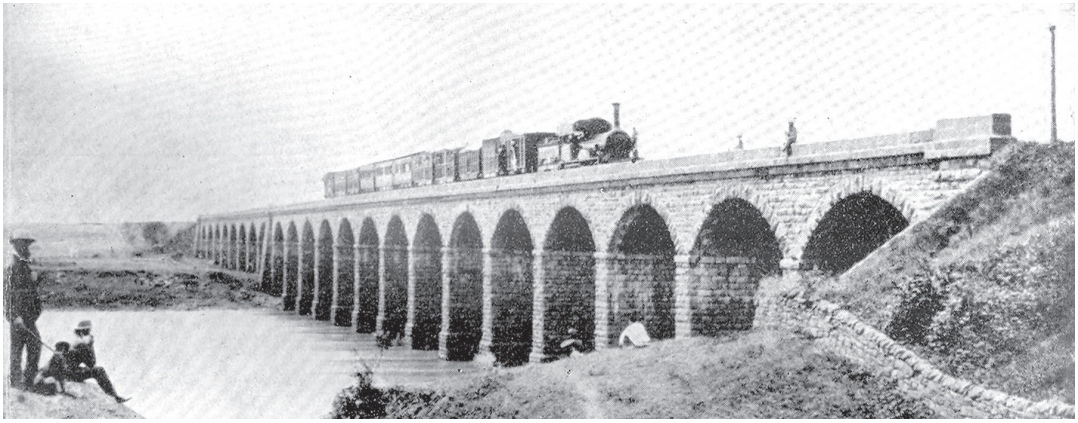


Fig. 1.4 *First Railway Bridge linking Bombay with Thane, 1854*

Indian people gained owing to the introduction of the railways, were thus outweighed by the country's huge economic loss.

Along with the development of roads and railways, the colonial dispensation also took measures for developing the inland trade and sea lanes. However, these measures were far from satisfactory. The inland waterways, at times, also proved uneconomical as in the case of the Coast Canal on the Orissa coast. Though the canal was built at a huge cost to the government exchequer, yet, it failed to compete with the railways, which soon traversed the region running parallel to the canal, and had to be ultimately abandoned. The introduction of the expensive system of electric telegraph in India, similarly, served the purpose of maintaining law and order. The postal services, on the other hand, despite serving a useful public purpose, remained all through



Fig.1.5 *Tata Airlines, a division of Tata and Sons, was established in 1932 inaugurating the aviation sector in India*



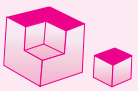
Work This Out

- There is a perception still going around that in many ways the British administration in India was quite beneficial. This perception needs an informed debate. How would you look at this perception? Argue this out in your class—'Was the British Raj good for India'?

inadequate. You will learn more about the present status of various infrastructure in Chapter 8.

1.9 Conclusion

By the time India won its independence, the impact of the two-century long British colonial rule was already showing on all aspects of the Indian economy. The agricultural sector was already saddled with surplus labour and extremely low productivity. The industrial sector was crying for modernisation, diversification, capacity building and increased public investment. Foreign trade was oriented to feed the Industrial Revolution in Britain. Infrastructure facilities, including the famed railway network, needed upgradation, expansion and public orientation. Prevalence of rampant poverty and unemployment required welfare orientation of public economic policy. In a nutshell, the social and economic challenges before the country were enormous.



Recap

- An understanding of the economy before independence is necessary to know and appreciate the level of development achieved during the post-independence period.
- Under the colonial dispensation, the economic policies of the government were concerned more with the protection and promotion of British economic interests than with the need to develop the economic condition of the colonised country and its people.
- The agricultural sector continued to experience stagnation and deterioration despite the fact that the largest section of Indian population depended on it for sustenance.
- The rule of the British-India government led to the collapse of India's world famous handicraft industries without contributing, in any significant manner, to its replacement by a modern industrial base.
- Lack of adequate public health facilities, occurrence of frequent natural calamities and famines pauperised the hapless Indian people and resulted in engendering high mortality rates.
- Some efforts were made by the colonial regime to improve infrastructure facilities but these efforts were spiced with selfish motives. However, the independent Indian government had to built on this base through planning.

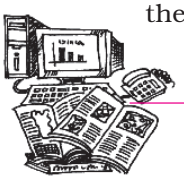


EXERCISES

1. What was the focus of the economic policies pursued by the colonial government in India? What were the impacts of these policies?
2. Name some notable economists who estimated India's per capita income during the colonial period.
3. What were the main causes of India's agricultural stagnation during the colonial period?
4. Name some modern industries which were in operation in our country at the time of independence.
5. What was the two-fold motive behind the systematic de-industrialisation effected by the British in pre-independent India?
6. The traditional handicrafts industries were ruined under the British rule. Do you agree with this view? Give reasons in support of your answer.
7. What objectives did the British intend to achieve through their policies of infrastructure development in India?
8. Critically appraise some of the shortfalls of the industrial policy pursued by the British colonial administration.
9. What do you understand by the drain of Indian wealth during the colonial period?
10. Which is regarded as the defining year to mark the demographic transition from its first to the second decisive stage?
11. Give a quantitative appraisal of India's demographic profile during the colonial period.
12. Highlight the salient features of India's pre-independence occupational structure.
13. Underscore some of India's most crucial economic challenges at the time of independence.
14. When was India's first official census operation undertaken?
15. Indicate the volume and direction of trade at the time of independence.
16. Were there any positive contributions made by the British in India? Discuss.

SUGGESTED ADDITIONAL ACTIVITIES

1. Prepare a list of goods and services that were available to people in pre-independence India in rural and urban areas. Compare it with the consumption pattern of such goods and services by the people today. Highlight the perceptible difference in the people's standard of living.
2. Find pictures of towns/villages, in your vicinity, of the pre-independence period and compare these with their present scenario. What changes can you mark? Are such changes for better or for worse? Discuss.
3. Rally around your teacher and organise a group discussion on 'Has the *zamindari* system really been abolished in India'? If the consensus is negative, then what measures would you think should be taken to banish it and why?
4. Identify the major occupations followed by the people of our country at the time of independence. What major occupations do the people follow today? In the light of reform policies, how would you visualise the occupational scenario in India 15 years from now—say, 2035?



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2



INDIAN ECONOMY
1950–1990

After studying this chapter, the learners will

- come to know the goals of India's five year plans
- know about the development policies in different sectors such as agriculture and industry from 1950-1990
- learn to think about the merits and limitations of a regulated economy.

The central objective of Planning in India... is to initiate a process of development which will raise the living standards and open out to the people new opportunities for a richer and more varied life.

First Five Year Plan

2.1 Introduction

On 15 August 1947, India woke to a new dawn of freedom. Finally we were masters of our own destiny after some two hundred years of British rule; the job of nation building was now in our own hands. The leaders of independent India had to decide, among other things, the type of **economic system** most suitable for our nation, a system which would promote the welfare of all rather than a few. There are different types of economic systems (see Box 2.1) and among them, socialism appealed to Jawaharlal Nehru the most. However, he was not in favour of the kind of socialism established in the former Soviet Union where all the means of production, i.e. all the factories and farms in the country, were owned by the government. There was no private property. It is not possible in a democracy like India for the government to change the ownership pattern of land and other properties of its citizens in the way that it was done in the former Soviet Union.

Nehru, and many other leaders and thinkers of the newly independent India, sought an alternative to the extreme versions of capitalism and socialism. Basically sympathising with the socialist outlook, they found the answer in an economic system which, in their view, combined the best features of socialism without its drawbacks. In this view, India would be a socialist society with a strong public sector but also with private property and democracy; the government would plan (see Box 2.2) for the



Work These Out

- Prepare a chart on the different types of economic systems prevalent in the world. List out the countries as capitalist, socialist and mixed economy.
- Plan a class trip to an agriculture farm. Divide the class into seven groups with each group to plan a specific goal, for example, the purpose of the visit, money expenditure involved, time taken, resources, people accompanying the group and who need to be contacted, possible places of visit, possible questions to be asked etc. Now, with the help of your teacher, compile these specific goals and compare with long-term goals of successful visit to an agricultural farm.

Box 2.1: Types of Economic Systems

Every society has to answer three questions

- What goods and services should be produced in the country?
- How should the goods and services be produced? Should producers use more human labour or more capital (machines) for producing things?
- How should the goods and services be distributed among people?

One answer to these questions is to depend on the **market forces** of supply and demand. In a market economy, also called capitalism, only those consumer goods will be produced that are in demand, i.e., goods that can be sold profitably either in the domestic or in the foreign markets. If cars are in demand, cars will be produced and if bicycles are in demand, bicycles will be produced. If labour is cheaper than capital, more labour-intensive methods of production will be used and vice-versa. In a **capitalist** society the goods produced are distributed among people not on the basis of what people need but on the basis of Purchasing Power—the ability to buy goods and services. That is, one has to have the money in the pocket to buy it. Low cost housing for the poor is much needed but will not count as demand in the market sense because the poor do not have the purchasing power to back the demand. As a result this commodity will not be produced and supplied as per market forces. Such a society did not appeal to Jawaharlal Nehru, our first prime minister, for it meant that the great majority of people of the country would be left behind without the chance to improve their quality of life.

A **socialist** society answers the three questions in a totally different manner. In a socialist society the government decides what goods are to be produced in accordance with the needs of society. It is assumed that the government knows what is good for the people of the country and so the desires of individual consumers are not given much importance. The government decides how goods are to be produced and how they should be distributed. In principle, distribution under socialism is supposed to be based on what people need and not on what they can afford to purchase. Unlike under capitalism, for example, a socialist nation provides free health care to all its citizens. Strictly, a socialist society has no private property since everything is owned by the state. In Cuba and China, for example, most of the economic activities are governed by the socialistic principles.

Most economies are **mixed economies,** i.e. the government and the market together answer the three questions of what to produce, how to produce and how to distribute what is produced. In a mixed economy, the market will provide whatever goods and services it can produce well, and the government will provide essential goods and services which the market fails to do.

Box 2.2: What is a Plan?

A plan spells out how the resources of a nation should be put to use. It should have some general goals as well as specific objectives which are to be achieved within a specified period of time; in India plans were of five years duration and were called five year plans (we borrowed this from the former Soviet Union, the pioneer in national planning). Our plan documents upto the year 2017 not only specify the objectives to be attained in the five years of a plan but also what is to be achieved over a period of twenty years. This long-term plan is called 'perspective plan'. The five year plans were supposed to provide the basis for the perspective plan.

It will be unrealistic to expect all the goals of a plan to be given equal importance in all the plans. In fact the goals may actually be in conflict. For example, the goal of introducing modern technology may be in conflict with the goal of increasing employment if the technology reduces the need for labour. The planners have to balance the goals, a very difficult job indeed. We find different goals being emphasised in different plans in India.

India's five year plans did not spell out how much of each and every good and service is to be produced. This is neither possible nor necessary (the former Soviet Union tried to do this and failed). It is enough if the plan is specific about the sectors where it plays a commanding role, for instance, power generation and irrigation, while leaving the rest to the market.

economy with the private sector being encouraged to be part of the plan effort. The 'Industrial Policy Resolution' of 1948 and the Directive Principles of the Indian Constitution reflected this outlook. In 1950, the **Planning Commission** was set up with the Prime Minister as its Chairperson. The era of five year plans had begun.

2.2 The Goals of Five Year Plans

A plan should have some clearly specified goals. The goals of the five year plans were: growth, modernisation, self-reliance and equity. This does not mean that all the plans have given equal importance to all these goals. Due to limited resources, a choice has to be made in each plan about which of the goals is to be given primary importance. Nevertheless, the planners have to ensure that, as far as possible, the policies of the plans do not contradict these four goals. Let us now learn about the goals of planning in some detail.

Growth: It refers to increase in the country's capacity to produce the output of goods and services within the country. It implies either a larger stock of productive capital, or a larger size of supporting services like transport and banking, or an increase in the efficiency of productive capital and services. A good indicator of economic growth, in the language of

Box 2.3: Mahalanobis: the Architect of Indian Planning

Many distinguished thinkers contributed to the formulation of India's five year plans. Among them, the name of the statistician, Prasanta Chandra Mahalanobis, stands out.

Planning, in the real sense of the term, began with the Second Five Year Plan. The Second Plan, a landmark contribution to development planning in general, laid down the basic ideas regarding goals of Indian planning; this plan was based on the ideas of Mahalanobis. In that sense, he can be regarded as the architect of Indian planning.



Mahalanobis was born in 1893 in Calcutta. He was educated at the Presidency College in Calcutta and at Cambridge University in England. His contributions to the subject of statistics brought him international fame. In 1945 he was made a Fellow (member) of Britain's Royal Society, one of the most prestigious organisations of scientists; only the most outstanding scientists are made members of this Society.

Mahalanobis established the Indian Statistical Institute (ISI) in Calcutta and started a journal, *Sankhya,* which still serves as a respected forum for statisticians to discuss their ideas. Both, the ISI and *Sankhya,* are highly regarded by statisticians and economists all over the world to this day.

During the second plan period, Mahalanobis invited many distinguished economists from India and abroad to advise him on India's economic development. Some of these economists became Nobel Prize winners later, which shows that he could identify individuals with talent. Among the economists invited by Mahalanobis were those who were very critical of the socialist principles of the second plan. In other words, he was willing to listen to what his critics had to say, the mark of a great scholar.

Many economists today reject the approach to planning formulated by Mahalanobis but he will always be remembered for playing a vital role in putting India on the road to economic progress, and statisticians continue to profit from his contribution to statistical theory.

Source: *Sukhamoy Chakravarty, 'Mahalanobis, Prasanta Chandra' in John Eatwell et.al, (Eds.) The New Palgrave Dictionary: Economic Development, W.W. Norton, New York and London.*

Box 2.4: The Service Sector

As a country develops, it undergoes 'structural change'. In the case of India, the structural change is peculiar. Usually, with development, the share of agriculture declines and the share of industry becomes dominant. At higher levels of development, the service sector contributes more to the GDP than the other two sectors. In India, the share of agriculture in the GDP was more than 50 per cent—as we would expect for a poor country. But by 1990 the share of the service sector was 40.59 per cent, more than that of agriculture or industry, like what we find in developed nations. This phenomenon of growing share of the service sector was accelerated in the post 1991 period (this marked the onset of globalisation in the country which will be discussed in chapter 3).

economics, is steady increase in the Gross Domestic Product (GDP). The GDP is the market value of all the final goods and services produced in the country during a year. You have studied this concept in Class X as well. You can think of the GDP as a cake and growth is increase in the size of the cake. If the cake is larger, more people can enjoy it. It is necessary to produce more goods and services if the people of India are to enjoy (in the words of the First Five Year Plan) a more rich and varied life.

The GDP of a country is derived from the different sectors of the economy, namely the agricultural sector, the industrial sector and the service sector. The contribution made by each of these sectors makes up the **structural composition** of the economy. In some countries, growth in agriculture contributes more to the GDP growth, while in some countries the growth in the service sector contributes more to GDP growth (see Box 2.4).

Modernisation: To increase the production of goods and services the producers have to adopt new technology. For example, a farmer can increase the output on the farm by using new seed varieties instead of using the old ones. Similarly, a factory can increase output by using a new type of machine. Adoption of new technology is called modernisation.

However, modernisation does not refer only to the use of new technology but also to changes in social outlook such as the recognition that women should have the same rights as men. In a traditional society, women are supposed to remain at home while men work. A modern society makes use of the talents of women in the work place — in banks, factories, schools etc. — and such a society in most occassions is also prosperous.

Self-reliance: A nation can promote economic growth and modernisation by using its own resources or by using resources imported from other nations. The first seven five year plans gave importance to self-reliance which means avoiding imports of those goods which could be



Work These Out

- Discuss in your class the changes in technology used for
  (a) Production of food grains
  (b) Packaging of products
  (c) Mass communication
- Find out and prepare a list of major items that India imported and exported during 1990-91 and 2018-19. (For this, see P. 192 also).
  (a) Observe the difference
  (b) Do you see the impact of self-reliance? Discuss.

For getting these details you may refer to *Economic Survey* of the latest year.

produced in India itself. This policy was considered a necessity in order to reduce our dependence on foreign countries, especially for food. It is understandable that people who were recently freed from foreign domination should give importance to self-reliance. Further, it was feared that dependence on imported food supplies, foreign technology and foreign capital may make India's sovereignty vulnerable to foreign interference in our policies.

Equity: Now growth, modernisation and self-reliance, by themselves, may not improve the kind of life which people are living. A country can have high growth, the most modern technology developed in the country itself, and also have most of its people living in poverty. It is important to ensure that the benefits of economic prosperity reach the poor sections as well instead of being enjoyed only by the rich. So, in addition to growth, modernisation and self-reliance, equity is also important. Every Indian should be able to meet his or her basic needs such as food, a decent house, education and health care and inequality in the distribution of wealth should be reduced.

Let us now see how the first seven five year plans, covering the period 1950-1990, attempted to attain these four goals and the extent to which they succeeded in doing so, with reference to agriculture, industry and trade. You will study the policies and developmental issues taken up after 1991 in Chapter 3.

2.3 Agriculture

You have learnt in Chapter 1 that during the colonial rule there was neither growth nor equity in the agricultural sector. The policy makers of independent India had to address these issues which they did through land reforms and promoting the use of 'High Yielding Variety' (HYV) seeds which ushered in a revolution in Indian agriculture.

Land Reforms: At the time of independence, the land tenure system was characterised by intermediaries

Box 2.5: Ownership and Incentives

The policy of 'land to the tiller' is based on the idea that the cultivators will take more interest — they will have more incentive — in increasing output if they are the owners of the land. This is because ownership of land enables the tiller to make profit from the increased output. Tenants do not have the incentive to make improvements on land since it is the landowner who would benefit more from higher output. The importance of ownership in providing incentives is well illustrated by the carelessness with which farmers in the former Soviet Union used to pack fruits for sale. It was not uncommon to see farmers packing rotten fruits along with fresh fruits in the same box. Now, every farmer knows that the rotten fruits will spoil the fresh fruits if they are packed together. This will be a loss to the farmer since the fruits cannot be sold. So why did the Soviet farmers do something which would so obviously result in loss for them? The answer lies in the incentives facing the farmers. Since farmers in the former Soviet Union did not own any land, they neither enjoyed the profits nor suffered the losses. In the absence of ownership, there was no incentive on the part of farmers to be efficient, which also explains the poor performance of the agricultural sector in the Soviet Union despite availability of vast areas of highly fertile land.

Source: *Thomas Sowell, Basic Economics: A Citizen's Guide to the Economy, New York: Basic Books, 2004, Second Edition.*

(variously called *zamindars, jagirdars* etc.) who merely collected rent from the actual tillers of the soil without contributing towards improvements on the farm. The low productivity of the agricultural sector forced India to import food from the United States of America (U.S.A.). Equity in agriculture called for **land reforms** which primarily refer to change in the ownership of landholdings. Just a year after independence, steps were taken to abolish intermediaries and to make the tillers the owners of land. The idea behind this move was that ownership of land would give incentives (see Box 2.5) to the tillers to invest in making improvements provided sufficient capital was made available to them.

Land ceiling was another policy to promote equity in the agricultural sector. This means fixing the maximum size of land which could be owned by an individual. The purpose of land ceiling was to reduce the concentration of land ownership in a few hands.

The abolition of intermediaries meant that some 200 lakh tenants came into direct contact with the government — they were thus freed from being exploited by the *zamindars*. The ownership conferred on tenants gave them the incentive to increase output and this contributed to growth in agriculture. However, the goal of equity was not fully served by abolition of intermediaries. In some areas the former *zamindars*

continued to own large areas of land by making use of some loopholes in the legislation; there were cases where tenants were evicted and the landowners claimed to be self-cultivators (the actual tillers), claiming ownership of the land; and even when the tillers got ownership of land, the poorest of the agricultural labourers (such as sharecroppers and landless labourers) did not benefit from land reforms.

The land ceiling legislation also faced hurdles. The big landlords challenged the legislation in the courts, delaying its implementation. They used this delay to register their lands in the name of close relatives, thereby escaping from the legislation. The legislation also had a lot of loopholes which were exploited by the big landholders to retain their land. Land reforms were successful in Kerala and West Bengal because these states had governments committed to the policy of land to the tiller. Unfortunately other states did not have the same level of commitment and vast inequality in landholding continues to this day.

The Green Revolution: At independence, about 75 per cent of the country's population was dependent on agriculture. Productivity in the agricultural sector was very low because of the use of old technology and the absence of required infrastructure for the vast majority of farmers. India's agriculture vitally depends on the monsoon and if the monsoon fell short the farmers were in trouble unless they had access to irrigation facilities which very few had. The stagnation in agriculture during the colonial rule was permanently broken by the green revolution. This refers to the large increase in production of food grains resulting from the use of high yielding variety (HYV) seeds especially for wheat and rice. The use of these seeds required the use of fertiliser and pesticide in the correct quantities as well as regular supply of water; the application of these inputs in correct proportions is vital. The farmers who could benefit from HYV seeds required reliable irrigation facilities as well as the financial resources to purchase fertiliser and pesticide. As a result, in the first phase of the green revolution (approximately mid 1960s upto mid 1970s), the use of HYV seeds was restricted to the more affluent states such as Punjab, Andhra Pradesh and Tamil Nadu. Further, the use of HYV seeds primarily benefited the wheat-growing regions only. In the second phase of the green revolution (mid-1970s to mid-1980s), the HYV technology spread to a larger number of states and benefited more variety of crops. The spread of green revolution technology enabled India to achieve self-sufficiency in food grains; India no longer had to be at the mercy of America, or any other nation, for meeting its food requirements.

Growth in agricultural output is important but it is not enough. If a large proportion of this increase is



consumed by the farmers themselves instead of being sold in the market, the higher output will not make much of a difference to the economy as a whole. If, on the other hand, a substantial amount of agricultural produce is sold in the market by the farmers, the higher output can make a difference to the economy. The portion of agricultural produce which is sold in the market by the farmers is called **marketed surplus**. A good proportion of the rice and wheat produced during the green revolution period (available as marketed surplus) was sold by the farmers in the market. As a result, the price of food grains declined relative to other items of consumption. The low-income groups, who spend a large percentage of their income on food, benefited from this decline in relative prices. The green revolution enabled the government to procure sufficient amount of food grains to build a stock which could be used in times of food shortage.

While the nation had immensely benefited from the green revolution, the technology involved was not free from risks. One such risk was the possibility that it would increase the disparities between small and big farmers—since only the big farmers could afford the required inputs, thereby reaping most of the benefits of the green revolution. Moreover, the HYV crops were also more prone to attack by pests and the small farmers who adopted this technology could lose everything in a pest attack.

Fortunately, these fears did not come true because of the steps taken by the government. The government provided loans at a low interest rate to small farmers and subsidised fertilisers so that small farmers could also have access to the needed inputs. Since the small farmers could obtain the required inputs, the output on small farms equalled the output on large farms in the course of time. As a result, the green revolution benefited the small as well as rich farmers. The risk of the small farmers being ruined when pests attack their crops was considerably reduced by the services rendered by research institutes established by the government. You should note that the green revolution would have favoured the rich farmers only if the state did not play an extensive role in ensuring that the small farmer also gains from the new technology.

The Debate Over Subsidies: The economic justification of subsidies in agriculture is, at present, a hotly debated question. It is generally agreed that it was necessary to use subsidies to provide an incentive for adoption of the new HYV technology by farmers in general and small farmers in particular. Any new technology will be looked upon as being risky by farmers. Subsidies were, therefore, needed to encourage farmers to test the new technology. Some economists believe that once the technology is found profitable and is widely adopted, subsidies should be phased out since their purpose has been served. Further, subsidies are meant to benefit the farmers but a substantial amount of fertiliser subsidy also benefits the fertiliser industry; and among farmers, the subsidy largely benefits the farmers in the more prosperous regions. Therefore, it is argued that there is no case for continuing with fertiliser subsidies; it does not benefit the target group and it is a huge burden on the government's finances (see also Box 2.6).

On the other hand, some believe that the government should continue with agricultural subsidies because farming in India continues to be a risky business. Most farmers are very poor and they will not be able to afford the required inputs without subsidies. Eliminating subsidies will increase the inequality between rich and poor farmers and violate the goal of equity. These experts argue that if subsidies are largely benefiting the fertiliser industry and big farmers, the correct policy is not to abolish subsidies but to take steps to ensure that only the poor farmers enjoy the benefits.

Thus, by the late 1960s, Indian agricultural productivity had increased sufficiently to enable the country to be self-sufficient in food grains. This is an achievement to be proud of. On the negative side, some 65 per cent of the country's population continued to be employed in agriculture even as late as 1990. Economists have found that as

Box 2.6: Prices as Signals

You would have learnt in an earlier class about how prices of goods are determined in the market. It is important to understand that prices are signals about the availability of goods. If a good becomes scarce, its price will rise and those who use this good will have the incentive to make efficient decisions about its use based on the price. If the price of water goes up because of lower supply, people will have the incentive to use it with greater care; for example, they may stop watering the garden to conserve water. We complain whenever the price of petrol increases and blame it on the government. But the increase in petrol price reflects greater scarcity and the price rise is a signal that less petrol is available—this provides an incentive to use less petrol or look for alternate fuels.

Some economists point out that subsidies do not allow prices to indicate the supply of a good. When electricity and water are provided at a subsidised rate or free, they will be used wastefully without any concern for their scarcity. Farmers will cultivate water intensive crops if water is supplied free, although the water resources in that region may be scarce and such crops will further deplete the already scarce resources. If water is priced to reflect scarcity, farmers will cultivate crops suitable to the region. Fertiliser and pesticide subsidies result in overuse of resources which can be harmful to the environment. Subsidies provide an incentive for wasteful use of resources. Think about subsidies in terms of incentives and ask yourself whether it is wise from the economic viewpoint to provide free electricity to farmers.

a nation becomes more prosperous, the proportion of GDP contributed by agriculture as well as the proportion of population working in the sector declines considerably. In India, between 1950 and 1990, the proportion of GDP contributed by agriculture declined significantly but not the population depending on it (67.5 per cent in 1950 to 64.9 per cent by 1990). Why was such a large proportion of the population engaged in agriculture although agricultural output could have grown with much less people working in the sector? The answer is that the industrial sector and the service sector did not absorb the people working in the agricultural sector. Many economists call this an important failure of our policies followed during 1950-1990.

2.4 Industry and Trade

Economists have found that poor nations can progress only if they have a good industrial sector. Industry provides employment which is more stable than the employment in agriculture; it promotes modernisation and overall prosperity. It is for this reason that the five year plans placed a lot of emphasis on industrial development. You might have studied in the previous chapter that, at the time of independence, the variety of industries was very narrow



Work These Out

- A group of students may visit an agricultural farm, prepare a case study on the method of farming used, that is, types of seeds, fertilisers, machines, means of irrigation, cost involved, marketable surplus and income earned. It will be beneficial if the changes in cultivation methods could be collected from an elderly member of the farming family
  (a) Discuss the findings in your class.
  (b) The different groups can then prepare a chart showing variations in cost of production, productivity, use of seeds, fertilisers, means of irrigation, time taken, marketable surplus and income of the family.
- Collect newspaper cuttings related to the World Bank, International Monetary Fund, World Trade Organisation (and meets of G7, G8, G10 countries). Discuss the views shared by the developed and developing countries on farm subsidies.
- Prepare pie charts on the occupational structure of the Indian economy available in the following table. Discuss the possible reasons for the change in the shape of pies.

Sector	*1950–51*	*1990–91*
Agriculture	72.1	66.8
Industry	10.7	12.7
Services	17.2	20.5

- Study the arguments for and against agricultural subsidies. What is your view on this issue?
- Some economists argue that farmers in other countries, particularly developed countries, are provided with high amount of subsidies and are encouraged to export their produce to other countries. Do you think our farmers will be able to compete with farmers from developed countries? Discuss.

— largely confined to cotton textiles and jute. There were two well-managed iron and steel firms — one in Jamshedpur and the other in Kolkata — but, obviously, we needed to expand the industrial base with a variety of industries if the economy was to grow.

Public and Private Sectors in Indian Industrial Development: The big question facing the policy makers was — what should be the role of the government and the private sector in industrial development? At the time of independence, Indian industrialists did not have the capital to undertake

investment in industrial ventures required for the development of Indian economy; nor was the market big enough to encourage industrialists to undertake major projects even if they had the capital to do so. It is principally for these reasons that the erstwhile governments had to play an extensive role in promoting the industrial sector. In addition, the decision to develop the Indian economy on socialist lines led to the policy of the government controlling the commanding heights of the economy, as the Second Five Year plan put it. This meant that the government would have complete control of those industries that were vital for the economy. The policies of the private sector would have to be complimentary to those of the public sector, with the public sector leading the way.

Industrial Policy Resolution 1956 (IPR 1956): In accordance with the goal of the state controlling the commanding heights of the economy, the Industrial Policy Resolution of 1956 was adopted. This resolution formed the basis of the Second Five Year Plan, the plan which tried to build the basis for a socialist pattern of society. This resolution classified industries into three categories. The first category comprised industries which would be exclusively owned by the government; the second category consisted of industries in which the private sector could supplement the efforts of the public sector, with the government taking the sole responsibility for starting new units; the third category consisted of the remaining industries which were to be in the private sector.

Although there was a category of industries left to the private sector, the sector was kept under state control through a system of licenses. No new industry was allowed unless a license was obtained from the government. This policy was used for promoting industry in backward regions; it was easier to obtain a license if the industrial unit was established in an economically backward area. In addition, such units were given certain concessions such as tax benefits and electricity at a lower tariff. The purpose of this policy was to promote **regional equality**.

Even an existing industry had to obtain a license for expanding output or for diversifying production (producing a new variety of goods). This was meant to ensure that the quantity of goods produced was not more than what the economy required. License to expand production was given only if the government was convinced that the economy required a larger quantity of goods.

Small-Scale Industry: In 1955, the Village and Small-Scale Industries Committee, also called the Karve Committee, noted the possibility of using small-scale industries for promoting rural development. A 'small-scale industry' is defined with reference to the maximum investment allowed on the assets of a unit. This limit has changed over a period of time. In 1950 a small-scale industrial unit was one which invested

a maximum of rupees five lakh; at present the maximum investment allowed is rupees one crore.

It is believed that small-scale industries are more 'labour intensive' i.e., they use more labour than the large-scale industries and, therefore, generate more employment. But these industries cannot compete with the big industrial firms; it is obvious that development of small-scale industry requires them to be shielded from the large firms. For this purpose, the production of a number of products was reserved for the small-scale industry; the criterion of reservation being the ability of these units to manufacture the goods. They were also given concessions such as lower excise duty and bank loans at lower interest rates.

2.5 Trade Policy: Import Substitution

The industrial policy that India adopted was closely related to the **trade policy**. In the first seven plans, trade was characterised by what is commonly called an inward looking trade strategy. Technically, this strategy is called **import substitution**. This policy aimed at replacing or substituting imports with domestic production. For example, instead of importing vehicles made in a foreign country, industries would be encouraged to produce them in India itself. In this policy the government protected the domestic industries from foreign competition. Protection from imports took two forms: tariffs and quotas. Tariffs are a tax on imported goods; they make imported goods more expensive and discourage their use. Quotas specify the quantity of goods which can be imported. The effect of tariffs and quotas is that they restrict imports and, therefore, protect the domestic firms from foreign competition.

The policy of protection was based on the notion that industries of developing countries were not in a position to compete against the goods produced by more developed economies. It was assumed that if the domestic industries were protected they would learn to compete in the course of time. Our planners also feared the possibility of foreign exchange being spent on import of luxury goods if no restrictions were placed on imports. Nor was any serious thought given to promote exports until the mid-1980s.

Effect of Policies on Industrial Development: The achievements of India's industrial sector during the first seven plans are impressive indeed. The proportion of GDP contributed by the industrial sector increased in the period from 13 per cent in 1950-51 to 24.6 per cent in 1990-91. The rise in the industry's share of GDP is an important indicator of development. The six per cent annual growth rate of the industrial sector during the period is commendable. No longer was Indian industry restricted largely to cotton textiles and jute; in fact, the industrial sector became well diversified by 1990, largely due to

Work These Out

➢ Construct a pie chart for the following table on sectoral contribution to GDP and discuss the difference in the contribution of the sectors in the light of effects of development during 1950-91.

Sector	*1950-51*	*1990-91*
Agriculture 	59.0	34.9
Industry 	13.0	24.6
Services 	28.0	40.5

➢ Conduct a debate in your classroom on the usefulness of Public Sector Undertakings (PSUs) by dividing the class into two groups. One group may speak in favour of PSUs and the other group against the motion (involve as many students as possible and encourage them to give examples).

the public sector. The promotion of small-scale industries gave opportunities to those people who did not have the capital to start large firms to get into business. Protection from foreign competition enabled the development of indigenous industries in the areas of electronics and automobile sectors which otherwise could not have developed.

In spite of the contribution made by the public sector to the growth of the Indian economy, some economists are critical of the performance of many public sector enterprises. It was proposed at the beginning of this chapter that initially public sector was required in a big way. It is now widely held that state enterprises continued to produce certain goods and services (often monopolising them) although this was no longer required. An example is the provision of telecommunication service. This industry continued to be reserved for the Public Sector even after it was realised that private sector firms could also provide it. Due to the absence of competition, even till the late 1990s, one had to wait for a long time to get a telephone connection. Another instance could be the establishment of Modern Bread, a bread-manufacturing firm, as if the

private sector could not manufacture bread! In 2001 this firm was sold to the private sector. The point is that after four decades of Planned development of Indian Economy no distinction was made between (i) what the public sector alone can do and (ii) what the private sector can also do. For example, even now only the public sector supplies national defense. And even though the private sector can manage hotels well, yet, the government also runs hotels. This has led some scholars to argue that the state should get out of areas which the private sector can manage and the government may concentrate its resources on important services which the private sector cannot provide.

Many public sector firms incurred huge losses but continued to function because it is difficult to close a government undertaking even if it is a drain on the nation's limited resources. This does not mean that private firms are always profitable (indeed, quite a few of the public sector firms were originally private firms which were on the verge of closure due to losses; they were then nationalised to protect the jobs of the workers). However, a loss-making private firm will not waste resources by being kept running despite the losses.

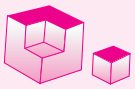
The need to obtain a license to start an industry was misused by industrial houses; a big industrialist would get a license not for starting a new firm but to prevent competitors from starting new firms. The excessive regulation of what came to be called the **permit license raj** prevented certain firms from becoming more efficient. More time was spent by industrialists in trying to obtain a license or lobby with the concerned ministries rather than on thinking about how to improve their products.

The protection from foreign competition was also being criticised on the ground that it continued even after it proved to do more harm than good. Due to restrictions on imports, the Indian consumers had to purchase whatever the Indian producers produced. The producers were aware that they had a captive market; so they had no incentive to improve the quality of their goods. Why should they think of improving quality when they could sell low quality items at a high price? Competition from imports forces our producers to be more efficient.

A few economists also point out that the public sector is not meant for earning profits but to promote the welfare of the nation. The public sector firms, on this view, should be evaluated on the basis of the extent to which they contribute to the welfare of people and not on the profits they earn. Regarding protection, some economists hold that we should protect our producers from foreign competition as long as the rich nations continue to do so. Owing to all these conflicts, economists called for a change in our policy. This, alongwith other problems, led the government to introduce a new economic policy in 1991.

2.6 Conclusion

The progress of the Indian economy during the first seven plans was impressive indeed. Our industries became far more diversified compared to the situation at independence. India became self- sufficient in food production thanks to the green revolution. Land reforms resulted in abolition of the hated *zamindari* system. In industrial sector, many economists became dissatisfied with the performance of many public sector enterprises. Excessive government regulation prevented growth of entrepreneurship. In the name of self-reliance, Indian producers were protected against foreign competition and this did not give them the incentive to improve the quality of goods that they produced. Indian policies were 'inward oriented' that failed to develop a strong export sector. The need for reform of economic policy was widely felt in the context of changing global economic scenario, and the new economic policy was initiated in 1991 to make Indian economy more efficient. This is the subject of the next chapter.



Recap

- After independence, India envisaged an economic system which combines the best features of socialism and capitalism—this culminated in the mixed economy model.
- All the economic planning has been formulated through five year plans.
- Common goals of five year plans are growth, modernisation, self-sufficiency and equity.
- The major policy initiatives in agriculture sector were land reforms and green revolution. These initiatives helped India to become self-sufficient in food grains production.
- The proportion of people depending on agriculture did not decline as expected.
- Import substitution policy initiatives in the industrial sector raised its contribution to GDP.
- One of the major drawbacks in the industrial sector was the inefficient functioning of the public sector as it started incurring losses leading to drain on the nation's limited resources.



EXERCISES

1. Define a plan.
2. Why did India opt for planning?
3. Why should plans have goals?
4. What are High Yielding Variety (HYV) seeds?
5. What is marketable surplus?
6. Explain the need and type of land reforms implemented in the agriculture sector.
7. What is Green Revolution? Why was it implemented and how did it benefit the farmers? Explain in brief.
8. Explain 'growth with equity' as a planning objective.
9. Does modernisation as a planning objective create contradiction in the light of employment generation? Explain.
10. Why was it necessary for a developing country like India to follow self-reliance as a planning objective?
11. What is sectoral composition of an economy? Is it necessary that the service sector should contribute maximum to GDP of an economy? Comment.
12. Why was public sector given a leading role in industrial development during the planning period?
13. Explain the statement that green revolution enabled the government to procure sufficient food grains to build its stocks that could be used during times of shortage.
14. While subsidies encourage farmers to use new technology, they are a huge burden on government finances. Discuss the usefulness of subsidies in the light of this fact.
15. Why, despite the implementation of green revolution, 65 per cent of India's population continued to be engaged in the agriculture sector till 1990?
16. Though public sector is very essential for industries, many public sector undertakings incur huge losses and are a drain on the economy's resources. Discuss the usefulness of public sector undertakings in the light of this fact.

17. Explain how import substitution can protect domestic industry.

18. Why and how was private sector regulated under the IPR 1956?

19. Match the following:

1. Prime Minister	A. Seeds that give large proportion of output
2. Gross Domestic Product	B. Quantity of goods that can be imported
3. Quota	C. Chairperson of the planning commission
4. Land Reforms	D. The money value of all the final goods and services produced within the economy in one year
5. HYV Seeds	E. Improvements in the field of agriculture to increase its productivity
6. Subsidy	F. The monetary assistance given by government for production activities.



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UNIT II

Economic Reforms Since 1991

After forty years of planned development, India has been able to achieve a strong industrial base and became self-sufficient in the production of food grains. Nevertheless, a major segment of the population continues to depend on agriculture for its livelihood. In 1991, a crisis in the balance of payments led to the introduction of economic reforms in the country. This unit is an appraisal of the reform process and its implications for India.

3



LIBERALISATION, PRIVATISATION AND GLOBALISATION: AN APPRAISAL

After studying this chapter, the learners will

- understand the background of the reform policies introduced in India in 1991
- understand the mechanism through which reform policies were introduced
- comprehend the process of globalisation and its implications for India
- be aware of the impact of the reform process in various sectors.

There is a consensus in the world today that economic development is not all and the GDP is not necessarily a measure of progress of a society.

K.R. Narayanan, Former President of India

3.1 Introduction

You have studied in the previous chapter that, since independence, India followed the mixed economy framework by combining the advantages of the capitalist economic system with those of the socialist economic system. Some scholars argue that, over the years, this policy resulted in the establishment of a variety of rules and laws, which were aimed at controlling and regulating the economy, ended up instead in hampering the process of growth and development. Others state that India, which started its developmental path from near stagnation, has since been able to achieve growth in savings, developed a diversified industrial sector which produces a variety of goods and has experienced sustained expansion of agricultural output which has ensured food security.

In 1991, India met with an economic crisis relating to its external debt — the government was not able to make repayments on its borrowings from abroad; **foreign exchange reserves**, which we generally maintain to import petroleum and other important items, dropped to levels that were not sufficient for even a fortnight. The crisis was further compounded by rising prices of essential goods. All these led the government to introduce a new set of policy measures which changed the direction of our developmental strategies. In this chapter, we will look at the background of the crisis, measures that the government has adopted and their impact on various sectors of the economy.

3.2 Background

The origin of the financial crisis can be traced from the inefficient management of the Indian economy in the 1980s. We know that for implementing various policies and its general administration, the government generates funds from various sources such as taxation, running of public sector enterprises etc. When expenditure is more than income, the government borrows to finance the **deficit** from banks and also from people within the country and from international **financial institutions**. When we import goods like petroleum, we pay in dollars which we earn from our exports.

Development policies required that even though the revenues were very low, the government had to overshoot its revenue to meet challenges like unemployment, poverty and population explosion. The continued spending on development programmes of the government did not generate additional revenue. Moreover, the government was not able to

generate sufficiently from internal sources such as taxation. When the government was spending a large share of its income on areas which do not provide immediate returns such as the social sector and defence, there was a need to utilise the rest of its revenue in a highly efficient manner. The income from public sector undertakings was also not very high to meet the growing expenditure. At times, our **foreign exchange**, borrowed from other countries and international financial institutions, was spent on meeting consumption needs. Neither was an attempt made to reduce such profligate spending nor sufficient attention was given to boost exports to pay for the growing imports.

In the late 1980s, government expenditure began to exceed its revenue by such large margins that meeting the expenditure through borrowings became unsustainable. Prices of many essential goods rose sharply. Imports grew at a very high rate without matching growth of exports. As pointed out earlier, foreign exchange reserves declined to a level that was not adequate to finance imports for more than two weeks. There was also not sufficient foreign exchange to pay the interest that needed to be paid to international lenders. Also no country or international funder was willing to lend to India.

India approached the **International Bank for Reconstruction and Development (IBRD)**, popularly known as **World Bank** and the **International Monetary Fund (IMF)**, and received $7 billion as loan to manage the crisis. For availing the loan, these international agencies expected India to liberalise and open up the economy by removing restrictions on the private sector, reduce the role of the government in many areas and remove trade restrictions between India and other countries.

India agreed to the conditionalities of World Bank and IMF and announced the **New Economic Policy (NEP)**. The NEP consisted of wide ranging economic reforms. The thrust of the policies was towards creating a more competitive environment in the economy and removing the barriers to entry and growth of firms. This set of policies can broadly be classified into two groups: the stabilisation measures and the structural reform measures. Stabilisation measures are short-term measures, intended to correct some of the weaknesses that have developed in the **balance of payments** and to bring **inflation** under control. In simple words, this means that there was a need to maintain sufficient foreign exchange reserves and keep the rising prices under control. On the other hand, structural reform policies are long-term measures, aimed at improving the efficiency of the economy and increasing its international competitiveness by removing the rigidities in various segments of the Indian economy. The government initiated a variety of policies which fall under three heads

viz., liberalisation, privatisation and globalisation.

3.3 Liberalisation

As pointed out in the beginning, rules and laws which were aimed at regulating the economic activities became major hindrances in growth and development. Liberalisation was introduced to put an end to these restrictions and open various sectors of the economy. Though a few liberalisation measures were introduced in 1980s in areas of industrial licensing, **export-import policy**, technology upgradation, **fiscal policy** and foreign investment, reform policies initiated in 1991 were more comprehensive. Let us study some important areas, such as the industrial sector, financial sector, tax reforms, **foreign exchange markets** and trade and investment sectors which received greater attention in and after 1991.

Deregulation of Industrial Sector: In India, regulatory mechanisms were enforced in various ways (i) industrial licensing under which every entrepreneur had to get permission from government officials to start a firm, close a firm or decide the amount of goods that could be produced (ii) private sector was not allowed in many industries (iii) some goods could be produced only in small-scale industries, and (iv) controls on price fixation and distribution of selected industrial products.

The reform policies introduced in and after 1991 removed many of these restrictions. Industrial licensing was abolished for almost all but product categories — alcohol, cigarettes, hazardous chemicals, industrial explosives, electronics, aerospace and drugs and pharmaceuticals. The only industries which are now reserved for the public sector are a part of atomic energy generation and some core activities in railway transport. Many goods produced by small-scale industries have now been **dereserved**. In most industries, the market has been allowed to determine the prices.

Financial Sector Reforms: Financial sector includes financial institutions, such as commercial banks, investment banks, **stock exchange** operations and foreign exchange market. The financial sector in India is regulated by the Reserve Bank of India (RBI). You may be aware that all banks and other financial institutions in India are regulated through various norms and regulations of the RBI. The RBI decides the amount of money that the banks can keep with themselves, fixes interest rates, nature of lending to various sectors, etc. One of the major aims of financial sector reforms is to reduce the role of RBI from regulator to facilitator of financial sector. This means that the financial sector may be allowed to take decisions on many matters without consulting the RBI.

The reform policies led to the establishment of private sector banks, Indian as well as foreign. Foreign investment limit in banks was raised to around 74 per cent. Those banks which fulfil certain conditions have been given freedom to set up new branches without the approval of the RBI and rationalise their existing branch networks. Though banks have been given permission to generate resources from India and abroad, certain managerial aspects have been retained with the RBI to safeguard the interests of the account-holders and the nation. **Foreign Institutional Investors (FII),** such as merchant bankers, mutual funds and pension funds, are now allowed to invest in Indian financial markets.

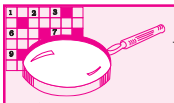
Tax Reforms: Tax reforms are concerned with the reforms in the government's taxation and public expenditure policies, which are collectively known as its **fiscal policy**. There are two types of taxes: direct and indirect. **Direct taxes** consist of taxes on incomes of individuals, as well as, profits of business enterprises. Since 1991, there has been a continuous reduction in the taxes on individual incomes as it was felt that high rates of income tax were an important reason for tax evasion. It is now widely accepted that moderate rates of income tax encourage savings and voluntary disclosure of income. The rate of **corporation tax**, which was very high earlier, has been gradually reduced. Efforts have also been made to reform the **indirect taxes**, taxes levied on commodities, in order to facilitate the establishment of a common national market for goods and commodities.

In 2016, the Indian Parliament passed a law, Goods and Services Tax Act 2016, to simplify and introduce a unified indirect tax system in India. This law came into effect from July 2017. This is expected to generate additional revenue for the government, reduce tax evasion and create 'one nation, one tax and one market'. Another component of reform in this area is simplification. In order to encourage better compliance on the part of taxpayers, many procedures have been simplified and the rates also substantially lowered.

Foreign Exchange Reforms: The first important reform in the external sector was made in the foreign exchange market. In 1991, as an immediate measure to resolve the balance of payments crisis, the rupee was **devalued** against foreign currencies. This led to an increase in the inflow of foreign exchange. It also set the tone to free the determination of rupee value in the foreign exchange market from government control. Now, more often than not, markets determine exchange rates based on the demand and supply of foreign exchange.

Trade and Investment Policy Reforms: Liberalisation of trade and investment regime was initiated to increase international competitiveness of industrial production and also foreign investments and technology into the economy. The aim was also to promote the efficiency of local industries and adoption of modern technologies.

In order to protect domestic industries, India was following a regime of **quantitative restrictions** on imports. This was encouraged through tight control over imports and by keeping the **tariffs** very high. These policies reduced efficiency and competitiveness which led to slow growth of the manufacturing sector. The trade policy reforms aimed at (i) dismantling of quantitative restrictions on imports and exports (ii) reduction of tariff rates and (iii) removal of licensing procedures for imports. **Import licensing** was abolished except in case of hazardous and environmentally sensitive industries. Quantitative restrictions on imports of manufactured consumer goods and agricultural products were also fully removed from April 2001. **Export duties** have been removed to increase the competitive position of Indian goods in the international markets.



Work These Out

- Give an example each of nationalised bank, private bank, private foreign bank, FII and a mutual fund.
- Visit a bank in your locality with your parents. Observe and find out the functions it performs. Discuss the same with your classmates and prepare a chart on it.
- Find out from your parents if they pay taxes. If yes, why do they do so and how?
- Do you know that for a very long time countries used to keep silver and gold as reserves to make payments abroad? Find out in what form do we keep our foreign exchange reserves and find out from newspapers, magazines and the Economic Survey how much foreign exchange reserves India had during the last year. Also find the foreign currency of the following countries and its latest rupee exchange rate.

Country	*Currency*	*Value of 1(one) unit of foreign currency in Indian rupee*
U.S.A. U.K. Japan China Korea Singapore Germany		

3.4 Privatisation

It implies shedding of the ownership or management of a government owned enterprise. Government companies are converted into private companies in two ways (i) by withdrawal of the government from ownership and management of public sector companies and or (ii) by outright sale of public sector companies.

Privatisation of the public sector enterprises by selling off part of the equity of PSEs to the public is known as **disinvestment**. The purpose of the sale, according to the government, was mainly to improve financial discipline and facilitate modernisation. It was also envisaged that private capital and managerial capabilities could be effectively utilised to improve the performance of the PSUs.

Box 3.1: *Navratnas* and Public Enterprise Policies

You must have read in your childhood about the famous *Navratnas* or Nine Jewels in the Imperial Court of King Vikramaditya who were eminent persons of excellence in the fields of art, literature and knowledge. In order to improve efficiency, infuse professionalism and enable them to compete more effectively in the liberalised global environment, the government identifies PSEs and declare them as *maharatnas, navratnas* and *miniratnas.* They were given greater managerial and operational autonomy, in taking various decisions to run the company efficiently and thus increase their profits. Greater operational, financial and managerial autonomy has also been granted to profit-making enterprises referred to as *miniratnas.*

The Central Public Sector Enterprises are designated with different status. A few examples of public enterprises with their status are as follows: (i) *Maharatnas* – (a) Indian Oil Corporation Limited, and (b) Steel Authority of India Limited, (ii) *Navratnas* – (a) Hindustan Aeronautics Limited, (b) Mahanagar Telephone Nigam Limited; and (iii) *Miniratnas* – (a) Bharat Sanchar Nigam Limited; (b) Airport Authority of India and (c) Indian Railway Catering and Tourism Corporation Limited.

Many of these profitable PSEs were originally formed during the 1950s and 1960s when self-reliance was an important element of public policy. They were set up with the intention of providing infrastructure and direct employment to the public so that quality end-product reaches the masses at a nominal cost and the companies themselves were made accountable to all stakeholders.

The granting of status resulted in better performance of these companies. Scholars allege that instead of facilitating public enterprises in their expansion and enabling them to become global players, the government partly privatised them through disinvesment. Of late, the government has decided to retain them in the public sector and enable them to expand themselves in the global markets and raise resources by themselves from financial markets.



Work These Out

- Some scholars refer to disinvestment as the wave of privatisation spreading all over the world to improve the performance of public sector enterprises whereas others call it as outright sale of public property to the vested interests. What do you think?
- Prepare a poster which contains 10-15 news clippings which you consider as important and relating to *navaratnas* from newspapers. Also collect the logos and advertisements of these PSEs. Put these on the notice board and discuss them in the classroom.
- Do you think only loss making companies should be privatised? Why?
- Losses incurred by public sector enterprises are to be met out of the public budget. Do you agree with this statement? Discuss.

The government envisaged that privatisation could provide strong impetus to the inflow of FDI.

The government has also made attempts to improve the efficiency of PSUs by giving them autonomy in taking managerial decisions. For instance, some PSUs have been granted special status as *maharatnas, navratnas* and *miniratnas* (see Box 3.1).

3.5 Globalisation

Although globalisation is generally understood to mean integration of the economy of the country with the world economy, it is a complex phenomenon. It is an outcome of the set of various policies that are aimed at transforming the world towards greater interdependence and integration. It involves creation of networks and activities transcending economic, social and geographical boundaries. Globalisation attempts to establish links in such a way that the happenings in India can be influenced by events happening miles away. It is turning the world into one whole or creating a borderless world.

Outsourcing: This is one of the important outcomes of the globalisation process. In outsourcing, a company hires regular service from external sources, mostly from other countries, which was previously provided internally or from within the country (like legal advice, computer service, advertisement, security — each provided by respective departments of the company). As a form of economic activity, outsourcing has intensified, in recent times, because of the growth of fast modes of communication, particularly the growth of Information Technology (IT). Many of the services such as voice-based business processes (popularly known as BPO or call centres), record keeping,

Box 3.2: Global Footprint!

Owing to globalisation, you might find many Indian companies have expanded their wings to many other countries. For example, ONGC Videsh, a subsidiary of the Indian public sector enterprise, Oil and Natural Gas Corporation engaged in oil and gas exploration and production has projects in 16 countries. Tata Steel, a private company established in 1907, is one of the top ten global steel companies in the world which have operations in 26 countries and sell its products in 50 countries. It employs nearly 50,000 persons in other countries. HCL Technologies, one of the top five IT companies in India has offices in 31 countries and employs about 15,000 persons abroad. Dr Reddy's Laboratories, initially was a small company supplying pharmaceutical goods to big Indian companies, today has manufacturing plants and research centres across the world.

Source: *www.rediff.com accessed on 14.10.2014.*

accountancy, banking services, music recording, film editing, book transcription, clinical advice or even teaching are being outsourced by companies in developed countries to India. With the help of modern telecommunication links including the Internet, the text, voice and visual data in respect of these services is digitised and transmitted in real time over continents and national boundaries. Most multinational corporations, and even small companies, are outsourcing their services to India where they can be availed at a cheaper cost with reasonable degree of skill and accuracy. The low wage rates and availability of skilled manpower in India have made it a destination for global outsourcing in the post-reform period.

World Trade Organisation (WTO): The WTO was founded in 1995 as the successor organisation to the General Agreement on Trade and Tariff (GATT). GATT was established in 1948 with 23 countries as the global trade organisation to administer all **multilateral trade agreements** by providing equal opportunities to all countries in the international market for trading purposes. WTO is expected to establish a rule-based trading regime in which nations cannot place arbitrary restrictions on trade. In addition, its purpose is also to enlarge

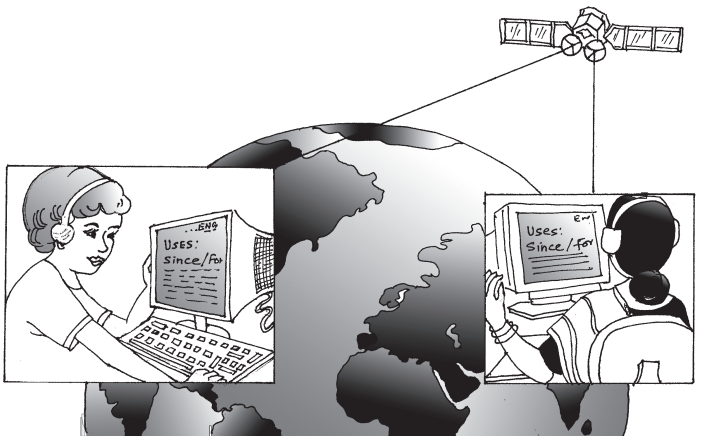
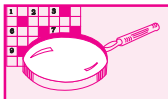


Fig. 3.1 *Outsourcing: a new employment opportunity in big cities*



Work These Out

- Many scholars argue that globalisation is a threat as it reduces the role of the government in many sectors. Some counter argue that it is an opportunity as it opens up markets to compete in and capture. Debate in the classroom.
- Prepare a chart consisting of a list of five companies that have BPO services in India, along with their turnover.
- Did you attend online classes or watched videos of your teachers or any other teacher taking classes during the last year through television, mobile phone or computers due to Covid 19 Pandemic? Share your experiences related to information technology.
- Is employment in call centres sustainable? What kinds of skills should people working in call centres acquire to get a regular income?
- If the multinational companies outsource many services to countries like India because of cheap manpower, what will happen to people living in the countries where the companies are located? Discuss.

production and trade of services, to ensure optimum utilisation of world resources and to protect the environment. The WTO agreements cover trade in goods as well as services to facilitate international trade (**bilateral** and multilateral) through removal of tariff as well as **non-tariff barriers** and providing greater market access to all member countries.

As an important member of WTO, India has been in the forefront of framing fair global rules, regulations and safeguards and advocating the interests of the developing world. India has kept its commitments towards liberalisation of trade, made in the WTO, by removing quantitative restrictions on imports and reducing tariff rates.

TABLE 3.1
Growth of GDP and Major Sectors (in %)

Sector	*1980-91*	*1992-2001*	*2002-07*	*2007-12*	*2012-13*	*2013-14*	*2014-15*
Agriculture	3.6	3.3	2.3	3.2	1.5	4.2	– 0.2*
Industry	7.1	6.5	9.4	7.4	3.6	5	7.0*
Services	6.7	8.2	7.8	10	8.1	7.8	9.8*
Total	**5.6**	**6.4**	**7.8**	**8.2**	**5.6**	**6.6**	**7.4**

Source: *Economic Survey for various years, Ministry of Finance, Government of India.*

Note: **Data pertaining to Gross Value Added (GVA). The GVA is estimated from GDP by adding subsidies on production and subtracting indirect taxes.*

Some scholars question the usefulness of India being a member of the WTO as a major volume of international trade occurs among the developed nations. They also say that while developed countries file complaints over agricultural subsidies given in their countries, developing countries feel cheated as they are forced to open their markets for developed countries but are not allowed access to the markets of developed countries. What do you think?



Fig. 3.2 *IT industry is seen as a major contributor to India's exports*

3.6 Indian Economy During Reforms: an Assessment

The reform process has completed three decades since its introduction. Let us now look at the performance of the Indian economy during this period. In economics, the growth of an economy is measured by the Gross Domestic Product. Look at Table 3.1. The post–1991 India witnessed a rapid growth in GDP on a continual basis for two decades. The growth of GDP increased from 5.6 per cent during 1980–91 to 8.2 per cent during 2007–12. During the reform period, the growth of agriculture has declined. While the industrial sector reported fluctuation, the growth of the service sector has gone up. This indicates that GDP growth is mainly driven by growth in the service sector. During 2012-15, there has been a setback in the growth rates of different sectors witnessed post–1991. While agriculture recorded a high growth rate during 2013–14, this sector witnessed negative growth in the subsequent year. While the service sector continued to witness a high level of growth — higher than the overall GDP growth in 2014–15, this sector witnessed the high growth rate of 9.8 per cent. The industrial sector witnessed a steep decline during 2012–13, in the subsequent years it began to show a continuous positive growth.

The opening of the economy has led to a rapid increase in foreign direct investment and foreign exchange reserves. The foreign investment, which includes **foreign direct investment (FDI)** and **foreign institutional investment (FII)**, has increased from about US $100 million in 1990-91 to US $ 30 billion in 2017-18. There has been an increase in the foreign exchange reserves from about US $ 6 billion in 1990-91 to about US $ 413 billion in 2018-19. India is one of the largest foreign exchange reserve holders in the world.

Since 1991, India is seen as a successful exporter of auto parts, pharmaceutical goods engineering goods, IT software and textiles. Rising prices have also been kept under control.



Work These Out

- In the previous chapter, you might have studied about subsidies in various sectors, including agriculture. Some scholars argue that subsidy in agriculture should be removed to make the sector internationally competitive. Do you agree? If so, how can it be done? Discuss in class.
- Read the following passage and discuss in class.

  Groundnut is a major oilseed crop in Andhra Pradesh. Mahadeva, who was a farmer in Anantpur district of Andhra Pradesh, used to spend Rs. 10,000 for growing groundnut on his plot of half an acre. The cost included expenditure on raw material (seeds, fertilisers, etc.), labour, bullock power and machinery used. On an average, Mahadeva used to get two quintals of groundnut, and each quintal was sold for Rs. 7,000. Mahadeva, thus, was spending Rs. 10,000 and getting an income of Rs. 14,000. Anantpur district is a drought-prone area. As a result of economic reforms, the government did not undertake any major irrigation project. Recently, groundnut crop in Anantpur is facing problems due to crop disease. Research and extension work has gone down due to lower government expenditure. Mahadeva and his friends brought this matter repeatedly to the notice of the government officials entrusted with this responsibility, but failed. Subsidy was reduced on material (seeds, fertilisers) which increased Mahadeva's cost of cultivation. Moreover, the local markets were flooded with cheap imported edible oils, which was a result of removal of restriction on imports. Mahadeva was not able to sell his groundnut in the market as he was not getting the price to cover his cost.

  What could be done to farmers like Mahadeva from incurring losses? Discuss in the class.

On the other hand, the reform process has been widely criticised for not being able to address some of the basic problems facing our economy especially in areas of employment, agriculture, industry, infrastructure development and fiscal management.

Growth and Employment: Though the GDP growth rate has increased in the reform period, scholars point out that the reform-led growth has not generated sufficient employment opportunities in the country. You will study the link between different aspects of employment and growth in the next unit.

Reforms in Agriculture: Reforms have not been able to benefit agriculture, where the growth rate has been decelerating.

Since 1991, public investment in agriculture sector especially in infrastructure, which includes irrigation, power, roads, market linkages and research and extension (which played a crucial role in the Green Revolution), has fallen. Further, the partial removal of fertiliser subsidy has led to increase in the cost of production, which has severely affected the small and marginal farmers. This sector has been experiencing a number of policy changes such as reduction in import duties on agricultural products, low minimum support price and lifting of quantitative restrictions on the imports of agricultural products. These have adversely affected Indian farmers as they now have to face increased international competition.

Moreover, because of export-oriented policy strategies in agriculture, there has been a shift from production for the domestic market towards production for the export market focusing on cash crops in lieu of production of food grains. This puts pressure on prices of food grains.

Reforms in Industry: Industrial growth has also recorded a slowdown. This is because of decreasing demand of industrial products due to various reasons such as cheaper imports, inadequate investment in infrastructure etc. In a globalised world, developing countries are compelled to open up their economies to greater flow of goods and capital from developed countries and rendering their industries vulnerable to imported goods. Cheaper imports have, thus, replaced the demand for domestic goods. Domestic manufacturers are facing competition from imports. The infrastructure facilities, including power supply, have remained inadequate due to lack of investment. Globalisation is, thus, often seen as creating conditions for the free movement of goods and services from foreign countries that adversely affect the local industries and employment opportunities in developing countries.

Moreover, a developing country like India still does not have the access to developed countries' markets because of high non-tariff barriers. For example, although all quota restrictions on exports of textiles and clothing have been removed in India, USA has not removed their quota restriction on import of textiles from India and China.

Disinvestment: Every year, the government fixes a target for disinvestment of PSEs. For instance, in 1991-92, it was targeted to mobilise Rs 2500 crore through disinvestment. The government was able to mobilise ₹ 3,040 crore more than the target. In 2017–18, the target was about ₹1,00,000 crore, and the achievement was about ₹ 1,00,057 crore. Critics

point out that the assets of PSEs have been undervalued and sold to the private sector. This means that there has been a substantial loss to the government and the outright sale of public assets! Moreover, the proceeds from disinvestment are used to offset the shortage of government revenues rather than using it for the development of PSEs and building social infrastructure in the country. Do you think selling a part of the properties of government companies is the best way to improve their efficiency?

Reforms and Fiscal Policies: Economic reforms have placed limits on the growth of public expenditure, especially in social sectors. The tax reductions in the reform period, aimed at yielding larger revenue and curb tax evasion, have not resulted in increase in tax revenue for the government. Also, the reform policies, involving tariff reduction, have curtailed the scope for raising revenue through custom duties. In order to attract foreign investment, tax incentives are provided to foreign investors which further reduced the scope for raising tax revenues. This has a negative impact on developmental and welfare expenditures.

3.7 Conclusion

The process of globalisation through liberalisation and privatisation policies has produced positive, as well as, negative results both for India and other countries. Some scholars argue that globalisation should be seen as an opportunity in terms of greater access to global markets, high technology and increased possibility of large industries of developing countries to become important players in the international arena.

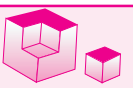
Box 3.3: Siricilla Tragedy!

Power sector reforms in many Indian states led to do away with the supply of electricity at subsidised rates and steep rise in power tariff. This has affected workers engaged in small industries. Powerloom textile industry in Andhra Pradesh is an example. Since the wages of the powerloom workers are linked to the production of cloth, power cut means cut in wages of weavers, who were already suffering from hike in tariff. A few years ago, this led to a crisis in the livelihood of the weavers and 50 powerloom workers committed suicide in a small town called 'Siricilla' in Andhra Pradesh.

- Do you think the power tariff should not be raised?
- What would be your suggestions to revive small industries affected by reforms?

On the contrary, the critics argue that globalisation is a strategy of the developed countries to expand their markets in other countries. According to them, it has compromised the welfare and identity of people belonging to poor countries. It has further been pointed out that market-driven globalisation has widened the economic disparities among nations and people.

Viewed from the Indian context, some studies have stated that the crisis that erupted in the early 1990s was basically an outcome of the deep-rooted inequalities in Indian society and the economic reform policies initiated as a response to the crisis by the government, with externally advised policy package, further aggravated the inequalities. Further, it has increased the income and quality of consumption of only high-income groups and the growth has been concentrated only in some select areas in the services sector such as telecommunication, information technology, finance, entertainment, travel and hospitality services, real estate and trade, rather than vital sectors such as agriculture and industry which provide livelihoods to millions of people in the country.



Recap

- The economy was facing problems of declining foreign exchange, growing imports without matching rise in exports and high inflation. India changed its economic policies in 1991 due to a financial crisis and pressure from international organisations like the World Bank and IMF.
- In the domestic economy, major reforms were undertaken in the industrial and financial sectors. Major external sector reforms included foreign exchange deregulations and import liberalisation.
- With a view to improving the performance of the public sector, there was a consensus on reducing its role and opening it up to the private sector. This was done through disinvestment and liberalisation measures.
- Globalisation is the outcome of the policies of liberalisation and privatisation. It means an integration of the economy of the country with the world economy.
- Outsourcing is emerging as a major activity in industrial and service sectors.
- The objective of the WTO is to establish a rule based trade regime to ensure optimum utilisation of world resources.
- During the reforms, growth of agriculture and industry has gone down but the service sector has registered growth.
- Reforms have not benefited the agriculture sector. There has also been a decline in public investment in this sector.
- Industrial sector growth has slowed down due to availability of cheaper imports and lower investment.



EXERCISES

1. Why were reforms introduced in India?
2. Why is it necessary to became a member of WTO?
3. Why did RBI have to change its role from controller to facilitator of financial sector in India?
4. How is RBI controlling the commercial banks?
5. What do you understand by devaluation of rupee?
6. Distinguish between the following
   (i) Strategic and Minority sale
   (ii) Bilateral and Multi-lateral trade
   (iii) Tariff and Non-tariff barriers.
7. Why are tariffs imposed?
8. What is the meaning of quantitative restrictions?
9. Those public sector undertakings which are making profits should be privatised. Do you agree with this view? Why?
10. Do you think outsourcing is good for India? Why are developed countries opposing it?
11. India has certain advantages which makes it a favourite outsourcing destination. What are these advantages?
12. Do you think the *navaratna* policy of the government helps in improving the performance of public sector undertakings in India? How?
13. What are the major factors responsible for the high growth of the service sector?
14. Agriculture sector appears to be adversely affected by the reform process. Why?
15. Why has the industrial sector performed poorly in the reform period?
16. Discuss economic reforms in India in the light of social justice and welfare.

SUGGESTED ADDITIONAL ACTIVITIES

1. The table given below shows the GDP growth rate at 2004-05 prices. You have studied about the techniques of presentation of data in your *Statistics for Economics* course. Draw a time series line graph based on the data given in the table and interpret the same.

Year	*GDP Growth Rate (%)*
2005-06	9.5
2006-07	9.6
2007-08	9.3
2008-09	6.7
2009-10	8.6
2010-11	8.9
2011-12	6.7
2012-13	5.4
2013-14	6.4
2014-15	7.4

2. Observe around you — you will find State Electricity Boards (SEBs), BSES and many public and private organisations supplying electricity in different states and union territories. There are private buses on roads alongside the goverment bus services and so on.

   (i) What do you think about this dual system of the co-existence of public and private sectors?

   (ii) What are the merits and demerits of such a dual system? Discuss.

3. With the help of your parents and grandparents prepare a list of multinational companies that existed in India at the time of independence. Now put a (✓) mark against those which are still growing and a ( ) against those which do not exist any more. Are there any companies whose names have changed? Find out the new names, the country of origin, nature of product, logo and prepare charts and display in your class.

4. Give appropriate examples for the following:

Nature of Product	*Name of a Foreign Company*
Biscuits	
Shoes	
Computers	
Cars	
TV and Refrigerators	
Stationery	

Now, find out if these companies which are mentioned above existed in India before 1991, or came after the New Economic Policy. For this, take the help of your teacher, parents, grandparents and shopkeepers.

5. Collect a few relevant newspaper cuttings and from the internet on meetings organised by WTO. Discuss the issues debated in these meetings, and find out how WTO facilitates world trade.

6. Was it necessary for India to introduce economic reforms at the behest of World Bank and International Monetary Fund? Was there no alternative for the government to solve the balance of payments crisis? Discuss in the classroom.



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UNIT III

Current Challenges Facing the Indian Economy

4



Human Capital Formation in India

After studying this chapter, the learners will understand

- the concepts of Human Resource, Human Capital Formation and Human Development
- the links between investment in human capital, economic growth and human development
- the need for government spending on education and health
- the state of India's educational attainment.

"... the wisdom of expending public and private funds on education is not to be measured by its direct fruits alone. It will be profitable as a mere investment, to give the masses of people much greater opportunities than they can generally avail themselves of. For by this means many, who would have died unknown, are enabled to get the start needed for bringing out their latent abilities".

Alfred Marshall

4.1 INTRODUCTION

Think of one factor that has made a great difference in the evolution of mankind. Perhaps it is human capacity to store and transmit knowledge which they have been doing through conversation, through songs and through elaborate lectures. But humans soon found out that we need a good deal of training and skill to do things efficiently. We know that the labour skill of an educated person is more than that of an uneducated person and hence the former is able to generate more income than the latter and his or her contribution to economic growth is, consequently, more.

Education is sought not only as it confers higher earning capacity on people but also for its other highly valued benefits: it gives one a better social standing and pride; it enables one to make better choices in life; it provides knowledge to understand the changes taking place in society; it also stimulates innovations. Moreover, the availability of educated labour force facilitates adaptation of new



Fig. 4.1 *Adequate education and training to farmers can raise productivity in farms*

technologies. Economists have stressed the need for expanding educational opportunities in a nation as it accelerates the development process.

4.2 What is Human Capital?

Just as a country can turn physical resources like land into physical capital like factories, similarly, it can also turn human resources like nurses, farmers, teachers, students into human capital like engineers and doctors. Societies need sufficient human capital in the first place—in the form of competent people who have themselves been educated and trained as professors and other professionals. In other words, we need good human capital to produce other human capital (say, nurses, farmers, teachers, doctors, engineers...). This means that we need investment in human capital to produce more human capital out of human resources.

Let us understand a little more of what human capital means by posing the following questions :

(i) What are the sources of human capital?
(ii) Is there any relation between human capital and economic growth of a country?
(iii) Is the formation of human capital linked to people's all-round development or, as it is now called, human development?
(iv) What role can the government play in human capital formation in India?

4.3 Sources of Human Capital

Investment in education is considered as one of the main sources of human capital. There are several other sources as well. Investments in health, on- the-job training, migration and information are the other sources of human capital formation.

Why do your parents spend money on education? Spending on education by individuals is similar to spending on capital goods by companies with the objective of increasing future profits over a period of time. Likewise, individuals invest in education with the objective of increasing their future income.



Work This Out

Identify and collect the data from three families from different strata (i) very poor (ii) middle class and (iii) affluent. Study the expenditure pattern of the families on education of male and female children.

Like education, health is also considered as an important input for the development of a nation as much as it is important for the development of an individual.

Who can work better—a sick person or a person with sound health? A sick labourer without access to medical facilities is compelled to abstain from work and there is loss of productivity. Hence, expenditure on health is an important source of human capital formation.



The amount of money spent on preventive medicine (vaccination), **curative medicine** (medical intervention during illness), **social medi-cine** (spread of health literacy) and provision of clean drinking water and good sanitation are the various forms of health expenditures. Health expenditure directly increases the supply of healthy labour force and is, thus, a source of human capital formation.



Firms spend on giving **on-the-job-training** to their workers. This may take different forms: one, the workers may be trained in the firm itself under the supervision of a skilled worker; two, the workers may be sent for off-campus training. In both these cases firms incur some expenses. Firms will, thus, insist that the workers should work for a specific period of time, after their on-the-job training, during which it can recover the benefits of the enhanced productivity owing to the training. Expenditure regarding on-the-job training is a source of human capital formation as the return of such expenditure in the form of enhanced labour productivity is more than the cost of it.

People **migrate** in search of jobs that fetch them higher salaries than what they may get in their native



places. Unemployment is the reason for the rural-urban migration in India. Technically qualified persons, like engineers and doctors, migrate to other countries because of higher salaries that they may get in such countries. Migration in both these cases involves cost of transport, higher cost of living in the migrated places and psychic costs of living in a strange socio-cultural setup. The enhanced earnings in the new place outweigh the costs of migration; hence, expenditure on migration is also a source of human capital formation.

People spend to acquire **information** relating to the labour market and other markets like education and health. For example, people want to know the level of salaries associated with various types of jobs, whether the educational institutions provide the right type of employable skills and at what cost. This information is necessary to make decisions regarding investments in human capital as well as for efficient utilisation of the acquired human capital stock. Expenditure incurred for acquiring information relating to the labour market and other markets is also a source of human capital formation.

Box 4.1: Physical and Human Capital

Both the forms of capital formation are outcomes of conscious investment decisions. Decision regarding investment in physical capital is taken on the basis of one's knowledge in this regard. The entrepreneur possesses knowledge to calculate the expected rates of return to a range of investments and then rationally decides which one of the investments should be made. The ownership of physical capital is the outcome of the conscious decision of the owner — the physical capital formation is mainly an economic and technical process. A substantial part of the human capital formation takes place in one's life when she/he is unable to decide whether it would maximise her/his earnings. Children are given different types of school education and health care facilities by their parents and the society. The peers, educators and society influence the decisions regarding human capital investments even at the tertiary level, that is, at the college level. Moreover, the human capital formation at this stage is dependent upon the already formed human capital at the school level. Human capital formation is partly a social process and partly a conscious decision of the possessor of the human capital.

You know that the owner of a physical capital, say a bus, need not be present in the place where it is used; whereas, a bus-driver, who possesses the knowledge and ability to drive the bus, should be present when the bus is used for transportation of people and materials. Physical capital is tangible and can be easily sold in the market like any other commodity. Human capital is intangible; it is endogenously built in the body and mind of its owner. Human capital is not sold in the market; only the services of the human capital are sold and, hence, there arises the necessity of the owner of the human capital to be present in the place of production. The physical capital is separable from its owner, whereas, human capital is inseparable from its owner.

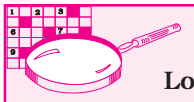
The two forms of capital differ in terms of mobility across space. Physical capital is completely mobile between countries except for some artificial trade restrictions. Human capital is not perfectly mobile between countries as movement is restricted by nationality and culture. Therefore, physical capital formation can be built even through imports, whereas human capital formation is to be done through conscious policy formulations in consonance with the nature of the society and economy and expenditure by the state and the individuals.

Both forms of capital depreciate with time but the nature of depreciation differs between the two. Continuous use of machine leads to depreciation and change of technology makes a machine obsolete. In the case of human capital, depreciation takes place with ageing but can be reduced, to a large extent, through continuous investment in education, health, etc. This investment also facilitates the human capital to cope with change in technology which is not the case with physical capital.

Nature of benefits flowing from human capital are different from that of physical capital. Human capital benefits not only the owner but also the society in general. This is called external benefit. An educated person can effectively take part in a democratic process and contribute to the socio-economic progress of a nation. A healthy person, by maintaining personal hygiene and sanitation, stops the spread of contagious diseases and epidemics. Human capital creates both private and social benefits, whereas physical capital creates only private benefit. That is, benefits from a capital good flow to those who pay the price for the product and services produced by it.

The concept of physical capital is the base for conceptualising human capital. There are some similarities between the two forms of capital; there are some striking dissimilarities as well. See Box 4.1.

Human Capital and Economic Growth: Who contributes more to national income — a worker in a factory or a software professional? We know that the labour skill of an educated person is more than that of an uneducated person and that the former generates more income than the latter. Economic growth means the increase in real national income of a country; naturally, the contribution of the educated person to economic growth is more than that of an illiterate person. If a healthy person could provide uninterrupted labour supply for a longer period of time, then health is also an important factor for economic growth. Thus, both education and health, along with many other factors like on-the-job training, job market information and migration, increase an



Look at Fig. 4.2 and discuss.

(a) What are the advantages of having proper 'classroom'?

(b) Do you think the children going to this schol are receiving quality education?

(c) Why these schools do not have buildings?

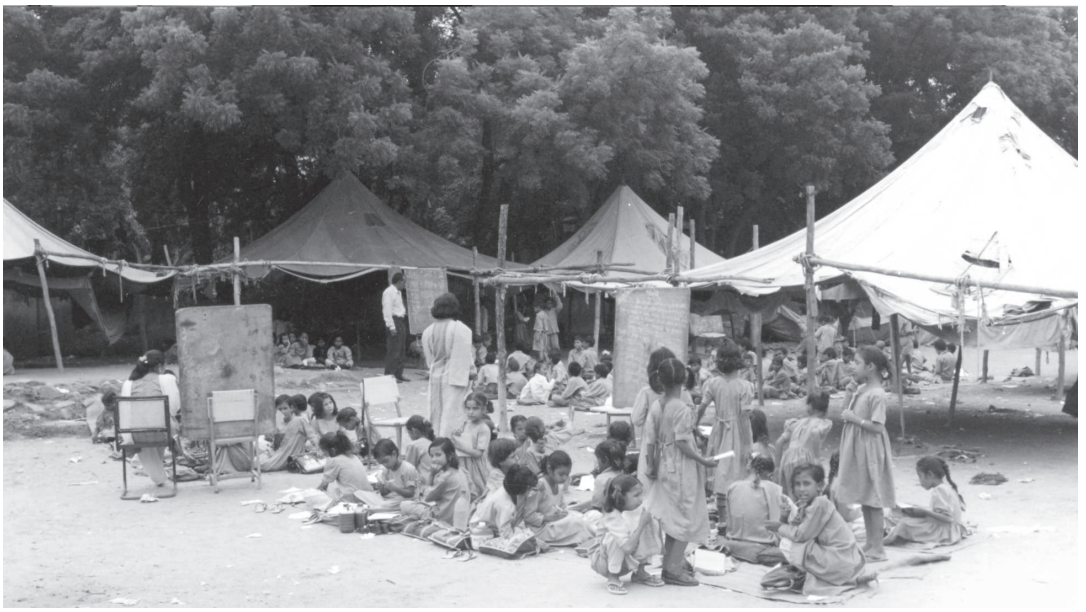


Fig. 4.2 *Creating human capital: a school being run in make shift premises in Delhi*

individual's income generating capacity.

This enhanced productivity of human beings or human capital contributes substantially not only towards increasing labour productivity but also stimulates innovations and creates ability to absorb new technologies. Education provides knowledge to understand changes in society and scientific advancements, thus, facilitate inventions and innovations. Similarly, the availability of educated labour force facilitates adaptation to new technologies.

Empirical evidence to prove that increase in human capital causes economic growth is rather nebulous. This may be because of measurement problems. For example, education measured in terms of years of schooling, teacher-pupil ratio and enrolment rates may not reflect the quality of education; health services measured in monetary terms, life expectancy and mortality rates may not reflect the true health status of the people in a country. Using the indicators mentioned above, an analysis of improvement in education and health sectors and growth in real per capita income in both developing and developed countries shows that there is convergence in the measures of human capital but no sign of convergence of per capita real income. In other words, the human capital growth in developing countries has been faster but the growth of per capita real income has not been that fast. There are reasons to believe that the



Fig. 4.3 *Scientific and technical manpower: a rich ingredient of human capital*

causality between human capital and economic growth flows in either directions. That is, higher income causes building of high level of human capital and vice versa, that is, high level of human capital causes growth of income.

India recognised the importance of human capital in economic growth long ago. The Seventh Five Year Plan says, "Human resources development (read human capital) has necessarily to be assigned a key role in any development strategy, particularly in a country with a large population. Trained and educated on sound lines, a large population can itself become an asset in accelerating economic growth and in ensuring social change in desired directions."

It is difficult to establish a relation of cause and effect from the growth of human capital (education and health) to economic growth but we can see in

TABLE 4.1

Select Indicators of Development in Education and Health Sectors

Particulars		*1951*	*1981*	*1991*	*2001*	*2016-17*
Real Per Capita Income (in Rs)		7,651	12,174	15,748	23,095	77,659
Crude Death Rate (Per 1,000 Population)		25.1	12.5	9.8	8.1	6.3
Infant Mortality Rate		146	110	80	63	33
Life Expectancy at Birth (in Years)	Male	37.2	54.1	59.7	63.9	67
	Female	36.2	54.7	60.9	66.9	70
Literacy Rate (%)		16.67	43.57	52.21	65.20	76

Source: *Economic Survey for various years, Ministry of Finance; National Statistical Office, Ministry of Statistics and Programme Implementation, Government of India.*

Table 4.1 that these sectors have grown simultaneously. Growth in each sector probably has reinforced the growth of every other sector.

The National Education Policy 2020 states that the world is undergoing rapid changes in the knowledge landscape. With various dramatic scientific and technological advances, such as the rise of big data, machine learning, and artificial intelligence, many unskilled jobs worldwide may be taken over by machines, while the need for a skilled workforce, particularly involving mathematics, computer science, and data science, in conjunction with multidisciplinary abilities across the sciences and social sciences, and humanities, will be increasingly in greater demand. With climate change, increasing pollution, and depleting natural resources, there will be a sizeable shift in how we meet the world's energy, water, food, and sanitation needs, again resulting in the need for new skilled labour, particularly in biology, chemistry, physics, agriculture, climate science, and social science. The growing emergence of epidemics and pandemics will also call for collaborative research in infectious disease management and development of vaccines and the resultant social issues heightens the need for multidisciplinary learning. There will be a growing demand for humanities and art, as India moves towards becoming a developed country as well as



Fig. 4.4 *Job on hand: transforming India into a knowledge economy*

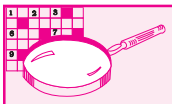
Box 4.2: India as a Knowledge Economy

The Indian software industry has been showing an impressive record over the past two decades. Entrepreneurs, bureaucrats and politicians are now advancing views about how India can transform itself into a knowledge-based economy by using information technology (IT). There have been some instances of villagers using e-mail which are cited as examples of such transformation. Likewise, e-governance is being projected as the way of the future. The value of IT depends greatly on the existing level of economic development. Do you think IT – based services in rural areas will lead to human development? Discuss.

among the three largest economies in the world. This policy vision suggests how human capital formation in India will move its economy to a higher growth trajectory based on knowledge landscape.

4.4 Human Capital and Human Development

The two terms sound similar but there is a clear distinction between them. Human capital considers education and health as a means to increase labour productivity. Human development is based on the idea that education and health are integral to human well-being because only when people have the ability to read and write and the ability to lead a long and healthy life, they will be able to make other choices which they value. Human capital treats human beings as a means to an end; the end being the increase in productivity. In this view, any investment in education and health is unproductive if it does not enhance output of goods and services. In the human development perspective, human beings are ends in themselves. Human welfare should be increased through investments in education and health even if such investments do not result in higher labour productivity. Therefore, basic education and basic health are important in themselves, irrespective of their contribution to labour productivity. In such a view, every individual has a right to get basic education and basic health care, that is, every individual has a right to be literate and lead a healthy life.



Work This Out

➢ If a construction worker, maid-servant, *dhobi* or a peon in school has absented herself/himself for long due to ill health, find out how it has affected her/his

(i) job security

(ii) wage/salary

What could be the possible reasons?

4.5 State of Human Capital Formation in India

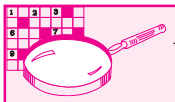
In this section we are going to analyse human capital formation in India. We have already learnt that human capital formation is the outcome of investments in education, health, on-the-job training, migration and information. Of these education and health are very important sources of human capital formation. We know that India is a federal country with a union government, state governments and local governments (Municipal Corporations, Municipalities and Village Panchayats). The Constitution of India mentions the functions to be carried out by each level of government. Accordingly, expenditures on both education and health are to be carried out simultaneously by all the three tiers of the government. Analysis of health sector is taken up in Chapter 8; hence, we will analyse only the education sector here.

Do you know who takes care of education and health in India? Before we take up the analysis of the education sector in India, we will look into the need for government intervention in education and health sectors. We do understand that education and health care services create both private and social benefits and this is the reason for the existence of both private and public institutions in the education and health service markets. Expenditures on education and health make substantial long-term impact and they cannot be easily reversed; hence, government intervention is essential. For instance, once a child is admitted to a school or health care centre where the required services are not provided, before the decision is taken to shift the child to another institution, substantial amount of damage would have been done. Moreover, individual consumers of these services do not have complete information about the quality of services and their costs. In this situation, the providers of education and health services acquire monopoly power and are involved in exploitation. The role of government in this situation is to ensure that the private providers of these services adhere to the standards stipulated by the government and charge the correct price.

In India, the ministries of education at the union and state level, departments of education and various organisations like National Council of Educational Research and Training (NCERT), University Grants Commission (UGC) and All India Council of Technical Education (AICTE) facilitate institutions which come under the education sector. Similarly, the ministries of health at the union and state level, departments of health and various organisations like National Medical Commission and Indian Council for Medical Research (ICMR) facilitate institutions which come under the health sector.

In a developing country like India, with a large section of the population living below the poverty line, many people cannot afford to access basic education and health care facilities. Moreover, a substantial section of India's population cannot afford

to reach super specialty health care and higher education. Furthermore, when basic education and health care is considered as a right of the citizens, then it is essential that the government should provide education and health services free of cost for the deserving citizens and those from the socially oppressed classes. Both, the union and state governments, have been stepping up expenditures in the education sector over the years in order to fulfil the objective of attaining cent per cent literacy and considerably increase the average educational attainment of Indians.



Work This Out

- Identify the objectives and functions of NCERT, UGC, AICTE and ICMR.

4.6 Education Sector in India

Growth in Government Expenditure on Education: Do you know how much the government spends on education? This expenditure by the government is expressed in two ways (i) as a percentage of 'total government expenditure' (ii) as a percentage of Gross Domestic Product (GDP).

The percentage of 'education expenditure of total government expenditure' indicates the importance of education in the scheme of things before the government. The percentage of 'education expenditure of GDP' expresses how much of people's income is being committed to the development of education in the country. During 1952-2014, education expenditure as percentage of total government expenditure increased from 7.92 to 15.7 and as percentage of GDP increased from 0.64 to 4.13. Throughout this period the increase in education expenditure has not been uniform and there has been irregular rise and fall. To this if we include the private expenditure incurred by individuals and by philanthropic institutions, the total education expenditure should be much higher.

Elementary education takes a major share of total education expenditure and the share of the higher/tertiary education (institutions of higher learning like colleges, polytechnics and universities) is the least. Though, on an average, the government spends less on tertiary education, 'expenditure per student' in tertiary education is higher than that of elementary. This does not mean that financial resources should be transferred from tertiary education to elementary education. As we expand school education, we need more teachers who are trained in the higher educational institutions; therefore, expenditure on all levels of education should be increased.

In 2014-15, the per capita public expenditure on elementary education differs considerably across states from as high as Rs 34,651 in Himachal Pradesh to as low as Rs 4088 in Bihar. This leads to differences in educational opportunities and attainments across states.



Fig. 4.5 *Investment in educational infrastructure is inevitable*

One can understand the inadequacy of the expenditure on education if we compare it with the desired level of education expenditure as recommended by the various commissions. The Education Commission (1964–66) had recommended that at least 6 per cent of GDP be spent on education so as to make a noticeable rate of growth in educational achievements. The Tapas Majumdar Committee, appointed by the Government of India in 1999, estimated an expenditure of around Rs 1.37 lakh crore over 10 years (1998-99 to 2006-07) to bring all Indian children in the age group of 6-14 years under the purview of school education. Compared to this desired level of education expenditure of around 6 per cent of GDP, the current level of a little over 4 per cent has been quite inadequate. In principle, a goal of 6 per cent needs to be reached—this has been accepted as a must for the coming years. In 2009, the Government of India enacted the Right of Children to Free and Compulsory Education Act to make free education a fundamental right of all children in the age group of 6-14 years.

Government of India has also started levying a 2 per cent 'education cess' on all Union taxes. The revenues from education cess has been earmarked for spending on elementary education. In addition to this, the government sanctions a large outlay for the promotion of higher education and

Work These Out

- Prepare case studies of dropouts at different levels of schooling, say
  - (i) Primary dropouts
  - (ii) Class VIII dropouts
  - (iii) Class X dropouts

  Find out the causes and discuss in the class.
- 'School dropouts are giving way to child labour'. Discuss how this is a loss to human capital.

TABLE 4.2

Educational Attainment in India

Sl.No.	*Particulars*	*1990*	*2000*	*2011*	*2017-18*
1.	Adult Literacy Rate (per cent of people aged 15+)				
	1.1 Male	61.9	68.4	79	82
	1.2 Female	37.9	45.4	59	66
2.	Primary completion rate (per cent of relevant age group)				
	2.1 Male	78	85	92	93
	2.2 Female	61	69	94	96
3.	Youth literacy rate (per cent of people aged 15+ to 24)				
	3.1 Male	76.6	79.7	90	93
	3.2 Female	54.2	64.8	82	90

new loan schemes for students to pursue higher education.

Educational Achievements in India: Generally, educational achievements in a country are indicated in terms of adult literacy level, primary education completion rate and youth literacy rate. These statistics for the last two decades are given above in Table 4.2.

4.7 Future Prospects

Education for All — Still a Distant Dream: Though literacy rates for both — adults as well as youth — have increased, still the absolute number of illiterates in India is as much as India's population was at the time of independence. In 1950, when the Constitution of India was passed by the Constituent Assembly, it was noted in the Directive Principles of the



Fig. 4.6 *School dropouts give way to child labour: a loss to human capital*

Constitution that the government should provide free and compulsory education for all children up to the age of 14 years within 10 years from the commencement of the Constitution. Had we achieved this, we would have cent per cent literacy by now.

Gender Equity — Better than Before: The differences in literacy rates between males and females are narrowing signifying a positive development in gender equity; still the need to promote education for women in India is imminent for various reasons such as improving economic independence and social status of women and also because women education makes a favourable impact on fertility rate and health care of women and children. Therefore, we cannot be complacent about the upward movement in the literacy rates and we have miles to go in achieving cent per cent adult literacy.

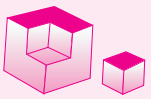


Fig. 4.7 *Higher Education: few takers*

Higher Education — a Few Takers: The Indian education pyramid is steep, indicating lesser and lesser number of people reaching the higher education level. Moreover, the level of unemployment among educated youth is the highest. As per NSSO data, in the year 2011-12, the rate of unemployment among youth males who studied graduation and above in rural areas was 19 per cent. Their urban counterparts had relatively less level of unemployment at 16 per cent. The most severely affected ones were young rural female graduates as nearly 30 per cent of them are unemployed. In contrast to this, only about 3-6 per cent of primary level educated youth in rural and urban areas were unemployed. The situation is yet to improve as indicated by the Periodic Labour Force Survey 2017-18. Therefore, the government should increase allocation for higher education and also improve the standard of higher education institutions, so that students are imparted employable skills in such institutions. When compared to less educated, a large proportion of educated persons are unemployed. Why?

4.8 Conclusion

The economic and social benefits of human capital formation and human development are well known. The union and state governments in India have been earmarking substantial financial outlays for development of education and health sectors. The spread of education and health services across different sectors of society should be ensured so as to simultaneously attain economic growth and equity. India has a rich stock of scientific and technical manpower in the world. The need of the hour is to better it qualitatively and provide such conditions so that they are utilised in India.



Recap

- Investments in education convert human beings into human capital; human capital represents enhanced labour productivity, which is an acquired ability and an outcome of deliberate investment decisions with an expectation that it will increase future income sources.
- Investments in education, on-the-job training, health, migration and information are the sources of human capital formation.
- The concept of physical capital is the base for conceptualising human capital. There are some similarities as well as dissimilarities between the two forms of capital formation.
- Investment in human capital formation is considered as efficient and growth enhancing.
- Human development is based on the idea that education and health are integral to human well-being because only when people have the ability to read and write and the ability to lead a long and healthy life, will they be able to make other choices which they value.
- The percentage of expenditure on education of the total government expenditure indicates the importance of education in the scheme of things for the government.



EXERCISES

1. What are the two major sources of human capital in a country?
2. What are the indicators of educational achievement in a country?
3. Why do we observe regional differences in educational attainment in India?
4. Bring out the differences between human capital and human development.
5. How is human development a broader term as compared to human capital?
6. What factors contribute to human capital formation?

7. How government organisations facilitate the functioning of schools and hospitals in India?

8. Education is considered to be an important input for the development of a nation. How?

9. Discuss the following as a source of human capital formation

   (i) Health infrastructure

   (ii) Expenditure on migration.

10. Establish the need for acquiring information relating to health and education expenditure for the effective utilisation of human resources.

11. How does investment in human capital contribute to growth?

12. 'There is a downward trend in inequality world-wide with a rise in the average education levels'. Comment.

13. Examine the role of education in the economic development of a nation.

14. Explain how investment in education stimulates economic growth.

15. Bring out the need for on-the-job-training for a person.

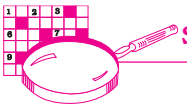
16. Trace the relationship between human capital and economic growth.

17. Discuss the need for promoting women's education in India.

18. Argue in favour of the need for different forms of government intervention in education and health sectors.

19. What are the main problems of human capital formation in India?

20. In your view, is it essential for the government to regulate the fee structure in education and health care institutions? If so, why?



SUGGESTED ADDITIONAL ACTIVITIES

1. Identify how Human Development Index is calculated. What is the position of India in the World Human Development Index?

2. Is India going to be a knowledge based economy in the near future? Discuss in the classroom.

3. Interpret the data given in Table 5.2.

4. As an educated person, what will be your contribution to the cause of education? (Example 'Each one — teach one').

5. Enlist the various sources that provide information regarding education, health and labour.

6. Read the annual reports of Union Ministries of Education and Health and Family welfare and make summaries. Read the chapter on social sector in the *Economic Survey*. These can be downloaded from websites of the respective Union Government Ministries.



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5



RURAL DEVELOPMENT

After studying this chapter, the learners will

- understand rural development and the major issues associated with it
- appreciate how crucial the development of rural areas is for India's overall development
- understand the critical role of credit and marketing systems in rural development
- learn about the importance of diversification of productive activities to sustain livelihoods
- understand the significance of organic farming in sustainable development.

"Only the tillers of the soil live by the right. The rest form their train and eat only the bread of dependence".

Thiruvalluvar

5.1 Introduction

In Chapter 4, we studied how poverty was a major challenge facing India. We also came to know that the majority of the poor live in rural areas where they do not have access to the basic necessities of life.

Agriculture is the major source of livelihood in the rural sector. Mahatma Gandhi once said that the real progress of India did not mean simply the growth and expansion of industrial urban centres but mainly the development of the villages. This idea of village development being at the centre of the overall development of the nation is relevant even today. Why is this so? Why should we attach such significance to rural development when we see around us fast growing cities with large industries and modern information technology hubs? It is because more than two-third of India's population depends on agriculture that is yet to become productive enough to provide for them; one-fourth of rural India still lives in abject poverty. That is the reason why we have to see a developed rural India if our nation has to realise real progress. What, then, does rural development imply?

5.2 What is Rural Development?

Rural development is a comprehensive term. It essentially focuses on action for the development of areas that are lagging behind in the overall development of the village economy. Some of the areas which are challenging and need fresh initiatives for development in rural India include

- Development of human resources including
  - literacy, more specifically, female literacy, education and skill development
  - health, addressing both sanitation and public health
- Land reforms
- Development of the productive resources of each locality
- Infrastructure development like electricity, irrigation, credit, marketing, transport facilities including construction of village roads and feeder roads to nearby highways, facilities for agriculture research and extension, and information dissemination
- Special measures for alleviation of poverty and bringing about significant improvement in the living conditions of the weaker sections of the population emphasising access to productive employment opportunities

All this means that people engaged in farm and non-farm activities in rural areas have to be provided with various means that help them increase the productivity. They also need to be given opportunities to diversify into various non-farm productive

activities such as food processing. Enabling them better and more affordable access to healthcare, sanitation facilities at workplaces and homes and education for all would also need to be given top priority for rapid rural development.

It was observed in an earlier chapter that although the share of agriculture sector's contribution to GDP was on a decline, the population dependent on this sector did not show any significant change. Further, after the initiation of reforms, the growth rate of agriculture sector decelerated to about 3 per cent per annum during the 1991-2012, which was lower than the earlier years. In recent years, this sector has become volatile. During 2014-15, the GVA growth rate of agriculture and its allied sectors was less than one per cent. Scholars identify decline in public investment since 1991 as the major reason for this. They also argue that inadequate infrastructure, lack of alternate employment opportunities in the industry or service sector, increasing casualisation of employment etc., further impede rural development. The impact of this phenomenon can be seen from the growing distress witnessed among farmers across different parts of India. Against this background, we will critically look at some of the crucial aspects of rural India like credit and marketing systems, agricultural diversification and the role of organic farming in promoting sustainable development.



Work These Out

- On a monthly basis, go through the newspapers of your region and identify the problems raised by them in relation to rural areas and the solutions offered. You could also visit a nearby village and identify the problems faced by people there. Discuss this in the classroom.
- Prepare a list of recent schemes and their objectives from the government website https://www.rural.nic.in. Collect the details of how anyone of these schemes implemented in your region/rural neighbourhood areas. Discuss your observations in class.

5.3 Credit and Marketing in Rural Areas

Credit: Growth of rural economy depends primarily on infusion of capital, from time to time, to realise higher productivity in agriculture and non-agriculture sectors. As the time gestation between crop sowing and realisation of income after production is quite long, farmers borrow from various sources to meet their initial investment on seeds, fertilisers, implements and other family expenses of marriage, death, religious ceremonies etc.

At the time of independence, moneylenders and traders exploited small and marginal farmers and landless labourers by lending to them on high interest rates and by manipulating the accounts to keep them in a debt-trap. A major change occurred after 1969 when India adopted social banking and multi-agency approach to adequately meet the needs of rural credit. Later, the

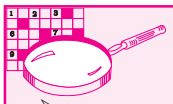
Box 5.1: The Poor Women's Bank

'Kudumbashree' is a women-oriented community-based poverty reduction programme being implemented in Kerala. In 1995, a thrift and credit society was started as a small savings bank for poor women with the objective to encourage savings. The thrift and credit society mobilised Rs 1 crore as thrift savings. These societies have been acclaimed as the largest informal banks in Asia in terms of participation and savings mobilised.

Source: *www.kudumbashree.org. Visit this website and explore various other initiatives undertaken by this organisation. Identify some factors which contributed to their successes and discuss in the class.*

National Bank for Agriculture and Rural Development (NABARD) was set up in 1982 as an apex body to coordinate the activities of all institutions involved in the rural financing system. The Green Revolution was a harbinger of major changes in the credit system as it led to the diversification of the portfolio of rural credit towards production-oriented lending.

The institutional structure of rural banking today consists of a set of multi-agency institutions, namely, commercial banks, regional rural banks (RRBs), cooperatives and land development banks. They are expected to dispense adequate credit at cheaper rates. Recently, Self-Help Groups (henceforth SHGs) have emerged to fill the gap in the formal credit system because the formal credit delivery mechanism has not only proven inadequate but has also not been fully integrated into the overall rural social and community development. Since some kind of collateral is required, vast proportion of poor rural households were automatically out of the credit network. The SHGs promote thrift in small proportions by a minimum contribution from each member. From the pooled money, credit is given to the needy members to be repayable in small instalments at reasonable interest rates. By May 2019, nearly 6 crore



Work These Out

- In your locality/neighbourhood, you might notice self-help groups providing credit. Attend few meetings of such self-help groups. Write a report on the profile of a self-help group. The profile may include — when it was started, the number of members, amount of savings and type of credit they provide and how borrowers use the loan.
- You might also find that those who take a loan for starting self-employment activities but use it for other purposes. Interact with few such borrowers. Identify the reasons for not starting self employment activities and discuss in the classroom.

women in India have become member in 54 lakh women SHGs. About ₹ 10-15,000 per SHG and another ₹2.5 lakhs per SHG as a Community Investment Support Fund (CISF) are provided as part of renovating fund to take up self employement for income generation. Such credit provisions are generally referred to as **micro-credit programmes**. SHGs have helped in the empowerment of women. It is alleged that the borrowings are mainly confined to consumption purposes. Why are borrowers not spending for productive purposes?

Rural Banking — a Critical Appraisal: Rapid expansion of the banking system had a positive effect on rural farm and non-farm output, income and employment, especially after the green revolution — it helped farmers to avail services and credit facilities and a variety of loans for meeting their prodution needs. Famines became events of the past; we have now achieved food security which is reflected in the abundant buffer stocks of grains. However, all is not well with our banking system.

With the possible exception of the commercial banks, other formal institutions have failed to develop a culture of deposit mobilisation — lending to worthwhile borrowers and effective loan recovery. Agriculture loan default rates have been chronically high. Why farmers failed to pay back loans? It is alleged that farmers are deliberately refusing to pay back loans. What could be the reasons?

Thus, the expansion and promotion of the rural banking sector has taken a backseat after reforms. To improve the situation, In recent years, all the adults are encouraged to open bank accounts as a part of a scheme known as Jan-Dhan Yojana. Those bank account holders can get Rs. 1-2 lakh accidental insurance coverage and overdraft facilities for Rs. 10,000 and get their wages if they get any government-related jobs and works under MNREGA; old age pension and other social security payments of the government are transferred



Work These Out

- In the last few years, you might have taken note — in your neighbourhood if you are living in rural areas or read in the newspapers or seen on TV — of farmers commiting suicides. Many such farmers had borrowed money for farming and other purposes. It was found that when they were unable to pay back due to crop failure, insufficient income and employment opportunities, they took such steps. Collect information relating to such cases and discuss in the classroom.
- Visit banks that cater to rural areas. They may be primary agricultural cooperative banks, land development banks, regional rural banks or district cooperative banks. Collect details such as how many rural households borrowed from them, amount generally borrowed, kinds of collateral used, interest rates and dues.
- If farmers who borrowed from cooperative banks could not pay back due to crop failure and other reasons, their loans should be waived otherwise they may take drastic decisions like committing suicides. Do you agree?

to bank accounts. There is no need to keep minimum bank balance. This has led to more than 40 crore people opening bank accounts; indirectly it has promoted thrift habit and efficient allocation of financial resources particularly in rural areas. Banks also could mobile funds of more than Rs. 1,40,000 crores through these accounts.

5.4 Agricultural Market System

Have you ever asked yourself how food grains, vegetables and fruits that we consume daily come from different parts of the country? The mechanism through which these goods reach different places depends on the market channels. Agricultural marketing is a process that involves the assembling, storage, processing, transportation, packaging, grading and distribution of different agricultural commodities across the country.

Prior to independence, farmers, while selling their produce to traders, suffered from faulty weighing and manipulation of accounts. Farmers who did not have the required information on prices prevailing in markets were often forced to sell at low prices. They also did not have proper storage facilities to keep back their produce for selling later at a better price. Do you know that even today, more than 10 per cent of goods produced in farms are wasted due to lack of storage? Therefore, government intervention became necessary to regulate the activities of the private traders.

Let us discuss four such measures that were initiated to improve the marketing aspect. The first step was regulation of markets to create orderly and transparent marketing conditions. By and large, this policy benefited farmers as well as consumers. However, there is still a need to develop about 27,000 rural periodic markets as regulated market places to realise the full potential of rural markets. Second component is provision of physical infrastructure facilities like roads, railways, warehouses, godowns, cold storages and processing units. The current infrastructure facilities are quite inadequate to meet the growing demand and need to be improved. Cooperative



Fig. 5.1 *Regulated market yards benefit farmers as well as consumers*



Work These Out

- Visit a nearby vegetable and fruit market. Observe and identify different characteristics of the market. Identify the place of origin of at least ten different fruits and vegetables and distance travelled to reach the market. Further, look at the modes of transport and its implication on prices.
- Most small towns have regulated market yards. Farmers can go to these markets and sell their produce. They can also store their goods in the yard. Visit one regulated market yard; collect the details of its functioning, kind of goods coming to the yard and how prices are fixed. Prepare a detailed report and discuss in the class.

marketing, in realising fair prices for farmers' products, is the third aspect of government initiative. The success of milk cooperatives in transforming the social and economic landscape of Gujarat and some other parts of the country is testimony to the role of cooperatives. However cooperatives have received a setback during the recent past due to inadequate coverage of farmer members, lack of appropriate link between marketing and processing cooperatives and inefficient financial management. The fourth element is the policy instruments like (i) assurance of minimum support prices (MSP) for agricultural products (ii) maintenance of buffer stocks of wheat and rice by Food Corporation of India and (iii) distribution of food grains and sugar through PDS. These instruments are aimed at protecting the income of the farmers and providing foodgrains at a subsidised rate to the poor. However, despite government intervention, private trade (by moneylenders, rural political elites, big merchants and rich farmers) predominates agricultural markets. The need for government intervention is imminent particularly when a large share of agricultural products, is handled by the private sector.

Agricultural marketing has come a long way with the intervention of the government in various forms. Some scholars argue that commercialisation of agriculture offers tremendous scope for farmers to earn higher incomes provided the government intervention is restricted. What do you think about this view?

Emerging Alternate Marketing Channels: It has been realised that if farmers directly sell their produce to consumers, it increases their incomes. Some examples of these channels are *Apni Mandi* (Punjab, Haryana and Rajasthan); *Hadaspar Mandi* (Pune); *Rythu Bazars* (vegetable and fruit markets in Andhra Pradesh and Telangana) and *Uzhavar Sandies* (farmers markets in Tamil Nadu). Further, several national and multinational fast food chains are increasingly entering into contracts/ alliances with farmers to encourage

them to cultivate farm products (vegetables, fruits, etc.) of the desired quality by providing them with not only seeds and other inputs but also assured procurement of the produce at pre-decided prices. It is argued that such arrangements will help in reducing the price risks of farmers and would also expand the markets for farm products. Do you think such arrangements raise incomes of small farmers.



Work This Out

➢Visit one such alternative marketing system which farmers in your locality, or in the neighbourhood rural areas, use. How are they different from regulated market yards? Should they be encouraged and supported by the government? Why and how? Discuss.

In 2020, Indian Parliament passed three laws to reform agriculture marketing system. While some section of farmers support these reforms, the rest of the farmers oppose and these Acts are still being debated. Collect the details of these laws, debate and discuss in the class.

5.5 Diversification into Productive Activities

Diversification includes two aspects – one relates to change in cropping pattern and the other relates to a shift of workforce from agriculture to other allied activities (livestock, poultry, fisheries etc.) and non-agriculture sector. The need for diversification arises from the fact that there is greater risk in depending exclusively on farming for livelihood. Diversification towards new areas is necessary not only to reduce the risk from agriculture sector but also to provide productive sustainable livelihood options to rural people. Much of the agricultural employment activities are concentrated in the Kharif season. But during the Rabi season, in areas where there are inadequate irrigation facilities, it becomes difficult to find gainful employment. Therefore expansion into other sectors is essential to provide supplementary gainful employment and in realising higher levels of income for rural people to overcome poverty and other tribulations. Hence, there is a need to focus on allied activities, non-farm employment and other emerging alternatives of livelihood, though there are many other options available for providing sustainable livelihoods in rural areas.



Fig. 5.2 *Jaggery making is an allied activity of the farming sector*

Box 5.2: Tamil Nadu Women in Agriculture (TANWA)

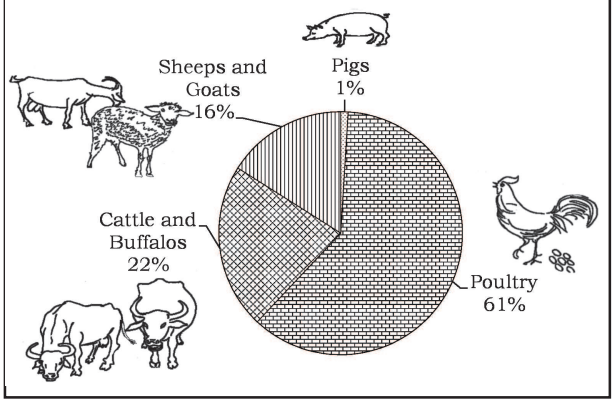
Tamil Nadu Women in Agriculture (TANWA) was a project initiated in the late 1980s in Tamil Nadu to train women in latest agricultural techniques and in organic farming. It encouraged women to actively participate in raising agricultural productivity and family income. At a Farm Women's Group in Thiruchirapalli, run by Anthoniammal, trained women are successfully making and selling vermicompost and earning money from this venture. Many other Farm Women's Groups are creating savings in their group by functioning like mini banks through a micro-credit system. With the accumulated savings, they promote small-scale household activities like mushroom cultivation, soap manufacture, doll making or other income-generating activities. There could be similar initiatives in your area or your neighborhood rural areas. Talk to organisation or women entrepreneurs and discuss the details in the class.

As agriculture is already overcrowded, a major proportion of the increasing labour force needs to find alternate employment opportunities in other non-farm sectors. Non-farm economy has several segments in it; some possess dynamic linkages that permit healthy growth while others are in subsistence, low productivity propositions. The dynamic sub-sectors include **agro-processing industries,** food processing industries, leather industry, tourism, etc. Those sectors which have the potential but seriously lack infrastructure and other support include traditional home-based industries like pottery, crafts, handlooms etc. Majority of rural women find employment in agriculture while men generally look for non-farm employment. In recent times, women have also begun looking for non-farm jobs (see Box 5.2).

Animal Husbandry: In India, the farming community uses the mixed crop-livestock farming system —cattle, goats, fowl are the widely held species. Livestock production provides increased stability in income, food security, transport, fuel and nutrition for the family without disrupting other food-producing activities. Today, livestock sector alone provides alternate livelihood options to over 70 million small and marginal farmers including landless labourers. A significant number of women also find employment in the livestock sector.

Chart 5.1 shows the distribution of livestock in India. Poultry accounts for the largest share with 61 per cent

Chart 5.1: Distribution of Poultry and Livestock in India, 2019



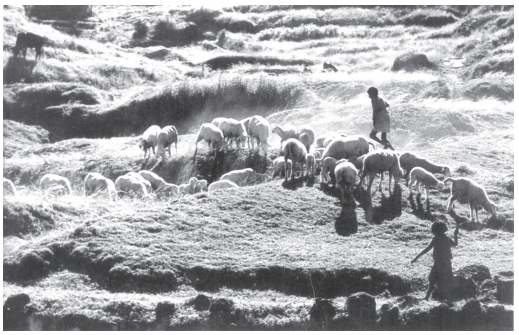


Fig. 5.3 *Sheep rearing — an important income augmenting activity in rural areas*

followed by others. Other animals which include camels, asses, horses, ponies and mules are in the lowest rung. India had about 303 million cattle, including 110 million buffaloes in 2019. Performance of the Indian dairy sector over the last three decades has been quite impressive. Milk production in the country has increased by about ten times between 1951-2016. This can be attributed mainly to the successful implementation of 'Operation Flood'. It is a system whereby all the farmers can pool their milk produced according to different grading (based on quality), processed and marketed to urban centres through cooperatives. In this system the farmers are assured of a fair price and income from the supply of milk to urban markets. As pointed out earlier Gujarat state is held as a success story in the efficient implementation of milk cooperatives which has been emulated by many states. Gujarat, Madhya Pradesh, Uttar Pradesh, Andhra Pradesh, Maharashtra, Punjab and Rajasthan, are major milk producing states. Meat, eggs, wool and other by- products are also emerging as important productive sectors for diversification.

Fisheries: The fishing community regards the water body as 'mother' or 'provider'. The water bodies consisting of sea, oceans, rivers, lakes, natural aquatic ponds, streams etc. are, therefore, an integral and life-giving source for the fishing community. In India, after progressive increase in budgetary allocations and introduction of new technologies in fisheries and aquaculture, the development of fisheries has come a long way. Presently, fish production from inland sources contributes about 65 per cent to the total value of fish production and the balance 35 per cent comes from the marine sector (sea and oceans). Today total fish production accounts for 0.9 per cent of the total GDP. In India, West Bengal, Andhra Pradesh, Kerala, Gujarat, Maharashtra and Tamil Nadu are major fish producing states. A large share of fishworker families are poor. Rampant underemployment, low per capita earnings, absence of mobility of labour to other sectors and a high rate of illiteracy and indebtedness are some of the major problems fishing community face today. Even though women are not involved in active fishing, about 60 per cent of the workforce in export marketing and 40 per cent in internal marketing are women. There is a need to increase credit facilities through cooperatives and SHGs for fisherwomen to meet the

working capital requirements for marketing.

Horticulture: Blessed with a varying climate and soil conditions, India has adopted growing of diverse horticultural crops such as fruits, vegetables, tuber crops, flowers, medicinal and aromatic plants, spices and plantation crops. These crops play a vital role in providing food and nutrition, besides addressing employment concerns. Horticulture sector contributes nearly one-third of the value of agriculture output and six per cent of Gross Domestic Product of India. India has emerged as a world leader in producing a variety of fruits like mangoes, bananas, coconuts, cashew nuts and a number of spices and is the second largest producer of fruits and vegetables. Economic condition of many farmers engaged in horticulture has improved and it has become a means of improving livelihood for many unprivileged classes. Flower harvesting, nursery



Fig. 5.5 *Women in rural households take up bee-keeping as an entrepreneurial activity*

maintenance, hybrid seed production and tissue culture, propagation of fruits and flowers and food processing are highly remunerative employment options for women in rural areas.

Though, in terms of numbers, our livestock population is quite impressive but its productivity is quite low as compared to other countries. It requires improved technology and promotion of good breeds of animals to enhance productivity. Improved veterinary care and credit facilities to small and marginal farmers and landless labourers would enhance sustainable livelihood options through livestock production. Production of fisheries has already increased substantially.

However problems related to over- fishing and pollution need to be regulated



Fig. 5.4 *Poultry has the largest share of total livestock in India*

and controlled. Welfare programmes for the fishing community have to be reoriented in a manner which can provide long-term gains and sustenance of livelihoods. Horticulture has emerged as a successful sustainable livelihood option and needs to be encouraged significantly. Enhancing its role requires investment in infrastructure like electricity, cold storage systems, marketing linkages, small-scale processing units and technology improvement and dissemination.

Other Alternate Livelihood Options: The IT has revolutionised many sectors in the Indian economy. There is broad consensus that IT can play a critical role in achieving sustainable development and food security in the twenty-first century. Governments can predict areas of food insecurity and vulnerability using appropriate information and software tools so that action can be taken to prevent or reduce the likelihood of an emergency. It also has a positive impact on the agriculture sector as it can disseminate information regarding emerging technologies and its applications, prices, weather and soil conditions for growing different crops etc. Though IT is, by itself, no catalyst of change but it can act as a tool for releasing the creative potential and knowledge embedded in the society. It also has potential of employment generation in rural areas. Experiments with IT and its application to rural development are carried out in different parts of India.

5.6 Sustainable Development and Organic Farming

In recent years, awareness of the harmful effect of chemical-based fertilisers and pesticides on our health is on a rise. Conventional agriculture relies heavily on chemical fertilisers and toxic pesticides etc., which enter the food supply, penetrate the water sources, harm the livestock, deplete the soil and devastate natural eco-systems. Efforts in evolving technologies which are eco-friendly are essential for sustainable development and one such technology which is eco-friendly is organic farming. In short, organic agriculture is a whole system of

Box 5.3: Adoption of Village by Parliamentarians

In October, 2014, The Government of India introduced a new scheme called Saansad Adarsh Gram Yojana (SAGY). Under this scheme, Members of India's Parliament need to identify and develop one village from their constituencies. To begin with, MPs can develop one village as a model village by 2016, and two more by 2019, covering over 2,500 villages in India. According to the scheme, the village can have a population of 3,000-5,000 in the plains and 1,000-3,000 in the hills and should not be MPs' own or their spouse's village. MPs are expected to facilitate a village development plan, motivate villagers to take up activities and built infrastructure in the areas of health, nutrition and education.

Source: *www.pib.nic.in accessed on 24 October 2014.*

Box 5.4: Organic Food

Organic food is growing in popularity across the world. Many countries have around 10 per cent of their food system under organic farming. There are many retail chains and supermarkets which are accorded with **green status** to sell organic food. Moreover, organic foods command higher price of around 10-100 per cent than conventional ones.

farming that restores, maintains and enhances the ecological balance. There is an increasing demand for organically grown food to enhance food safety throughout the world (see Box 5.4).

Benefits of Organic Farming: Organic agriculture offers a means to substitute costlier agricultural inputs (such as HYV seeds, chemical fertilisers, pesticides etc.) with locally produced organic inputs that are cheaper and thereby generate good returns on investment. Organic agriculture also generates income through exports as the demand for organically grown crops is on a rise. Studies across countries have shown that organically grown food has more nutritional value than chemical farming thus providing us with healthy foods. Since organic farming requires more labour input than conventional farming, India will find organic farming an attractive proposition. Finally, the produce is pesticide-free and produced in an environmentally sustainable way (see Box 5.5).

Popularising organic farming requires awareness and willingness on the part of farmers to adapt to new technology. Inadequate infrastructure and the problem of marketing the products are major concerns which need to be addressed apart from an

Box 5.5: Organically Produced Cotton in Maharashtra

In 1995, when Kisan Mehta of *Prakruti* (an NGO) first suggested that cotton, the biggest user of chemical pesticides, could be grown organically, the then Director of the Central Institute for Cotton Research, Nagpur, famously remarked, "Do you want India to go naked?" At present, as many as 130 farmers have committed 1,200 hectares of land to grow cotton organically on the International Federation of Organic Agriculture Movement's standards. The produce was later tested by the German Accredited Agency, AGRECO, and found to be of high quality. Kisan Mehta feels that about 78 per cent of Indian farmers are marginal farmers owning about less than 0.8 hectare but accounting for 20 per cent of India's cultivable land. For such farmers, organic agriculture is more profitable in terms of money and soil conservation in the long run. Visit a farm in your locality which uses organic manure and discuss the uses associated with the farming practices. Present the report in your class.

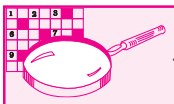
Source: *Lyla Bavadam, A Green Alternative, Frontline, 29 July 2005.*

appropriate agriculture policy to promote organic farming. It has been observed that the yields from organic farming are less than modern agricultural farming in the initial years. Therefore, small and marginal farmers may find it difficult to adapt to large-scale production. Organic produce may also have more blemishes and a shorter shelf life than sprayed produce. Moreover choice in production of off-season crops is quite limited in organic farming. Nevertheless, organic farming helps in sustainable development of agriculture and India has a clear advantage in producing organic products for both domestic and international markets. Do you think food and non-food items cultivated using organic farming methods will be cheaper?

5.7 Conclusion

It is clear that until and unless some spectacular changes occur, the rural sector might continue to remain backward. There is a greater need today to make rural areas more vibrant through diversification into dairying, poultry, fisheries, vegetables and fruits and linking up the rural production centres with the urban and foreign (export) markets to realise higher returns on the investments for the products. Moreover, infrastructure elements like credit and marketing, farmer-friendly agricultural policies and a constant appraisal and dialogue between farmers' groups and state agricultural departments are essential to realise the full potential of the sector.

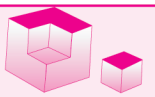
Today we cannot look at the environment and rural development as two distinct subjects. There is need to invent or procure alternate sets of eco-friendly technologies that lead to sustainable development in different circumstances. From these, each rural



Work These Out

- Make a list of five popular items that are organically produced in India.
- Visit a nearby super market, vegetable shop and/or a departmental shop. Identify a few products. Prepare a chart comparing a few goods that are produced organically and in the normal way on the basis of their prices, shelf life, quality and the kind of advertisement through which they are popularised.
- Visit a horticultural farm in the nearby locality. Collect the details of goods that they cultivate on the farm. They could have diversified their cropping patterns. Discuss with them the merits and demerits of the diversification.

community can choose whatever will suit its purpose. We need to learn from, and also try out when found relevant, practices from the available set of 'best practice' illustrations (which means success stories of rural development experiments that have already been carried out in similar conditions in different parts of India), to speed up this process of 'learning by doing'.



Recap

- Rural development is quite a comprehensive term but it essentially means a plan of action for the development of rural areas which are lagging behind in socio-economic development.
- There is a need for improving the quantity and quality of infrastructure in rural areas such as banking, marketing, storage, transport and communications etc. to realise its true potential.
- Diversification towards new areas such as livestock, fisheries and other non-agricultural activities is necessary not only to reduce the risk from agriculture sector but also to provide productive sustainable livelihood options to our rural people.
- The importance of organic farming as an environmentally sustainable production process is on a rise and needs to be promoted.



EXERCISES

1. What do you mean by rural development? Bring out the key issues in rural development.
2. Discuss the importance of credit in rural development.
3. Explain the role of micro-credit in meeting credit requirements of the poor.
4. Explain the steps taken by the government in developing rural markets.
5. Why is agricultural diversification essential for sustainable livelihoods?
6. Critically evaluate the role of the rural banking system in the process of rural development in India.

7. What do you mean by agricultural marketing?
8. Mention some obstacles that hinder the mechanism of agricultural marketing.
9. What are the alternative channels available for agricultural marketing? Give some examples.
10. Distinguish between 'Green Revolution' and 'Golden Revolution'.
11. Do you think various measures taken by the government to improve agricultural marketing are sufficient? Discuss.
12. Explain the role of non-farm employment in promoting rural diversification.
13. Bring out the importance of animal husbandry, fisheries and horticulture as a source of diversification.
14. 'Information technology plays a very significant role in achieving sustainable development and food security' — comment.
15. What is organic farming and how does it promote sustainable development?
16. Identify the benefits and limitations of organic farming.
17. Enlist some problems faced by farmers during the initial years of organic farming.
18. "Jan-Dhan-Yojna helps in the rural development." Do you agree with this statement? Explain.



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6



EMPLOYMENT : GROWTH, INFORMALISATION AND OTHER ISSUES

After studying this chapter, the learners will

- understand a few basic concepts relating to employment such as economic activity, worker, workforce and unemployment
- understand the nature of participation of men and women in various economic activities in various sectors
- know the nature and extent of unemployment
- assess the initiatives taken by the government in generating employment opportunities in various sectors and regions.

What I object to, is the 'craze' for machinery, not machinery as such. The craze is for what they call labour-saving machinery. Men go on 'saving labour' till thousands are without work and thrown on the open streets to die of starvation...

Mahatma Gandhi

6.1 Introduction

People do a variety of work. Some work on farms, in factories, banks, shops and many other workplaces; yet a few others work at home. Work at home includes not only traditional work like weaving, lace making or variety of handicrafts but also modern jobs like programming work in the IT industry. Earlier factory work meant working in factories located in cities whereas now technology has enabled people to produce those factory-based goods at home in villages. During Covid 19 pandemic in 2020-21, millions of workers delivered their products and services through work from home.



Fig. 6.1 *Multinational companies sell footballs made in the houses of Jalandhar, Punjab*

Why do people work? Work plays an important role in our lives as individuals and as members of society. People work for 'earning' a living. Some people get, or have, money by inheriting it, not working for it. This does not completely satisfy anybody. Being employed in work gives us a sense of self-worth and enables us to relate ourselves meaningfully with others. Every working person is actively contributing to national income and hence, the development of the country by engaging in various economic activities — that is the real meaning of 'earning' a living. We do not work only for ourselves; we also have a sense of accomplishment when we work to meet the requirements of those who are dependent on us. Having recognised the importance of work, Mahatma Gandhi insisted upon education and training through a variety of works including craft.

Studying about working people gives us insights into the quality and nature of employment in a country and helps in understanding and planning our human resources. It helps us to analyse the contribution made by different industries and sectors towards national income. It also helps us to address many social issues such as exploitation of marginalised sections of the society, child labour, etc.

6.2 Workers and Employment

What is employment? Who is a worker? When a farmer works on fields, he or she produces food grains and raw materials for industries. Cotton becomes cloth in textile mills and in powerlooms. Lorries transport goods from one place to another. We know that the total money value of all such final goods and services produced in a country in a year is called its gross domestic product for that year. When we also consider what we pay for our imports and get from our exports we find that there is a net earning for the country which may be positive (if we have exported more in value terms than imported) or negative (if imports exceeded exports in value terms) or zero (if exports and imports were of the same value). When we add this earning (plus or minus) from foreign transactions, what we get is called the country's gross national product for that year.

Those activities which contribute to the gross national product are called economic activities. All those who are engaged in economic activities, in whatever capacity — high or low, are workers. Even if some of them temporarily abstain from work due to illness, injury or other physical disability, bad weather, festivals, social or religious functions, they are also workers. Workers also include all those who help the main workers in these activities. We generally think of only those who are paid by an employer for their work as workers. This is not so. Those who are self-employed are also workers.

The nature of employment in India is multifaceted. Some get employment throughout the year; some others get employed for only a few months in a year. Many workers do not get fair wages for their work. While estimating the number of workers, all those who are engaged in economic activities are included as employed. You might be interested in knowing the number of people actively engaged in various economic activities. During 2017-18, India had about a 471 million strong workforce. Since majority of our people reside in rural areas, the proportion of workforce residing there is higher. The



Work This Out

➤ In your house or neighbourhood, you might come across many women who, even though they have technical degrees and diplomas and also free time to go to work, do not go to work. Ask them the reasons for not going to work. List all of them and discuss in the classroom whether they should go for work and why, and also ways by which they could be sent for work. Some social scientists argue that housewives working at home without getting paid for that work must also be regarded as contributing to the gross national product and therefore, as engaged in an economic activity. Would you agree?

rural workers constitute about two-thirds of this 471 million. Men form the majority of workforce in India. About 77 per cent of the workers are men and the rest are women (men and women include child labourers in respective sexes). Women workers account for one-fourth of the rural workforce whereas in urban areas, they are just one-fifth of the workforce. Women carry out works like cooking, fetching water and fuelwood and participate in farm labour. They are not paid wages in cash or in the form of grains; at times they are not paid at all. For this reason, these women are not categorised as workers. Economists argue that these women should also be called workers. What do you think?

6.3 Participation of People in Employment

Worker-population ratio is an indicator which is used for analysing the employment situation in the country. This ratio is useful in knowing the proportion of population that is actively contributing to the production of goods and services of a country. If the ratio is higher, it means that the engagement of people is greater; if the ratio for a country is medium, or low, it means that a very high proportion of its population is not involved directly in economic activities.

You might have already studied, in lower classes, the meaning of the term 'population'. Population is defined as the total number of people who reside in a particular locality at a particular point of time. If you want to know the worker-population ratio for India, divide the total number of workers in India by the population in India and multiply it by 100, you will get the worker-population ratio for India.

If you look at Table 6.1, it shows the different levels of participation of people in economic activities. For every 100 persons, about 35 (by rounding off 34.7) are workers in India. In urban areas, the proportion is about 34, whereas in rural India, the ratio is about 35. Why is there such a difference? People in rural areas have limited resources to earn a higher income and participate more in the employment market. Many do not go to schools, colleges and other training institutions. Even if some go, they discontinue in the middle to join the workforce; whereas, in urban areas, a considerable section is able to study in various educational institutions. Urban people have a variety of employment opportunities. They look for the appropriate job to suit their qualifications and skills. In rural areas, people cannot stay at home as their economic condition may not allow them to do so.

TABLE 6.1

Worker-Population Ratio in India, 2017-2018

Sex	*Worker-Population Ratio*		
	Total	*Rural*	*Urban*
Men	52.1	51.7	53.0
Women	16.5	17.5	14.2
Total	**34.7**	**35.0**	**33.9**

Work These Out

- Any study of employment must start with a review of the worker-population ratios — why?
- In some communities, you might have noticed that even if the males do not earn a high income, they do not send women to work. Why?

Compared to females, more males are found to be working. The difference in participation rates is very large in urban areas: for every 100 urban females, only about 14 are engaged in some economic activities. In rural areas, for every 100 rural women about 18 participate in the employment market. Why are women, in general, and urban women, in particular, not working? It is common to find that where men are able to earn high incomes, families discourage female members from taking up jobs.

Going back to what has already been mentioned above, many household activities done by women are not recognised as productive work. This narrow definition of work leads to non-recognition of women's work and, therefore, to the underestimation of the number of women workers in the country. Think of the women actively engaged in many activities within the house and at family farms who are not paid for such work. As they certainly contribute to the maintenance of the household and farms, do you think that their number should be added to the number of women workers?

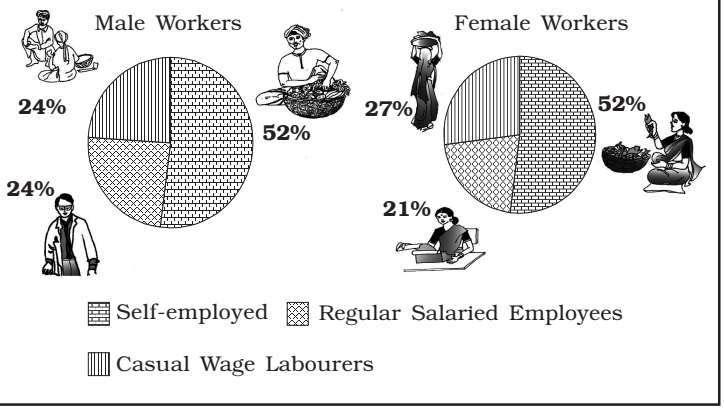
6.4 Self-employed and Hired Workers

Does the worker-population ratio say anything about workers' status in society or about the working conditions? By knowing the status with which a worker is placed in an enterprise, it may be possible to know one dimension — quality of employment in a country. It also



Fig. 6.2 *Brick-making: a form of casual work*

Chart 6.1 : Distribution of Employment

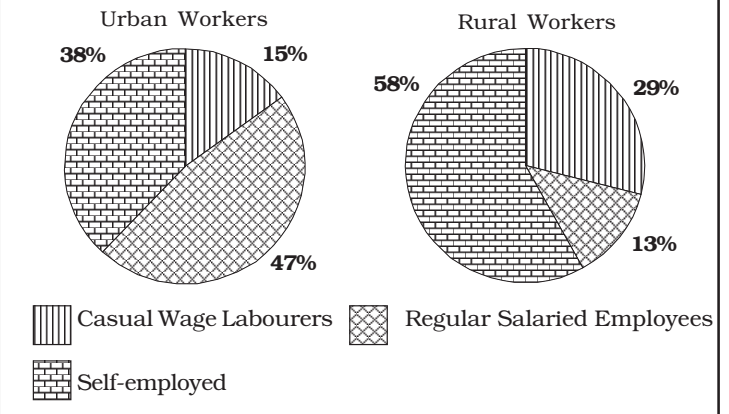


enables us to know the attachment a worker has with his or her job and the authority she or he has over the enterprise and over other co-workers.

Let us take three workers from the construction industry — a cement shop owner, a construction worker and a civil engineer of a construction company. Since the status of each one of them is different from another, they are also called differently. Workers who own and operate an enterprise to earn their livelihood are known as **self-employed**. Thus the cement shop owner is self-employed. About 52 per cent workforce in India belongs to this category. The construction workers are known as **casual wage labourers**; they account for about 25 per cent of India's workforce. Such labourers are casually engaged in others' farms and, in return, get a remuneration for the work done. Workers like the civil engineer working in the construction company account for 23 per cent of India's workforce. When a worker is engaged by someone or an enterprise and paid his or her wages on a regular basis, they are known as **regular salaried employees** (see table 6.3).

Look at Chart 6.1 you will notice that self-employment is a major source of livelihood for both men and women as this category accounts for more than 50 per cent of the workforce. Casual wage work is the second major source for both men and women,

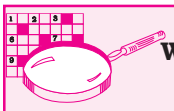
Chart 6.2 : Distribution of Employment by Region



a little more so for the latter (24-27 per cent). When it comes to regular salaried employment, both women and men are found to be so engaged in greater proportion. Men form 23 per cent whereas women form 21 per cent. The gap between men and women is very less.

When we compare the distribution of workforce in rural and urban areas in Chart 6.2 you will notice that the self-employed and casual wage labourers are found more in rural areas than in urban areas. In the latter, both self-employment and regular wage salaried jobs are greater. In the former, since majority of those depending on farming own plots of land and cultivate independently, the share of self-employed is greater.

The nature of work in urban areas is different. Obviously everyone cannot run factories, shops and offices of various types. Moreover enterprises in urban areas require workers on a regular basis.



Work These Out

- We generally think that only those who are doing paid work regularly or casually such as agricultural labourers, factory workers, those who work in banks and other offices as assistants and clerks are workers. From the above discussion, you must have understood that those who are self- employed such as pavement vegetable vendors, professionals such as lawyers, doctors and engineers are also workers. Mark (a), (b) and (c) against self-employed, regular salaried employees and casual wage labourers respectively:
  1. Owner of a saloon
  2. Worker in a rice mill who is paid on daily basis but employed regularly
  3. Cashier in State Bank of India
  4. Typist working in a state government office on a daily wage basis but paid monthly
  5. A handloom weaver
  6. Loading worker in wholesale vegetable shop
  7. Owner of a cool drinks shop which sells Pepsi, Coca Cola and Mirinda
  8. Nurse in a private hospital who gets monthly salary and has been working regularly for the past 5 years.
- Economists point out that casual wage labourers are the most vulnerable among the three categories. Could you locate who these workers are and where they are found and why?
- Can we say that the self-employed earn more than the casual wage labourers or regular salaried employees? Identify a few other indicators of quality of employment.

6.5 Employment in Firms, Factories and Offices



Fig. 6.3 *Garment workers: upcoming factory employment for women*

In the course of economic development of a country, labour flows from agriculture and other related activities to industry and services. In this process, workers migrate from rural to urban areas. Eventually, at a much later stage, the industrial sector begins to lose its share of total employment as the service sector enters a period of rapid expansion. This shift can be understood by looking at the distribution of workers by industry. Generally, we divide all economic activities into eight different industrial divisions. They are (i) Agriculture (ii) Mining and Quarrying (iii) Manufacturing (iv) Electricity, Gas and Water Supply (v) Construction (vi) Trade (vii) Transport and Storage and (viii) Services. For simplicity, all the working persons engaged in these divisions can be clubbed into three major sectors viz., (a) primary sector which includes (i) and (ii), (b) secondary sector which includes (iii), (iv) and (v) and (c) service sector which includes divisions (vi), (vii) and (viii). Table 6.2 shows the distribution of working persons in different industries during the year 2017-18.

Primary sector is the main source of employment for majority of workers

TABLE 6.2

Distribution of Workforce by Industry, 2017-2018

Industrial Category	*Place of Residence*		*Sex*		*Total*
	Rural	*Urban*	*Men*	*Female*	
Primary Sector	59.8	6.6	40.7	57.1	44.6
Secondary Sector	20.4	34.3	26.5	17.7	24.4
Tertiary / Service Sector	19.8	59.1	32.8	25.2	31.0
Total	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**



Work This Out

➢ All newspapers have one section meant for job opportunities. Some also devote an entire supplement in a day or every week like **Opportunities** in *The Hindu* or **Ascent** in The *Times of India*. Many companies advertise vacancies for various positions. Cut those sections. Develop a table which contains four columns: whether the company is private or public, name of the post, number of posts, sector — primary, secondary or tertiary — and qualification required. Analyse the table in the classroom about jobs advertised in the newspapers.

in India. Secondary sector provides employment to only about 24 per cent of workforce. About 31 per cent of workers are in the service sector. Table 6.2 also shows that about 60 per cent of the workforce in rural India depends on agriculture, forestry and fishing. About 20 per cent of rural workers are working in manufacturing industries, construction and other industrial activities. Service sector provides employment to about 20 per cent of rural workers. Agriculture is not a major source of employment in urban areas where people are mainly engaged in the service sector. About 60 per cent of urban workers are in the service sector. The secondary sector gives employment to about one-third of urban workforce.

Though both men and women workers are concentrated in the primary sector, women workers' concentration is very high there. About 57 per cent of the female workforce is employed in the primary sector whereas less than half of males work in that sector. Men get opportunities in both secondary and service sectors.

6.6 Growth and Changing Structure of Employment

In Chapters 2 and 3, you might have studied about the planning strategies in detail. Here we will look at two developmental indicators — growth of employment and GDP. Nearly seventy years of planned development have been aimed at expansion of the economy through increase in national output and employment.

During the period 1950–2010, Gross Domestic Product (GDP) of India grew positively and was higher than the employment growth. However, there was always fluctuation in the growth of GDP. During this period, employment grew at the rate of not more than 2 per cent.

Chart 6.3 also points at another disheartening development in the late 1990s: employment growth started declining and reached the level of growth that India had in the early stages of planning. During these years, we also find a widening gap between the growth of GDP and employment. This means that in the Indian economy, without generating employment, we have been able to

produce more goods and services. Scholars refer to this phenomenon as jobless growth.

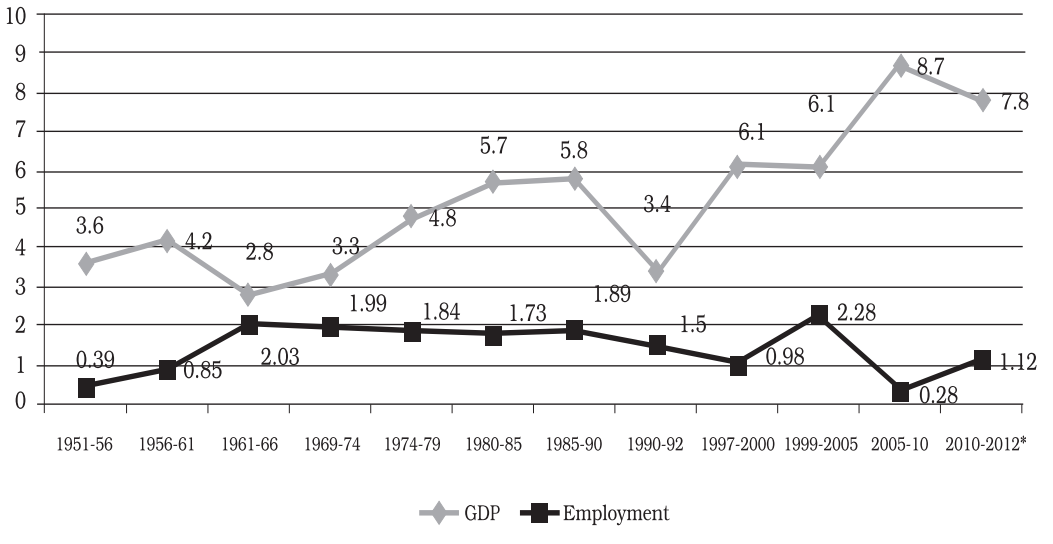
So far we have seen how employment has grown in comparison to GDP. Now it is necessary to know how the growth pattern of employment and GDP affected different sections of workforce. From this we will also be able to understand what types of employment are generated in our country.

Let us look at two indicators that we have seen in the preceding sections — employment of people in various industries and their status. We know that India is an agrarian nation; a major section of population lives in rural areas and is dependent on agriculture as their main livelihood. Developmental strategies in many countries, including India, have aimed at reducing the proportion of people depending on agriculture.

Distribution of workforce by industrial sectors shows substantial shift from farm work to non-farm work (see Table 6.3). In 1972-73, about 74 per cent of workforce was engaged in primary sector and in 2011-12, this proportion has declined to about 50 per cent. Secondary and service sectors are showing promising future for the Indian workforce. You may notice that the shares of these sectors have increased from 11 to 24 per cent and 15 to 27 per cent, respectively.

The distribution of workforce in different status indicates that over the

Chart 6.3: Growth of Employment and Gross Domestic Product, 1951–2012 (%)



Note: **This is the period for which comparable and authentic data are available.*

TABLE 6.3

Trends in Employment Pattern (Sector-wise and Status-wise), 1972-2018 (in %)

Item	*1972-73*	*1983*	*1993-94*	*2011-2012*	*2017-2018*
			Sector		
Primary	74.3	68.6	64	48.9	44.6
Secondary	10.9	11.5	16	24.3	24.4
Services	14.8	16.9	20	26.8	31.0
Total	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
			Status		
Self-employed	61.4	57.3	54.6	52.0	52.2
Regular Salaried Employees	15.4	13.8	13.6	18.0	22.8
Casual Wage Labourers	23.2	28.9	31.8	30.0	25.0
Total	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**

last five decades (1972-2018), people have moved from self-employment and regular salaried employment to casual wage work. Yet self-employment continues to be the major employment provider. Look at the last column of table 6.3. How do you understand the stagnation of secondary sector and moderate rise in self-employement during 2011-18? Discuss in the class. Scholars call the process of moving from self-employment and regular salaried employment to casual wage work noticed during 1972-94 as **casualisation of workforce**. This makes the workers highly vulnerable. How? Look at the case study of Ahmedabad in the preceding section.

In the mean time you also notice in 2017-18 a moderate rise in the share of regular salaried employees. How do you explain this phenomenon?



Work These Out

- Do you know that maintaining employment growth at 2 per cent for a country like India is not an easy thing? Why?
- What will happen if there is no additional employment generated in the economy even though we are able to produce goods and services in the economy? How could jobless growth happen?
- Economists say that if casualisation increases the earning of the people, such phenomenon should be welcomed. Suppose a marginal farmer becomes a full-time agricultural labourer, do you think he will be happy even if he earns more in his daily wage work? Or will a permanent and regular worker of the pharmaceutical industry be happy if he becomes a daily wage labourer, even if his or her overall earnings increase? Discuss in the classroom.

6.7 Informalisation of Indian Workforce

In the previous section we have found that the proportion of hired work jobs - working for others has been increasing. One of the objectives of development planning in India, since India's independence, has been to provide decent livelihood to its people. It has been envisaged that the industrialisation strategy would bring surplus workers from agriculture to industry with better standard of living as in developed countries. We have seen in the preceding section, that even after 70 years of planned development, more than half of the Indian workforce depends on farming as the major source of livelihood.

Economists argue that, over the years, the quality of employment has been deteriorating. Even after working for more than 10-20 years, why do some workers not get maternity benefit, provident fund, gratuity and pension? Why does a person working in the private sector get a lower salary as compared to another person doing the same work but in the public sector?

You may find that a small section of Indian workforce is getting regular income. The government, through its **labour laws**, enable them to protect their rights in various ways. This section of the workforce forms **trade unions**, bargains with employers for better wages and other **social security** measures. Who are they? To know this we classify workforce into two categories: workers in formal and informal sectors, which are also referred to as organised and unorganised sectors. All the **public sector establishments** and those **private sector establishments** which employ 10 hired workers or more are called formal sector establishments and those who work in such establishments are formal sector workers. All other enterprises and workers working in those enterprises form the informal sector. Thus, informal sector includes millions of farmers, agricultural labourers, owners of small enterprises and people working in those enterprises as also the self-employed who do not have any hired workers. It also includes all non-farm casual wage labourers who work for more than one employer such as construction workers and headload workers. You may note that this is one of the ways of classifying workers. There could be other ways of classification as well. Discuss the possible ways in the class.

Box 6.1: Formal Sector Employment

The information relating to employment in the formal sector is collected by the Union Ministry of Labour through employment exchanges located in different parts of the country. Do you know who is the major employer in the formal sector in India? In 2012, out of about 30 million formal sector workers, about 18 million workers were employed by the public sector. Here also men form the majority, as women constitute only about one-sixth of the formal sector workforce. Economists point out that the reform process initiated in the early 1990s resulted in a decline in the number of workers employed in the formal sector. What do you think?

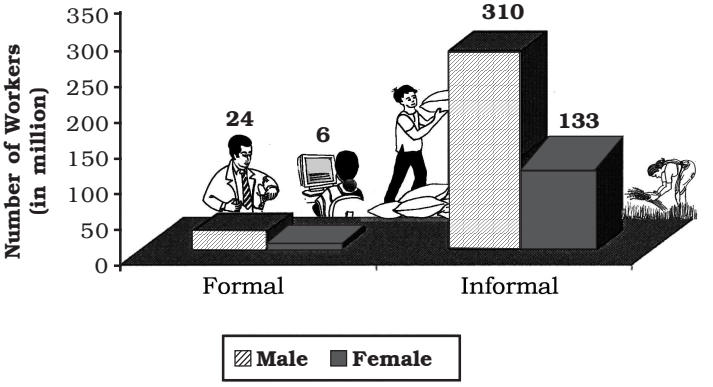


Fig. 6.4 *Road side vending: an increasing variety of informal sector employment*

Those who are working in the formal sector enjoy social security benefits. They earn more than those in the informal sector. Developmental planning envisaged that as the economy grows, more and more workers would become formal sector workers and the proportion of workers engaged in the informal sector would dwindle. But what has happened in India? Look at the following chart which gives the distribution of workforce in formal and informal sectors.

We learnt that in 2011-12 there were about 473 million workers in India. There were about 30 million workers in the formal sector. Can you estimate the percentage of people employed in the formal sectors in the country? About only six per cent (30/473×100)! Thus, the rest 94 per cent are in the informal sector. In 2011-12, the year for which gender wise data on formal-informal sector employment is available (Chart 6.4), about 20 per cent of formal sector and 30 per cent of informal sector workers are women.

Chart 6.4: Workers in Formal and Informal Sector, 2011-12



Box 6.2: Informalisation in Ahmedabad

Ahmedabad is a prosperous city with its wealth based on the produce of more than 60 textile mills with a labour force of 1,50,000 workers employed in them. These workers had, over the course of the century, acquired a certain degree of income security. They had secure jobs with a living wage; they were covered by social security schemes protecting their health and old age. They had a strong trade union which not only represented them in disputes but also ran activities for the welfare of workers and their families. In the early 1980s, textile mills all over the country began to close down. In some places, such as Mumbai, the mills closed rapidly. In Ahmedabad, the process of closure was long drawn out and spread over 10 years. Over this period, approximately over 80,000 permanent workers and over 50,000 non-permanent workers lost their jobs and were driven to the informal sector. The city experienced an economic recession and public disturbances, especially communal riots. A whole class of workers was thrown back from the middle class into the informal sector, into poverty. There was widespread alcoholism and suicides, children were withdrawn from school and sent to work.



Change in the balance of power in a house: an unemployed mill worker peeling garlic whereas his wife has a new job of beedi rolling.

Source: *Renana Jhabvala, Ratna M. Sudarshan and Jeemol Unni (Ed.) Informal Economy at Centre Stage: New Structures of Employment, Sage Publications, New Delhi, 2003, pp.265.*

Since the late 1970s, many developing countries, including India, started paying attention to enterprises and workers in the informal sector as employment in the formal sector is not growing. Workers and enterprises in the informal sector do not get regular income; they do not have any protection or regulation from the government. Workers are dismissed without any compensation. Technology used in the informal sector enterprises is outdated; they also do not maintain any accounts. Workers of this sector live in slums and are squatters. Of late, owing to the efforts of the International Labour Organisation (ILO), the Indian government has

Work These Out

Tick (✓) mark against those which are in the informal sector

- Worker in a hotel which has seven hired workers and three family workers
- A private school teacher in a school which has 25 teachers
- A police constable
- Nurse in a government hospital
- Cycle-rickshaw puller
- The owner of a textile shop employing nine workers
- Driver of a bus company which has more than 10 buses with 20 drivers, conductors and other workers
- Civil engineer working in a construction company which has 10 workers
- Computer operator in the state government office working on a temporary basis
- A clerk in the electricity office.

initiated the modernisation of informal sector enterprises and provision of social security measures to informal sector workers.

6.8 Unemployment

You might have seen people looking for jobs in newspapers. Some look for a job through friends and relatives. In many cities, you might find people standing in some select areas looking for people to employ them for that day's work. Some go to factories and offices and give their bio-data and ask whether there is any vacancy in their factory or office. Many in the rural areas do not go out and ask for a job but stay home when there is no work. Some go to



Fig. 6.5 *Unemployed mill workers waiting for casual jobs*



Fig. 6.6 *Sugar cane cutters: disguised unemployment is common in farm works*

employment exchanges and register themselves for vacancies notified through employment exchanges. The National Statistical Office (Previously it was known as National Sample Survey Organisation) defines unemployment as a situation in which all those who, owing to lack of work, are not working but either seek work through employment exchanges, intermediaries, friends or relatives or by making applications to prospective employers or express their willingness or availability for work under the prevailing condition of work and remunerations. There are a variety of ways by which an unemployed person is identified. Economists define unemployed person as one who is not able to get employment of even one hour in half a day.

There are three sources of data on unemployment : Reports of Census of India, National Statistical Office's Reports of Employment and Unemployment Situation, Annual Reports of Periodic Labour Force Survey, and Directorate General of Employment and Training data of Registration with Employment Exchanges. Though they provide different estimates of unemployment, they do provide us with the attributes of the unemployed and the variety of unemployment prevailing in our country.

Do we have different types of unemployment in our economy? The situation described in the first paragraph of this section is called open unemployment. Economists call unemployment prevailing in Indian farms as disguised unemployment. What is disguised unemployment? Suppose a farmer has four acres of land and he actually needs only two workers and himself to carry out various operations on his farm in a year, but if he employs five workers and his family members such as his wife and children, this situation is known as disguised unemployment. One study conducted in the late 1950s showed about one-third of agriculture workers in India as disguisedly unemployed.

You may have noticed that many people migrate to an urban area, pick up a job and stay there for some time, but come back to their home villages as soon as the rainy season begins. Why do they do so? This is because work in agriculture is seasonal; there are no employment opportunities in the village for all months in the year. When there is no work to do on farms, people go to urban areas and look for jobs.



Fig. 6.7 *Dam construction work is a direct way of employment generation by the government*

This kind of unemployment is known as seasonal unemployment. This is also a common form of unemployment prevailing in India.

Though we have witnessed slow growth of employment, have you seen people being unemployed over a very long time? Scholars say that in India, people cannot remain completely unemployed for very long because their desperate economic condition would not allow them to be so. You will rather find them being forced to accept jobs that nobody else would do, unpleasant or even dangerous jobs in unclean, or unhealthy surroundings. The Central and State governments take initiatives and generate employment to facilitate a decent living for low income families through various measures. These will be discussed in the following section.

6.9 Government and Employment Generation

You may recall about the Mahatma Gandhi National Rural Employment Guarantee Act 2005. It promises 100 days of guaranteed wage employment to all rural households who volunteer to do unskilled manual work. The scheme based on this Act is one of the many measures the government has implemented to generate employment for those who are in need of jobs in rural areas.

Since Independence, the Union and State governments have played an important role in generating employment or creating opportunities for employment generation. Their efforts can be broadly categorised into two — direct and indirect. In the first category, as you have seen in the preceding section, the government employs people in various departments for administrative purposes. It also runs industries, hotels and transport companies, and hence, provides employment directly to workers. When the output of goods and services from government enterprises increases, then private enterprises which receive raw materials from government enterprises will also raise their output and hence increase the number of employment opportunities in the economy. For example, when a government owned

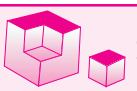
steel company increases its output, it will result in direct increase in employment in that government company. Simultaneously, private companies, which purchase steel from it, will also increase their output and thus employment. This is the indirect generation of employment opportunities by the government initiatives in the economy.

Many programmes that the governments implement, aimed at alleviating poverty, are through employment generation. They are also known as employment generation programmes. All these programmes aim at providing not only employment but also services in areas such as primary health, primary education, rural drinking water, nutrition, assistance for people to buy income and employment generating assets, development of community assets by generating wage employment, construction of houses and sanitation, assistance for constructing houses, laying of rural roads, development of wastelands/ degraded lands.

6.10 Conclusion

There has been a change in the structure of workforce in India. Newly emerging jobs are found mostly in the service sector. The expansion of the service sector and the advent of high technology now frequently permit a highly competitive existence for efficient small scale and often individual enterprises or specialist workers side by side with the multinationals. Outsourcing of work is becoming a common practice. It means that a big firm finds it profitable to close down some of its specialist departments (for example, legal or computer programming or customer service sections) and hand over a large number of small piecemeal jobs to very small enterprises or specialist individuals, sometimes situated even in other countries. The traditional notion of the modern factory or office, as a result, has been altered in such a manner that for many the home is becoming the workplace. All of this change has not gone in favour of the individual worker. The nature of employment has become more informal with only limited availability of social security measures to the workers.

In the last few decades, there has been rapid growth in the gross domestic product, but without simultaneous increase in employment opportunities. This has forced the government to take up initiatives in generating employment opportunities particularly in the rural areas.



Recap

- All those persons who are engaged in various economic activities and hence contribute to gross national product are workers.
- About two-fifth of the total population in the country is engaged in various economic activities.
- Men particularly rural men, form the major section of workforce in India.
- Majority of workers in India are self-employed. Casual wage labourers and regular salaried employees together account for less than half the proportion of India's workforce.
- About three-fifth of India's workforce depends on agriculture and other allied activities as the major source of livelihood.
- In recent years, the growth of employment has decelerated.
- During post-reform period, India has been witness to employment opportunities in the service sector. These new jobs are found mostly in the informal sector and the nature of jobs is also mostly casual.
- Government is the major formal sector employer in the country.
- Disguised unemployment is a common form of unemployment in rural India.
- There has been a change in the structure of the workforce in India.
- Through various schemes and policies, the government takes initiatives to generate employment directly and indirectly.



EXERCISES

1. Who is a worker?
2. Define worker-population ratio.
3. Are the following workers — a beggar, a thief, a smuggler, a gambler? Why?

4. Find the odd man out (i) owner of a saloon (ii) a cobbler (iii) a cashier in Mother Dairy or Milk Cooperative Society of your area (iv) a tuition master (v) transport operator (vi) construction worker.

5. The newly emerging jobs are found mostly in the _____ sector (service/manufacturing).

6. An establishement with four hired workers is known as _____ (formal/informal) sector establishment.

7. Raj is going to school. When he is not in school, you will find him working in his farm. Can you consider him as a worker? Why?

8. Compared to urban women, more rural women are found working. Why?

9. Meena is a housewife. Besides taking care of household chores, she works in the cloth shop which is owned and operated by her husband. Can she be considered as a worker? Why?

10. Find the odd man out (i) rickshaw puller who works under a rickshaw owner (ii) mason (iii) mechanic shop worker (iv) shoeshine boy.

11. The following table shows distribution of workforce in India for the year 1972-73. Analyse it and give reasons for the nature of workforce distribution. You will notice that the data is pertaining to the situation in India about 50 years ago!

Place of Residence	*Workforce (in millions)*		
	Male	*Female*	*Total*
Rural	125	69	195
Urban	32	7	39

12. The following table shows the population and worker population ratio for India in 1999-2000. Can you estimate the workforce (urban and total) for India?

Region	*Estimates of Population (in crores)*	*Worker Population Ratio*	*Estimated No. of Workers (in crores)*
Rural	71.88	41.9	$\frac{71.88}{100}$ 41.9 = 30.12
Urban	28.52	33.7	?
Total	100.40	39.5	?

13. Why are regular salaried employees more in urban areas than in rural areas?

14. Why are less women found in regular salaried employment?

15. Analyse the recent trends in sectoral distribution of workforce in India.

16. Compared to the 1970s, there has hardly been any change in the distribution of workforce across various industries. Comment.

17. Do you think that during 1950-2010 employment generated in the country is commensurate with the growth of GDP in India? How?

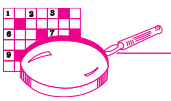
18. Is it necessary to generate employment in the formal sector rather than in the informal sector? Why?

19. Victor is able to get work only for two hours in a day. Rest of the day, he is looking for work. Is he unemployed? Why? What kind of jobs could persons like Victor be doing?

20. You are residing in a village. If you are asked to advice the village panchayat, what kinds of activities would you suggest for the improvement of your village which would also generate employment.

21. Who is a casual wage labourer?

22. How will you know whether a worker is working in the informal sector?



SUGGESTED ADDITIONAL ACTIVITIES

1. Select a region, say a street or colony, and divide it into 3-4 sub-regions. Conduct a survey by which you can collect the details of activity each person living there is engaged in. Derive the worker-population ratio for all the regions. Interpret the results for differences in worker-population ratio for the different sub-regions.

2. Suppose 3-4 groups of students are given different regions of a state. One region is mainly engaged in cultivation of paddy. In another region, coconut is the main plantation. The third region is a coastal region where fishing is the main activity. The fourth region has a river nearby with a lot of livestock rearing activities. Ask all the four groups to develop a report on what kind of employment could be generated in the four regions.

3. Visit the local library and ask for *Employment News,* a weekly published by the Government of India. Go through each issue for the last two months. There will be seven issues. Select 25 advertisements and

fill in the following table (expand the table as needed). Discuss the nature of jobs in the classroom.

Items	*Advertisement 1*	*Advertisement 2*
1. Name of Office		
2. Department/company		
3. Private/public/joint venture		
4. Name of the post		
5. Sector—primary/seconday/ service		
6. Number of posts/vacancies		
7. Qualification required		

4. You might notice, in your locality, a variety of works being done by the government, for example laying of roads, desilting of tanks, construction of school buildings, hospital and other government offices, construction of check dams and houses for the poor etc. Prepare a critical assessment report on one such activity. The issues covered could be the following (i) how the work was identified (ii) amount sanctioned (iii) contribution of local people, if any (iv) number of persons involved — both men and women (v) wages paid (vi) is it really required in that area and other critical comments on the implementation of the scheme under which the work is being carried out.
5. In recent years, you may have noticed that many voluntary organisations also take initiatives to generate employment in hilly and dry land regions. If you find such initiatives in your locality, visit and prepare a report.



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7



Environment and Sustainable Development

After studying this chapter, the learners will

- understand the concept of environment
- analyse the causes and effects of 'environmental degradation' and 'resource depletion'
- understand the nature of environmental challenges facing India
- relate environmental issues to the larger context of sustainable development.

The environment, left to itself, can continue to support life for millions of years. The single most unstable and potentially disruptive element in the scheme is the human species. Human beings, with modern technology, have the capacity to bring about, intentionally or unintentionally, far-reaching and irreversible changes in the enviornment.

Anonymous

7.1 Introduction

In the earlier chapters we have discussed the main economic issues faced by the Indian economy. The economic development that we have achieved so far has come at a very heavy price—at the cost of environmental quality. As we step into an era of globalisation that promises higher economic growth, we have to bear in mind the adverse consequences of the past developmental path on our environment and consciously choose a path of sustainable development. To understand the unsustainable path of development that we have taken and the challenges of sustainable development, we have to first understand the significance and contribution of environment to economic development. With this in mind, this chapter is divided into three sections. The first part deals with the functions and role of environment. The second section discusses the state of India's environment and the third section deals with steps and strategies to achieve sustainable development.

7.2 Environment — Definition and Functions

Environment is defined as the total planetary inheritance and the totality of all resources. It includes all the biotic and abiotic factors that influence each other. While all living elements—the birds, animals and plants, forests, fisheries etc.—are biotic elements, abiotic elements include air, water, land etc. Rocks and sunlight are examples of abiotic elements of the environment. A study of the environment then calls for a study of the inter-relationship between these biotic and abiotic components of the environment.

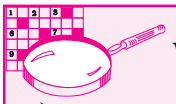
Functions of the Environment: The environment performs four vital functions (i) it supplies resources: resources here include both renewable and non-renewable resources. Renewable resources are those which can be used without the possibility of the resource becoming depleted or exhausted. That is, a continuous supply of the resource remains available. Examples of renewable resources are the trees in the forests and the fishes in the ocean. Non-renewable resources, on the other hand, are those which get exhausted with extraction and use, for example, fossil fuel (ii) it assimilates waste (iii) it sustains life by providing genetic and bio diversity and (iv) it also provides aesthetic services like scenery etc.

The environment is able to perform these functions without any interruption as long as the demand on these



Fig. 7.1 *Water bodies: small, snow-fed Himalayan streams are the few fresh-water sources that remain unpolluted.*

functions is within its **carrying capacity**. This implies that the resource extraction is not above the rate of regeneration of the resource and the wastes generated are within the assimilating capacity of the environment. When this is not so, the environment fails to perform its third and vital function of life sustenance and this results in an environmental crisis. This is the situation today all over the world. The rising population of the developing countries and the affluent consumption and production standards of the developed world have placed a huge stress on the environment in terms of its first two functions. Many resources have become extinct and the wastes generated are beyond the absorptive capacity of the environment. **Absorptive capacity** means the ability of the environment to absorb degradation. The result — we are today at the threshold of environmental crisis. The past development has polluted and dried up rivers and other aquifers making water an economic good. Besides, the intensive and extensive extraction of both renewable and non-renewable resources has exhausted some of these



Work These Out

- Why has water become an economic commodity? Discuss.
- Fill in the following table with some common types of diseases and illnesses that are caused due to air, water and noise pollution.

Air Pollution	*Water Pollution*	*Noise Pollution*
Asthma	Cholera	

Box 7.1: Global Warming

Global warming is a gradual increase in the average temperature of the earth's lower atmosphere as a result of the increase in greenhouse gases since the **Industrial Revolution**. Much of the recent observed and projected global warming is human-induced. It is caused by man-made increases in carbon dioxide and other **greenhouse gases** through the burning of **fossil fuels** and **deforestation**. Adding carbon dioxide, methane and such other gases (that have the potential to absorb heat) to the atmosphere with no other changes will make our planet's surface warmer. The atmospheric concentrations of carbon dioxide and CH_4 have increased by 31 per cent and 149 per cent respectively above pre-industrial levels since 1750. During the past century, the atmospheric temperature has risen by 1.1 F (0.6 C) and sea level has risen several inches. Some of the longer-term results of global warming are melting of polar ice with a resulting rise in sea level and coastal flooding; disruption of drinking water supplies dependent on snow melts; extinction of species as ecological niches disappear; more frequent tropical storms; and an increased incidence of tropical diseases.

Among factors that may be contributing to global warming are the burning of coal and petroleum products (sources of carbon dioxide, methane, nitrous oxide, ozone); deforestation, which increases the amount of carbon dioxide in the atmosphere; methane gas released in animal waste; and increased cattle production, which contributes to deforestation, methane production, and use of fossil fuels. A UN Conference on Climate Change, held in Kyoto, Japan, in 1997, resulted in an international agreement to fight global warming which called for reductions in emissions of greenhouse gases by industrialised nations.

Source: *www.wikipedia.org*

vital resources and we are compelled to spend huge amounts on technology and research to explore new resources. Added to these are the health costs of degraded environmental quality — decline in air and water quality (seventy per cent of water in India is polluted) have resulted in increased incidence of respiratory and water-borne diseases. Hence the expenditure on health is also rising. To make matters worse, global environmental issues such as **global warming** and **ozone depletion** also contribute to increased financial commitments for the government. Thus, it is clear that the **opportunity costs** of negative environmental impacts are high.

The biggest question that arises is: are environmental problems new to this century? If so, why? The answer to this question requires some elaboration. In the early days when civilisation just began, or before this phenomenal increase in population, and before countries took to industrialisation, the demand for environmental resources and services was much less than their supply. This meant that pollution was within the absorptive capacity of the

Box 7.2: Ozone Depletion

Ozone depletion refers to the phenomenon of reductions in the amount of ozone in the **stratosphere**. The problem of ozone depletion is caused by high levels of chlorine and bromine compounds in the stratosphere. The origins of these compounds are chlorofluorocarbons (CFC), used as cooling substances in air-conditioners and refrigerators, or as aerosol propellants, and bromofluorocarbons (halons), used in fire extinguishers. As a result of depletion of the ozone layer, more ultraviolet (UV) radiation comes to Earth and causes damage to living organisms. UV radiation seems responsible for skin cancer in humans; it also lowers production of phytoplankton and thus affects other aquatic organisms. It can also influence the growth of terrestrial plants. A reduction of approximately 5 per cent in the ozone layer was detected from 1979 to 1990. Since the **ozone layer** prevents most harmful wavelengths of ultraviolet light from passing through the **Earth's atmosphere**, observed and projected decreases in ozone have generated worldwide concern. This led to the adoption of the **Montreal Protocol** banning the use of chlorofluorocarbon (CFC) compounds, as well as other ozone depleting chemicals such as carbon tetrachloride, trichloroethane (also known as methyl chloroform), and bromine compounds known as **halons.**

Source: *www.ceu.hu*

environment and the rate of resource extraction was less than the rate of regeneration of these resources. Hence environmental problems did not arise.



Fig. 7.2 *Damodar Valley is one of India's most industrialised regions. Pollutants from the heavy industries along the banks of the Damodar river are converting it into an ecological disaster*

But with population explosion and with the advent of industrial revolution to meet the growing needs of the expanding population, things changed. The result was that the demand for resources for both production and consumption went beyond the rate of regeneration of the resources; the pressure on the absorptive capacity of the environment increased tremendously — this trend continues even today. Thus what has happened is a reversal of supply-demand relationship for environmental quality — we are now faced with increased demand for environmental resources and services but their supply is limited due to overuse

and misuse. Hence the environmental issues of waste generation and pollution have become critical today.



Fig. 7.3 *Deforestation leads to land degradation, biodiversity loss and air pollution*

7.3 State of India's Environment

India has abundant natural resources in terms of rich quality of soil, hundreds of rivers and tributaries, lush green forests, plenty of mineral deposits beneath the land surface, vast stretch of the Indian Ocean, ranges of mountains, etc. The black soil of the Deccan Plateau is particularly suitable for cultivation of cotton, leading to concentration of textile industries in this region. The Indo-Gangetic plains — spread from the Arabian Sea to the Bay of Bengal — are one of the most fertile, intensively cultivated and densely populated regions in the world. India's forests, though unevenly distributed, provide green cover for a majority of its population and natural cover for its wildlife. Large deposits of iron-ore, coal and natural gas are found in the country. India accounts for nearly 8 per cent of the world's total iron-ore reserves. Bauxite, copper, chromate, diamonds, gold, lead, lignite, manganese, zinc, uranium, etc. are also available in different parts of the country. However, the developmental activities in India have resulted in pressure on its finite natural resources, besides creating impacts on human health and well-being. The threat to India's environment poses a dichotomy—threat of poverty-induced environmental degradation and, at the same time, threat of pollution from affluence and a rapidly growing industrial sector. Air pollution, water contamination, soil erosion, deforestation and wildlife extinction are some of the most pressing environmental concerns of India. The priority issues identified are (i) land degradation (ii) biodiversity loss (iii) air pollution with special reference to vehicular pollution in urban cities (iv) management of fresh water and (v) solid waste management. Land in India suffers from varying degrees and types of degradation stemming mainly from unstable use and inappropriate management practices.

Box. 7.3: Chipko or Appiko — What's in a Name?

You may be aware of the Chipko Movement, which aimed at protecting forests in the Himalayas. In Karnataka, a similar movement took a different name, 'Appiko', which means to hug. On 8 September 1983, when the felling of trees was started in Salkani forest in Sirsi district, 160 men, women and children hugged the trees and forced the woodcutters to leave. They kept vigil in the forest over the next six weeks. Only after the forest officials assured the volunteers that the trees will be cut scientifically and in accordance with the working plan of the district, did they leave the trees.



When commercial felling by contractors damaged a large number of natural forests, the idea of hugging the trees gave the people hope and confidence that they can protect the forests. On that particular incident, with the felling discontinued, the people saved 12,000 trees. Within months, this movement spread to many adjoining districts.

Indiscriminate felling of trees for fuelwood and for industrial use has led to many environmental problems. Twelve years after setting up of a paper mill in Uttar Kanara area, bamboo has been wiped out from that area. "Broad-leaved trees which protected the soil from the direct onslaught of rain have been removed, the soil washed away, and bare laterite soil left behind. Now nothing grows but a weed", says a farmer. Farmers also complain that rivers and rivulets dry up quicker, and that rainfall is becoming erratic. Diseases and insects earlier unknown are now attacking the crops.

Appiko volunteers want the contractors and forest officials to follow certain rules and restrictions. For instance, local people should be consulted when trees are marked for felling and trees within 100 metres of a water source and on a slope of 30 degrees or above should not be felled.

Do you know that the government allocates forestlands to industries to use forest materials as industrial raw material? Even if a paper mill employs 10,000 workers and a plywood factory employs 800 people but if they deprive the daily needs of a million people, is it acceptable? What do you think?

Source: *Excerpts from 'State of India's Environment 2: The Second Citizens' Report 1984-85', Centre for Science and Environment, 1996, New Delhi.*

Some of the factors responsible for land degradation are (i) loss of vegetation occuring due to deforestation (ii) unsustainable fuel wood and fodder extraction (iii) shifting cultivation (iv) encroachment into forest lands (v) forest fires and over grazing (vi) non-adoption of adequate soil conservation measures (vii) improper crop rotation (viii) indiscriminate use of agro-chemicals such as fertilisers and pesticides (ix) improper planning and management of irrigation systems (x) extraction of ground water in excess of the recharge capacity (xi) open access resource and (xii) poverty of the agriculture-dependent people.

India supports approximately 17 per cent of the world's human and 20 per cent of livestock population on a mere 2.5 per cent of the world's geographical area. The high density of population and livestock and the competing uses of land for forestry, agriculture, pastures, human settlements and industries exert an enormous pressure on the country's finite land resources.

The per capita forest land in the country is only 0.06 hectare against the requirement of 0.47 hectare to meet basic needs, resulting in an excess felling of about 15 million cubic metre forests over the permissible limit.

Estimates of soil erosion show that soil is being eroded at a rate of 5.3 billion tonnes a year for the entire country as a result of which the country loses 0.8 million tonnes of nitrogen, 1.8 million tonnes of phosphorus and 26.3 million tonnes of potassium every year. According to the Government of India, the quantity of nutrients lost due to erosion each year ranges from 5.8 to 8.4 million tonnes.



Work These Out

- In order to enable the students to appreciate the contribution of environment to economic development, the following game can be introduced. One student may name a product used by any enterprise and the other student may trace out its roots to nature and earth.

trucks	←	steel and rubber						
steel	←	iron	←	mineral	←	earth		
rubber	←	trees	←	forests	←	earth		
books	←	paper	←	trees	←	forest	←	earth
cloth	←	cotton	←	plant	←	nature		
petrol	←	earth						
machinery	←	iron	←	mineral	←	earth		

- A truck driver had to pay Rs 10,000 as challan as his truck was emitting black soot. Why do you think he was penalised? Was it justified? Discuss.

Box 7.4 : Pollution Control Boards

In order to address two major environmental concerns in India, viz. water and air pollution, the government set up the Central Pollution Control Board (CPCB) in 1974. This was followed by states establishing their own state level boards to address all the environmental concerns. They investigate, collect and disseminate information relating to water, air and land pollution, lay down standards for sewage/trade effluent and emissions. These boards provide technical assistance to governments in promoting cleanliness of streams and wells by prevention, control and abatement of water pollution, and improve the quality of air and to prevent, control or abate air pollution in the country.

These boards also carry out and sponsor investigation and research relating to problems of water and air pollution and for their prevention, control or abatement. They organise, through mass media, a comprehensive mass awareness programme for the same. The PCBs prepare manuals, codes and guidelines relating to treatment and disposal of sewage and trade effluents.

They assess the air quality through regulation of industries. In fact, state boards, through their district level officials, periodically inspect every industry under their jurisdiction to assess the adequacy of treatment measures provided to treat the effluent and gaseous emissions. It also provides background air quality data needed for industrial siting and town planning.

The pollution control boards collect, collate and disseminate technical and statistical data relating to water pollution. They monitor the quality of water in 125 rivers (including the tributaries), wells, lakes, creeks, ponds, tanks, drains and canals.

- Visit a nearby factory/irrigation department and collect the details of measures that they adopt to control water and air pollution.
- You might be seeing advertisements in newspapers, radio and television or billboards in your locality on awareness programmes relating to water and air pollution. Collect a few news-clippings, pamphlets and other information and discuss them in the classroom.

In India, air pollution is widespread in urban areas where vehicles are the major contributors and in a few other areas which have a high concentration of industries and thermal power plants. Vehicular emissions are of particular concern since these are ground level sources and, thus, have the maximum impact on the general population. The number of motor vehicles has increased from about 3 lakh in 1951 to 30 crores in 2019. In 2016, personal transport vehicles (two-wheeled vehicles and cars only) constituted about 85 per cent of the total number of registered vehicles thus contributing significantly to total air pollution load.

India is one of the ten most industrialised nations of the world. But this status has brought with it unwanted and unanticipated consequences such as unplanned urbanisation, pollution and the risk of accidents. The CPCB (Central Pollution

Control Board) has identified seventeen categories of industries (large and medium scale) as significantly polluting (See Box 7.4).



Work This Out

➢ You can see a column on the measure of air pollution in any national daily. Cut out the news item a week before Diwali, on the day of Diwali and two days after Diwali. Do you observe a significant difference in the value? Discuss in your class.

The above points highlight the challenges to India's environment. The various measures adopted by the Ministry of Environment and the central and state pollution control boards may not yield reward unless we consciously adopt a path of sustainable development. The concern for future generations alone can make development last forever. Development to enhance our current living styles, without concern for posterity, will deplete resources and degrade environment at a pace that is bound to result in both environmental and economic crisis.

7.4 Sustainable Development

Environment and economy are interdependent and need each other. Hence, development that ignores its repercussions on the environment will destroy the environment that sustains life forms. What is needed is sustainable development: development that will allow all future generations to have a potential average quality of life that is at least as high as that which is being enjoyed by the current generation. The concept of sustainable development was emphasised by the United Nations Conference on Environment and Development (UNCED), which defined it as: 'Development that meets the need of the present generation without compromising the ability of the future generation to meet their own needs'.

Read the definition again. You will notice that the term 'need' and the phrase 'future generations' in the definition are the catch phrases. The use of the concept 'needs' in the definition is linked to distribution of resources. The seminal report — ***Our Common Future*** — that gave the above definition explained sustainable development as 'meeting the basic needs of all and extending to all the opportunity to satisfy their aspirations for a better life'. Meeting the needs of all requires redistributing resources and is hence a moral issue. Edward Barbier defined sustainable development as one which is directly concerned with increasing the material standard of living of the poor at the grass root level — this can be quantitatively measured in terms of increased income, real income, educational services, health care, sanitation, water supply etc. In more specific terms, sustainable development aims at decreasing the absolute poverty of the poor by providing lasting and secure livelihoods that minimise resource depletion, environmental

degradation, cultural disruption and social instability. Sustainable development is, in this sense, a development that meets the basic needs of all, particularly the poor majority, for employment, food, energy, water, housing, and ensures growth of agriculture, manufacturing, power and services to meet these needs.

The **Brundtland Commission** emphasises on protecting the future generation. This is in line with the argument of the environmentalists who emphasise that we have a moral obligation to hand over the planet earth in good order to the future generation; that is, the present generation should bequeath a better environment to the future generation. At least we should leave to the next generation a stock of 'quality of life' assets no less than what we have inherited.

The present generation can promote development that enhances the natural and built environment in ways that are compatible with (i) conservation of natural assets (ii) preservation of the regenerative capacity of the world's natural ecological system (iii) avoiding the imposition of added costs or risks on future generations.

According to Herman Daly, a leading environmental economist, to achieve sustainable development, the following needs to be done (i) limiting the human population to a level within the carrying capacity of the environment. The carrying capacity of the environment is like a'plimsoll line' of the ship which is its load limit mark. In the absence of the plimsoll line for the economy, human scale grows beyond the carrying capacity of the earth and deviates from sustainable development (ii) technological progress should be input efficient and not input consuming (iii) renewable resources should be extracted on a sustainable basis, that is, rate of extraction should not exceed rate of regeneration (iv) for non-renewable resources rate of depletion should not exceed the rate of creation of renewable substitutes and (v) inefficiencies arising from pollution should be corrected. In 2015, the UN formulated 17 Sustainable Development Goals (SDGs) intended to be achieved by the year 2030. Collect the details of those goals and discuss them in the context of India.

7.5 Strategies for Sustainable Development

Use of Non-conventional Sources of Energy: India, as you know, is hugely dependent on thermal and hydro power plants to meet its power needs. Both of these have adverse environmental impacts. Thermal power plants emit large quantities of carbon dioxide which is a green house gas. It also produces fly ash which, if not used properly, can cause pollution of water bodies, land and other components of the environment. Hydroelectric projects inundate forests and interfere with the natural flow of water in catchment areas and the river basins. Wind power and solar rays are good examples of conventional. In recent years, some efforts are being taken to tap these energy resources. Collect the details of one such unit set up in your area if any, and discuss in the class.

LPG, Gobar Gas in Rural Areas: Households in rural areas generally use wood, dung cake or other biomass as fuel. This practice has several adverse implications like deforestation, reduction in green cover, wastage of cattle dung and air pollution. To rectify the situation, subsidised LPG is being provided. In addition, gobar gas plants are being provided through easy loans and subsidy. As far as liquefied petroleum gas (LPG) is concerned, it is a clean fuel — it reduces household pollution to a large extent. Also, energy wastage is minimised. For the gobar gas plant to function, cattle dung is fed to the plant and gas is produced which is used as fuel while the slurry which is left over is a very good organic fertiliser and soil conditioner.

CNG in Urban Areas: In Delhi, the use of Compressed Natural Gas (CNG) as fuel in public transport system has significantly lowered air pollution and the air has become cleaner. In the last few years many other Indian cities also began to use CNG.

Work This Out

- In Delhi, buses and other public transport vehicles uses CNG as fuel instead of petrol or diesel; some vehicles use convertible engines; solar energy is being used to light up the streets. What do you think about these changes? Delhi also adopted odd/even scheme to restrict the use of vehicles with registration ending with odd/even numbers on alternative days, for specific period in a year. Organise a debate in class on the need for sustainable development practices in India.

Wind Power: In areas where speed of wind is usually high, wind mills can provide electricity without any adverse impact on the environment. Wind turbines move with the wind and electricity is generated. No doubt, the initial cost is high. But the benefits are such that the high cost gets easily absorbed.

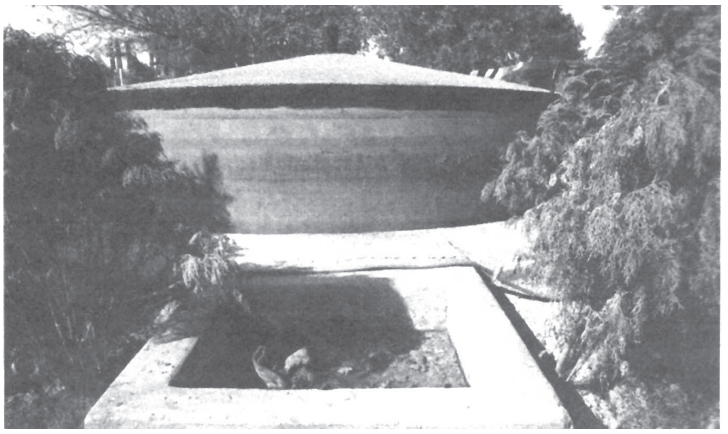


Fig.7.4 *Gobar Gas Plant uses cattle dung to produce energy*

Solar Power through Photovoltaic Cells: India is naturally endowed with a large quantity of solar energy in the form of sunlight. We use it in different ways. For example, we dry our clothes, grains, other agricultural products as well as various items made for daily use. We also use sunlight to warm ourselves in winter. Plants use solar energy to perform photosynthesis. Now, with the help of photovoltaic cells, solar energy can be converted into electricity. These cells use special kind of materials to capture solar energy and then convert the energy into electricity. This technology is extremely useful for remote areas and for places where supply of power through grid or power lines is either not possible or proves very costly. This technique is also totally free from pollution. In recent years India is taking efforts to increase the power generation through solar. India is also leading an International body called International Solar Alliance (ISA).

Mini-hydel Plants: In mountainous regions, streams can be found almost everywhere. A large percentage of such streams are perennial. Mini-hydel plants use the energy of such streams to move small turbines. The turbines generate electricity which can be used locally. Such power plants are more or less environment-friendly as they do not change the land use pattern in areas where they are located; they generate enough power to meet local demands. This means that they can also do away with the need for large scale transmission towers and cables and avoid transmission loss.

Traditional Knowledge and Practices: Traditionally, Indian people have been close to their environment. They have been more a component of the environment and not its controller. If we look back at our agriculture system, healthcare system, housing, transport etc., we find that all practices have been environment friendly. Only recently have we drifted away from the traditional systems and caused large scale damage to the environment and also our rural heritage. Now, it is time to go back. One apt example is in healthcare. India is very much privileged to have about 15,000 species of plants which have medicinal properties. About 8,000 of these are in regular use in various systems of treatment including the folk tradition. With the sudden onslaught of the western system of treatment, we ignored our traditional systems such as Ayurveda, Unani, Tibetan and folk systems. These healthcare systems are in great demand again for treating chronic health problems. Now a days every cosmetic produce — hair oil, toothpaste, body lotion, face cream and what not — is herbal in composition. Not only are these products environment friendly, they are relatively free from side effects and do not involve large-scale industrial and chemical processing.

Biocomposting: In our quest to increase agricultural production during the last five decades or so, we almost totally neglected the use of compost and completely switched over to chemical fertilisers. The result is that large tracts of productive land have

been adversely affected, water bodies including ground water system have suffered due to chemical contamination and demand for irrigation has been going up year after year.

Farmers, in large numbers all over the country, have again started using compost made from organic wastes of different types. In certain parts of the country, cattle are maintained only because they produce dung which is an important fertiliser and soil conditioner.

Earthworms can convert organic matter into compost faster than the normal composting process. This process is now being widely used. Indirectly, the civic authorities are benefited too as they have to dispose reduced quantity of waste.

Biopest Control: With the advent of green revolution, the entire country entered into a frenzy to use more and more chemical pesticides for higher yield. Soon, the adverse impacts began to show; food products were contaminated, soil, water bodies and even ground water were polluted with pesticides. Even milk, meat and fishes were found to be contaminated.

To meet this challenge, efforts are on to bring in better methods of pest control. One such step is the use of pesticides based on plant products. Neem trees are proving to be quite useful. Several types of pest controlling chemicals have been isolated from neem and these are being used. Mixed cropping and growing different crops in consecutive years on the same land have also helped farmers.

In addition, awareness is spreading about various animals and birds which help in controlling pests. For example, snakes are one of the prime group of animals which prey upon rats, mice and various other pests. Similarly, large varieties of birds, for example, owls and peacocks, prey upon vermin and pests. If these are allowed to dwell around the agricultural areas, they can clear large varieties of pests including insects. Lizards are also important in this regard. We need to know their value and save them.

Sustainable development has become a catch phrase today. It is 'indeed' a paradigm shift in development thinking. Though it has been interpreted in a number of ways, adherence to this path ensures lasting development and non-declining welfare for all.

7.6 Conclusion

Economic development, which aimed at increasing the production of goods and services to meet the needs of a rising population, puts greater pressure on the environment. In the initial stages of development, the demand for environmental resources was less than that of supply. Now the world is faced with increased demand for environmental resources but their supply is limited due to overuse and misuse. Sustainable development aims at promoting the kind of development that minimises environmental problems and meets the needs of the present generation without compromising the ability of the future generation to meet their own needs.

Recap

- Environment performs four functions: supplies resources, assimilates wastes, sustains life by providing genetic and bio diversity and provides aesthetic services.
- Population explosion, affluent consumption and production have placed a huge stress on the environment.
- Developmental activities in India have put immense pressure on its finite natural resources, besides creating impact on human health and well-being.
- The threat to India's environment is of two dimensions—threat of poverty induced environmental degradation and the threat of pollution from affluence and a rapidly growing industrial sector.
- Though the government, through various measures, attempts to safeguard the environment, it is also necessary to adopt a path of sustainable development.
- Sustainable development is development that meets the need of the present generation without compromising the ability of the future generation to meet their own needs.
- Promotion of natural resources, conservation, preserving regenerative capacity of ecological system and avoiding the imposition of environmental risks on future generations would lead to sustainable development.

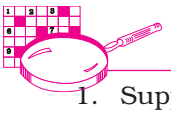


EXERCISES

1. What is meant by environment?
2. What happens when the rate of resource extraction exceeds that of their regeneration?
3. Classify the following into renewable and non-renewable resources (i) trees (ii) fish (iii) petroleum (iv) coal (v) iron-ore (vi) water.
4. Two major environmental issues facing the world today are _____ and _____.
5. How do the following factors contribute to the environmental crisis in India? What problem do they pose for the government?
   (i) Rising population
   (ii) Air pollution

(iii) Water contamination
(iv) Affluent consumption standards
(v) Illiteracy
(vi) Industrialisation
(vii) Urbanisation
(viii) Reduction of forest coverage
(ix) Poaching, and
(x) Global warming.

6. What are the functions of the environment?
7. Identify six factors contributing to land degradation in India.
8. Explain how the opportunity costs of negative environmental impact are high.
9. Outline the steps involved in attaining sustainable development in India.
10. India has abundant natural resources—substantiate the statement.
11. Is environmental crisis a recent phenomenon? If so, why?
12. Give two instances of
    (a) Overuse of environmental resources
    (b) Misuse of environmental resources.
13. State any four pressing environmental concerns of India.
14. Correction for environmental damages involves opportunity costs—explain.
15. Explain how the supply-demand reversal of environmental resources account for the current environmental crisis.
16. Highlight any two serious adverse environmental consequences of development in India. India's environmental problems pose a dichotomy — they are poverty induced and, at the same time, due to affluence in living standards—is this true?
17. What is sustainable development?
18. Keeping in view your locality, describe any four strategies of sustainable development.
19. Explain the relevance of intergenerational equity in the definition of sustainable development.



SUGGESTED ADDITIONAL ACTIVITIES

1. Suppose 70 lakh cars are added every year to the roads of metropolitans. Which type of resources do you think are undergoing depletion? Discuss.
2. Make a list of items that can be recycled.

3. Prepare a chart on the causes and remedies of soil erosion in India.
4. How does population explosion contribute to the environmental crisis? Debate in the classroom.
5. The nation has to pay heavily for correcting environmental damages—discuss.
6. A paper factory is to be set up in your village. Arrange a role play consisting of an activist, an industrialist and a group of villagers.



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UNIT IV

Development Experiences of India: A Comparison with Neighbours

In today's globalised world, where geographical boundaries are slowly becoming meaningless, it is important for neighbouring countries in the developing world to understand the development strategies being pursued by their neighbours. This is more so because they share the relatively limited economic space in world markets. In this unit, we will compare India's developmental experiences with two of its important and strategic neighbours — Pakistan and China.

8



Comparative Development Experiences of India and its Neighbours

After studying this chapter, the learners will

- figure out comparative trends in various economic and human development indicators of India and its neighbours, China and Pakistan
- assess the strategies that these countries have adopted to reach their present state of development.

Geography has made us neighbours. History has made us friends. Economics has made us partners, and necessity has made us allies. Those whom God has so joined together, let no man put asunder.

John F. Kennedy

8.1 Introduction

In the preceding units we studied the developmental experience of India in detail. We also studied the kind of policies India adopted, which had varying impacts in different sectors. Over the last two decades or so, the economic transformation that is taking place in different countries across the world, partly because of the process of globalisation, has both short as well as long-term implications for each country, including India. Nations have been primarily trying to adopt various means which will strengthen their own domestic economies. To this effect, they are forming regional and global economic groupings such as the **SAARC, European Union, ASEAN, G-8, G-20, BRICS** etc. In addition, there is also an increasing eagerness on the parts of various nations to try and understand the developmental processes pursued by their neighbouring nations as it allows them to better comprehend their own strengths and weaknesses vis-à-vis their neighbours. In the unfolding process of globalisation, this is particularly considered essential by developing countries as they face competition not only from developed nations but also amongst themselves in the relatively limited economic space enjoyed by the developing world. Besides, an understanding of the other economies in our neighbourhood is also required as all major common economic activities in the region impinge on overall human development in a shared environment.

In this chapter we will compare the developmental strategies pursued by India and the largest two of its neighbouring economies—Pakistan and China. It has to be remembered that despite being endowed with vast natural resources, there is little similarity between the political power setup of India – the largest democracy of the world which is wedded to a secular and deeply liberal Constitution for more than half a century, and the militarist political power structure of Pakistan or the command economy of China that has only recently started moving towards a democratic system and more liberal economic restructuring respectively.

8.2 Developmental Path—A Snapshot View

Do you know that India, Pakistan and China have many similarities in their developmental strategies? All the three nations have started towards their developmental path at the same time. While India and Pakistan became independent nations in 1947, People's Republic of China was established in 1949. In a speech at that time,

Jawaharlal Nehru had said, "These new and revolutionary changes in China and India, even though they differ in content, symbolise the new spirit of Asia and new vitality which is finding expression in the countries in Asia."

All three countries had started planning their development strategies in similar ways. While India announced its first Five Year Plan for 1951–56, Pakistan announced its first five year plan, now called the Medium Term Development Plan, in 1956. China announced its First Five Year Plan in 1953. Since 2018, Pakistan is working on the basis of 12th Five Year Development Plan (2018–23), whereas, China is working on 14th Five Year Plan (2021–25). Until March 2017, India has been following Five Year Plan- based development model. India and Pakistan adopted similar strategies, such as creating a large public sector and raising public expenditure on social development. Till the 1980s, all the three countries had similar growth rates and per capita incomes. Where do they stand today in comparison to one another? Before we answer this question, let us trace the historical path of developmental policies in China and Pakistan. After studying the last three units, we already know what policies India has been adopting since its Independence.

China: After the establishment of People's Republic of China under one-party rule, all critical sectors of the economy, enterprises and lands owned and operated by individuals were brought under government control. The **Great Leap Forward (GLF)** campaign initiated in 1958 aimed at industrialising the country on a massive scale. People were encouraged to set up industries in their backyards. In rural areas, communes were started. Under the **Commune** system, people collectively cultivated lands. In 1958, there were 26,000 communes covering almost all the farm population.

GLF campaign met with many problems. A severe drought caused havoc in China killing about 30 million people. When Russia had conflicts with China, it withdrew its professionals who had earlier been sent to China to help in the industrialisation process. In 1965, Mao introduced the **Great Proletarian Cultural Revolution** (1966–76) under which students and professionals were sent to work and learn from the countryside.

The present day fast industrial growth in China can be traced back to the reforms introduced in 1978. China introduced reforms in phases. In the initial phase, reforms were initiated in agriculture, foreign trade and investment sectors. In agriculture, for instance, commune lands were divided into small plots, which were allocated (for use not ownership) to individual households. They were allowed to keep all income from the land after paying stipulated taxes. In the later phase, reforms were initiated in the industrial sector. Private sector firms, in general, and township and village enterprises, i.e., those enterprises which were owned and operated by local collectives, in particular, were allowed to produce



Fig. 8.1 *Wagah Border is not only a tourist place but also used for trade between India and Pakistan*

goods. At this stage, enterprises owned by government (known as State Owned Enterprises—SOEs), which we, in India, call public sector enterprises, were made to face competition. The reform process also involved dual pricing. This means fixing the prices in two ways; farmers and industrial units were required to buy and sell fixed quantities of inputs and outputs on the basis of prices fixed by the government and the rest were purchased and sold at market prices. Over the years, as production increased, the proportion of goods or inputs transacted in the market also increased. In order to attract foreign investors, **special economic zones** were set up.

Pakistan: While looking at various economic policies that Pakistan adopted, you will notice many similarities with India. Pakistan also follows the mixed economy model with co-existence of public and private sectors. In the late 1950s and 1960s, Pakistan introduced a variety of regulated policy framework (for import substitution-based industrialisation). The policy combined tariff protection for manufacturing of consumer goods together with direct import controls on competing imports. The introduction of Green Revolution led to mechanisation and increase in public investment in infrastructure in select areas, which finally led to a rise in the production of foodgrains. This changed the agrarian structure dramatically. In the 1970s, **nationalisation** of capital goods industries took place. Pakistan then shifted its policy orientation in the late 1970s and 1980s when the major thrust areas were denationalisation and encouragement of private sector. During this period, Pakistan also received financial support from western nations and remittances from

continuously increasing outflow of emigrants to the Middle-east. This helped the country in stimulating economic growth. The then government also offered incentives to the private sector. All this created a conducive climate for new investments. In 1988, reforms were initiated in the country.

Having studied a brief outline of the developmental strategies of China and Pakistan, let us now compare some of the developmental indicators of India, China and Pakistan.

8.3 Demographic Indicators

If we look at the global population, out of every six persons living in this world, one is an Indian and another a Chinese. We shall compare some demographic indicators of India, China and Pakistan. The population of Pakistan is very small and accounts for roughly about one-tenth of China or India.

Though China is the largest nation and geographically occupies the largest area among the three nations, its density is the lowest. Table 8.1 shows the population growth as being the highest in Pakistan, followed by India and China. Scholars point out the one-child norm introduced in China in the late 1970s as the major reason for low population growth. They also state that this measure led to a decline in the sex ratio, the proportion of females per 1000 males. However, from the table, you will notice that the sex ratio is low and biased against females in all three countries. Scholars cite son preference prevailing in all these countries as the reason. In recent times, all three countries are adopting various measures to improve the situation. One-child norm and the resultant arrest in the growth of population also have other implications. For instance, after a few decades, in China, there will be more elderly people in proportion to young people. This led China to allow couples to have two children.

The fertility rate is also low in China and very high in Pakistan. Urbanisation is high in China with India having 34 per cent of its people living in urban areas.

TABLE 8.1

Select Demographic Indicators, 2017-18

Country	*Estimated Population (in million)*	*Annual Growth of Population*	*Density (per sq. km)*	*Sex Ratio*	*Fertility Rate*	*Urbanisation*
India	1352	1.03	455	924	2.2	34
China	1393	0.46	148	949	1.7	59
Pakistan	212	2.05	275	943	3.6	37

Source: *World Development Indicators 2019, www.worldbank.org*

8.4 Gross Domestic Product and Sectors

One of the much-talked issues around the world about China is its growth of Gross Domestic Product. China has the second largest GDP (PPP) of $22.5 trillion in the world, whereas, India's GDP (PPP) is $9.03 trillion and Pakistan's GDP is $ 0.94 trillion, roughly about 11 per cent of India's GDP. India's GDP is about 41 per cent of China's GDP.

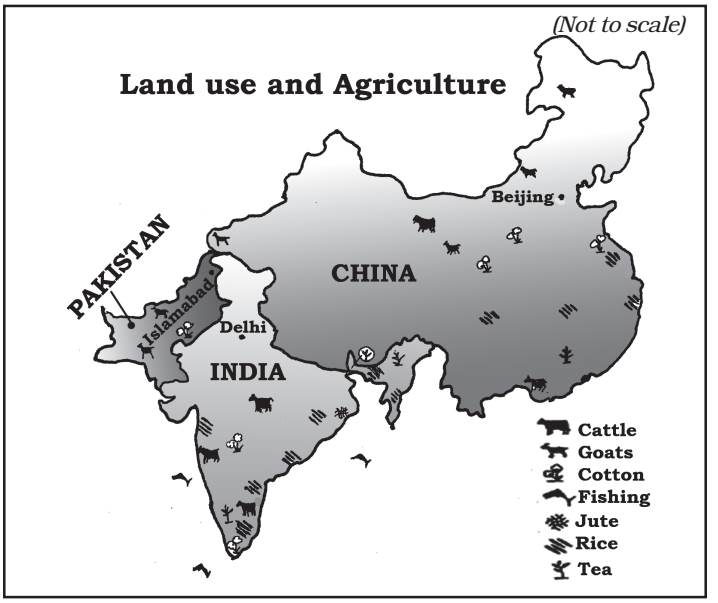


Fig. 8.2 *Land use and agriculture in India, China and Pakistan*

When many developed countries were finding it difficult to maintain a growth rate of even 5 per cent, China was able to maintain near double-digit growth during 1980s as can be seen from Table 8.2. Also, notice that in the 1980s, Pakistan was ahead of India; China was having double-digit growth and India was at the bottom. In 2015–17, there has been a decline in Pakistan

TABLE 8.2

Annual Growth of Gross Domestic Product (%), 1980–2017

Country	*1980–90*	*2015–2017*
India	5.7	7.3
China	10.3	6.8
Pakistan	6.3	5.3

Source: *Key Indicators for Asia and Pacific 2016, Asian Development Bank, Philippines; World Development Indicators 2018*



Work These Out

- Does India follow any population stabilisation measures? If so, collect the details and discuss in the classroom. You may refer to the latest Economic Survey, annual reports or website of the Ministry of Health and Family Welfare (http://mohfw.nic.in).
- Scholars find son preference as a common phenomenon in many developing countries, including India, China and Pakistan. Do you find this phenomenon in your family or neighbourhood? Why do people practise discrimination between male and female children? What do you think about it? Discuss it in the classroom.

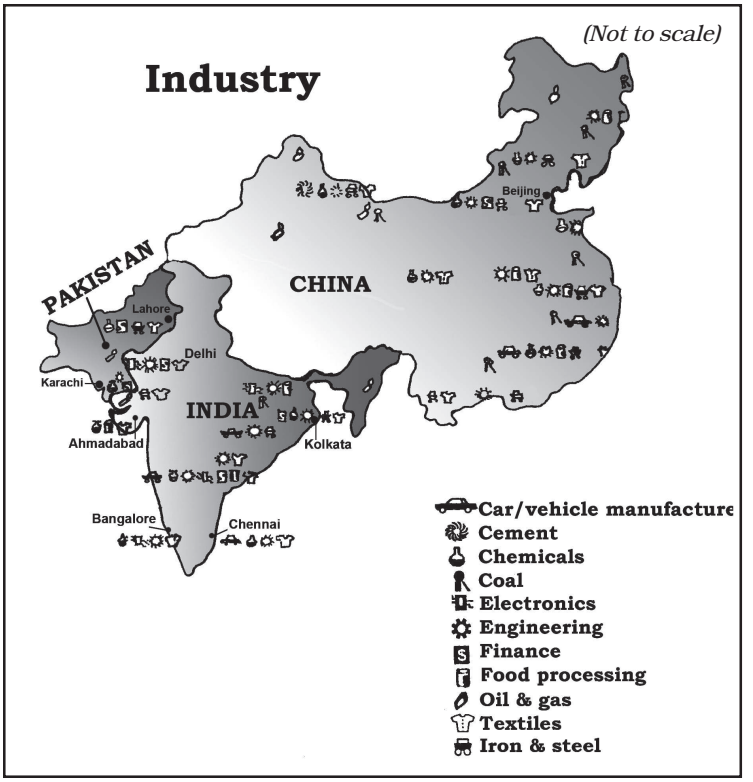


Fig. 8.3 *Industry in India, China and Pakistan*

and China's growth rates, whereas, India met with moderate increase in growth rates. Some scholars hold the reform processes introduced in Pakistan and political instability over a long period as reasons behind the declining growth rate in Pakistan. We will study in a later section which sector contributed to different growth rates in these countries.

First, look at how people engaged in different sectors contribute to the Gross Domestic Product now called as Gross Value Added. It was pointed out in the previous section that China and Pakistan have more proportion of urban population than India. In China, due to topographic and climatic conditions, the area suitable for cultivation is relatively small — only about 10 per cent of its total land area. The total cultivable area in China accounts for 40 per cent of the cultivable area in India. Until the 1980s, more than 80 per cent of the people in China were dependent on farming as their sole source of livelihood. Since then, the government encouraged people to leave their fields and pursue other activities such as handicrafts, commerce and transport. In 2018–19, with 26 per cent of its workforce engaged in agriculture, its contribution to the GVA in China is 7 per cent (see Table 8.3).

In both India and Pakistan, the contribution of agriculture to GVA were 16 and 24 per cent, respectively, but the proportion of workforce that works in this sector is more in India. In Pakistan, about 41 per cent of people work in agriculture, whereas, in India, it is 43 per cent. Twenty four per cent of Pakistan workforce is engaged in industry but it produces 19 per cent of GVA. In India, industry workforce

TABLE 8.3

Sectoral Share of Employment and GVA (%) in 2018–2019

Sector	*Contribution to GVA*			*Distribution of Workforce*		
	India	*China*	*Pakistan*	*India*	*China*	*Pakistan*
Agriculture	16	7	24	43	26	41
Industry	30	41	19	25	28	24
Services	54	52	57	32	46	35
Total	**100**	**100**	**100**	**100**	**100**	**100**

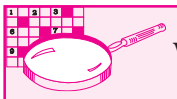
Source: *Human Development Report 2019 ; Key Indicators of Asia and Pacific 2019.*

account for 25 per cent but produces goods worth 30 per cent of GVA. In China, industries contribute to GVA at 41, and employ 28 per cent of workforce. In all the three countries, service sector contributes highest share of GVA.

In the normal course of development, countries first shift their employment and output from agriculture to Industry and then to services. This is what is happening in China as can be seen from Table 8.3. The proportion of workforce engaged in industry in India and Pakistan were low at 25 per cent and 24 per cent respectively. The contribution of industries to GVA is at 30 per cent in India and 19 per cent in Pakistan. In these countries, the shift is taking place directly to the service sector.

Thus, in all the three countries the service sector is emerging as a major player of development. It contributes more to GVA and, at the same time, emerges as a prospective employer. If we look at the proportion of workforce in the1980s, Pakistan was faster in shifting its workforce to service sector than India and China. In the 1980s, India, China and Pakistan employed 17, 12 and 27 per cent of its workforce in the service sector respectively. In 2019, it has reached the level of 32, 46 and 35 per cent, respectively.

In the last five decades, the growth of agriculture sector, which employs the largest proportion of workforce in all the three countries, has declined. In the industrial sector, China has maintained a near double-digit growth rate in 1980s but began showing decline in recent years,



Work These Out

- Do you think it is necessary for India and Pakistan to concentrate on the manufacturing sector as China does? Why?
- Scholars argue that the service sector should not be considered as an engine of growth whereas India and Pakistan have raised their share of output mainly in this sector only. What do you think?

TABLE 8.4

Trends in Output Growth in Different Sectors, 1980–2015

Country	*1980–90*			*2014–18*		
	Agriculture	*Industry*	*Service*	*Agriculture*	*Industry*	*Service*
India	3.1	7.4	6.9	3.1	6.9	7.6
China	5.9	10.8	13.5	3.1	5.3	7.1
Pakistan	4	7.7	6.8	1.7	4.8	5.0

whereas, for India and Pakistan growth rate has declined. In case of service sector, China was able to maintain its rate of growth during 1980–1990,while there was a positive and increasing growth of India's service sector output. Thus, China's growth is contributed by the manufacturing and service sectors and India's growth by the service sector. During this period, Pakistan has shown deceleration in all three sectors.

8.5 Indicators of Human Development

You might have studied about the importance of human development indicators in the lower classes and the position of many developed and developing countries. Let us look how India, China and Pakistan have performed in some of the select indicators of human development. Look at Table 8.5.

TABLE 8.5

Some Selected Indicators of Human Development, 2017-2019

Item	*India*	*China*	*Pakistan*
Human Development Index (Value)	0.645	0.761	0.557
Rank (based on HDI)	130	87	154
Life Expectancy at Birth (years)	69.7	76.9	67.3
Mean years of Schooling (% aged 15 and above)	6.5	8.1	5.2
Gross National Income per capita (PPP US$)	6,681	16,057	5,005
Percentage of People living Below Poverty Line (National)	21.9*	1.7**	24.3*
Infant Mortality Rate (per 1000 live births)	29.9	7.4	57.2
Maternal Mortality Rate (per 1 lakh births)	133	29	140
Population using at least basic Sanitation (%)	60	75	60
Population using at least basic drinking Water Source (%)	93	96	91
Percentage of Undernourished Children	37.9	8.1	37.6

Note: * *for the year 2011; for the years 2015.*
Sources: *Human Development Report 2019 and 2020 and World Development Indicators (www.worldbank.org); Key indicators for Asia and the Pacific 2019, Asian Development Bank (ADB).*

Table 8.5 shows that China is moving ahead of India and Pakistan. This is true for many indicators — income indicator such as GDP per capita, or proportion of population below poverty line or health indicators such as mortality rates, access to sanitation, literacy, life expectancy or malnourishment. China and Pakistan are ahead of India in reducing proportion of people below the poverty line and also their performance in sanitation. But India and Pakistan have not been able to save women from maternal mortality. In China, for one lakh births, only 29 women die whereas in India and Pakistan, about 133 and 140 women die respectively. Surprisingly all the three countries report providing improved drinking water sources for most of its population. China has the smallest share of poor among the three countries. Find out for yourself how these differences occur.

In dealing with or making judgements on such questions, however, we should also note a problem while using the human development indicators given above with conviction. This occurs because these are all extremely important indicators; but these are not sufficient. Along with these, we also need what may be called 'liberty indicators'. One such indicator has actually been added as a measure of 'the extent of democratic participation in social and political decision-making' but it has not been given any extra weight. Some obvious 'liberty indicators' like measures of 'the extent of Constitutional protection given to rights of citizens' or 'the extent of constitutional protection of the Independence of the Judiciary and the Rule of Law' have not even been introduced so far. Without including these (and perhaps some more) and giving them overriding importance in the list, the construction of a human development index may be said to be incomplete and its usefulness limited.

8.6 Development Strategies — an Appraisal

It is common to find developmental strategies of a country as a model to others for lessons and guidance for their own development. It is particularly evident after the introduction of the reform process in different parts of the world. In order to learn from economic performance of our neighbouring countries, it is necessary to have an understanding of the roots of their successes and failures. It is also necessary to distinguish between, and contrast, the different phases of their strategies. Though countries go through their development phases differently, let us take the initiation of reforms as a point of reference. We know that reforms were initiated in China in 1978, Pakistan in 1988 and India in 1991. Let us briefly assess their achievements and failures in pre- and post-reform periods.

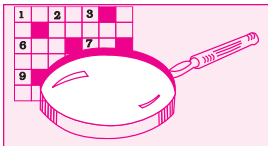
Why did China introduce structural reforms in 1978? China did not have any compulsion to introduce

reforms as dictated by the World Bank and International Monetary Fund to India and Pakistan. The new leadership at that time in China was not happy with the slow pace of growth and lack of modernisation in the Chinese economy under the Maoist rule. They felt that Maoist vision of economic development based on decentralisation, self sufficiency and shunning of foreign technology, goods and capital had failed. Despite extensive land reforms, collectivisation, the Great Leap Forward and other initiatives, the per capita grain output in 1978 was the same as it was in the mid-1950s.

It was found that establishment of infrastructure in the areas of education and health, land reforms, long existence of decentralised planning and existence of small enterprises had helped positively in improving the social and income indicators in the post reform period. Before the introduction of reforms, there had already been massive extension of basic health services in rural areas. Through the commune system, there was more equitable distribution of food grains. Experts also point out that each reform measure was first implemented at a smaller level and then extended on a massive scale. The experimentation under decentralised government enabled to assess the economic, social and political costs of success or failure. For instance, when reforms were made in agriculture, as pointed out earlier by handing over plots of land to individuals for cultivation, it brought prosperity to a vast number of poor people. It created conditions for the subsequent phenomenal growth in rural industries and built up a strong support base for more reforms. Scholars quote many such examples on how reform measures led to rapid growth in China.

Scholars argue that in Pakistan the reform process led to worsening of all the economic indicators. We have seen in an earlier section that compared to 1980s, the growth rate of GDP and its sectoral constituents have not yet improved.

Though the data on international poverty line for Pakistan is quite healthy, scholars using the official data of Pakistan indicate rising poverty there. The proportion of poor in 1960s was more than 40 per cent which declined to 25 per cent in 1980s and started rising again in the recent decades. The reasons for the slow-down of growth and re-emergence of poverty in Pakistan's economy, as scholars put it, are agricultural growth and food supply situation were based not on an institutionalised process of technical change but on good harvest. When there was a good harvest, the economy was in good condition, when it was not, the economic indicators showed stagnation or negative trends. You will recall that India had to borrow from the IMF and World Bank to set right its balance of payments crisis; foreign exchange is an essential component for any country and it is



Work These Out

- While India has performed relatively well vis-à-vis other developing countries (including its Asian neighbours) in terms of economic growth, India is yet to show the world that there is considerable progress in terms of human development indicators. Where India went wrong? Why did we not take care of our human resources? Discuss in the classroom.

- There is a general perception going around in India that there is sudden increase in dumping of Chinese goods into India which have implications for manufacturing sector in India and also that we do not engage ourselves in trading with our neighbouring nations. Look at the following table, which shows exports from India to, and imports from, Pakistan and China. From newspapers and websites and listening to news, collect the details of goods and services transacted in trading with our neighbours. In order to get detailed information relating to international trade, you can log on to the website: http://dgft.gov.in.

Country	*Exports from India* (Rs in crore)			*Imports to India* (Rs in crore)		
	2004-2005	*2018-2019*	*Annual Rate of Growth (%)*	*2004-2005*	*2018-2019*	*Annual Rate of Growth (%)*
Pakistan	2,341	14,426	3.7	427	3476	5.1
China	25,232	1,17,289	2.6	31,892	4,92,079	10.3

- Calculate exports as a % of imports for both the years and discuss the probable reasons for the trend in the class.

important to know how it can be earned. If a country is able to build up its foreign exchange earnings by sustainable export of manufactured goods, it need not worry. In Pakistan most foreign exchange earnings came from remittances from Pakistani workers in the Middle-east and the exports of highly volatile agricultural products; there was also growing dependence on foreign loans on the one hand and increasing difficulty in paying back the loans on the other.

However, during the last few years, Pakistan has recovered its economic growth and has been sustaining. In 2017-18, the Annual Plan 2019-20 reports that, the GDP registered a growth of 5.5 per cent, highest when compared to the previous decade. While agriculture recorded growth rate far from satisfactory level, industrial and service sectors grew at 4.9 and 6.2 per cent respectively. Many macroeconomic indicators also began to show stable and positive trends.

8.7 Conclusion

What are we learning from the developmental experiences of our neighbours? India, China and Pakistan have travelled seven decades of developmental path with varied results. Till the late 1970s, all of them were maintaining the same level of low development. The last three decades have taken these countries to different levels. India, with democratic institutions, performed moderately, but a majority of its people still depend on agriculture. India has taken many initiatives to develop the infrastructure and improve the standard of living. Scholars are of the opinion that political instability, over-dependence on remittances and foreign aid along with volatile performance of agriculture sector are the reasons for the slowdown of the Pakistan economy. Yet, last five years, many macroeconomic indicators began showing positive and moderate growth rates reflecting the economic recovery. In China, the lack of political freedom and its implications for human rights are major concerns; yet, in the last four decades, it used the 'market system without losing political commitment' and succeeded in raising the level of growth alongwith alleviation of poverty. You will also notice that unlike India and Pakistan, which are attempting to privatise their public sector enterprises, China has used the market mechanism to 'create additional social and economic opportunities'. By retaining collective ownership of land and allowing individuals to cultivate lands, China has ensured social security in rural areas. Public intervention in providing social infrastructure even prior to reforms has brought about positive results in human development indicators in China.



Recap

- With the unfolding of the globalisation process, developing countries are keen to understand the developmental processes pursued by their neighbours as they face competition from developed nations as also amongst themselves.
- India, Pakistan and China have similar physical endowments but totally different political systems.
- All the three countries follow the similar planned pattern of development. However, the structures established to implement developmental policies are quite different.
- Till the early 1980s, the developmental indicators of all the three countries, such as growth rates and sectoral contribution towards national income, were similar.
- Reforms were introduced in 1978 in China, in 1988 in Pakistan and in 1991 in India.
- China introduced structural reforms on its own initiative while they were forced upon India and Pakistan by international agencies.
- The impact of policy measures were different in these countries — for instance, one-child norm has arrested the population growth in China whereas in India and Pakistan, a major change is yet to take place.
- Even after seventy years of planned development, majority of the workforce in all the countries depends on agriculture. The dependency is greater in India.
- Though China has followed the classical development pattern of gradual shift from agriculture to manufacturing and then to services, India and Pakistan's shift has been directly from agriculture to service sector.
- China's industrial sector has maintained a high growth rate while it is not so in both India and Pakistan. This led to rapid increase of the GDP per capita in China than in India and Pakistan.
- China is ahead of India and Pakistan on many human development indicators. However these improvements were attributed not to the reform process but the strategies that China adopted in the pre-reform period.
- While assessing the developmental indicators, one also has to consider the liberty indicators.



EXERCISES

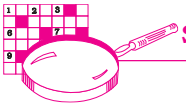
1. Why are regional and economic groupings formed?
2. What are the various means by which countries are trying to strengthen their own domestic economies?
3. What similar developmental strategies have India and Pakistan followed for their respective developmental paths?
4. Explain the Great Leap Forward campaign of China as initiated in 1958.
5. China's rapid industrial growth can be traced back to its reforms in 1978. Do you agree? Elucidate.
6. Describe the path of developmental initiatives taken by Pakistan for its economic development.
7. What is the important implication of the 'one child norm' in China?
8. Mention the salient demographic indicators of China, Pakistan and India.
9. Compare and contrast India and China's sectoral contribution towards GVA/GDP. What does it indicate?
10. Mention the various indicators of human development.
11. Define the liberty indicator. Give some examples of liberty indicators.
12. Evaluate the various factors that led to the rapid growth in economic development in China.
13. Group the following features pertaining to the economies of India, China and Pakistan under three heads
    - One-child norm
    - Low fertility rate
    - High degree of urbanisation
    - Mixed economy
    - Very high fertility rate
    - Large population
    - High density of population
    - Growth due to manufacturing sector
    - Growth due to service sector.

14. Give reasons for the slow growth and re-emergence of poverty in Pakistan.

15. Compare and contrast the development of India, China and Pakistan with respect to some salient human development indicators.

16. Comment on the growth rate trends witnessed in China and India in the last two decades.

17. Fill in the blanks
    (a) First Five Year Plan of _____ commenced in the year 1956. (Pakistan/China)
    (b) Maternal mortality rate is high in _____. (China/Pakistan)
    (c) Proportion of people below poverty line is more in _____. (India/Pakistan)
    (d) Reforms in _____ were introduced in 1978. (China/Pakistan)



SUGGESTED ADDITIONAL ACTIVITIES

1. Organise a class debate on the issue of free trade between India and China and India and Pakistan.

2. You are aware that cheap Chinese goods are available in the market, for example, toys, electronic goods, clothes, batteries etc. Do you think that these products are comparable in quality and price with their Indian counterparts? Do they create a threat to our domestic producers? Discuss.

3. Do you think India can introduce the one-child norm like China to reduce population growth? Organise a debate on the policies that India can follow to reduce population growth.

4. China's growth is mainly contributed by the manufacturing sector and India's growth by the service sector—prepare a chart showing the relevance of this statement with respect to the structural changes in the last decade in the respective countries.

5. How is China able to lead in all the Human Development Indicators? Discuss in the classroom. Use Human Development Report of the latest year.



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GLOSSARY

The Association of South-East Asian Nations (ASEAN) : It is a political, economic, and cultural organisation of countries located in South-east Asia—Thailand, Indonesia, Malaysia, Singapore, the Philippines, Brunei Darussalam, Cambodia, Laos, Myanmar and Vietnam.

Balance of Payments (BOP) : It is a statistical statement summarising all the external transactions (receipts and payments) on current and capital account in which a country is involved over a period of time, say, a year. As the BOP shows the total assets and obligations over a time-period, it always balances.

Barriers to Entry : This refers to the factors which make it disadvantageous for new entrants to enter an industry as compared with the firms already established within the industry.

Better Compliance : Obeying or complying with the Government regulation. It is referred to usually in case of payment of taxes and dues to the Government.

Bilateral Trade Agreements : The agreements relating to exchange of commodities or services between two countries.

Brundtland Commission : A Commission established by United Nations Organisation in 1983 to study the world's environmental problems and propose agenda for addressing them. It came out with a report. The definition provided by the Commission for the term, 'sustainable development', is very popular and widely cited all over the world.

Budgetary Deficit : A situation when the government's income and tax receipts fail to cover its expenditures.

Bureau of Energy Efficiency (BEE) : It is a government organisation that aims to develop policies and strategies with a thrust on self regulation and market principles. It promotes energy conservation in different sectors of the economy and undertakes measures against the wasteful uses of electricity.

Business Process Outsourcing (BPO) : Outsourcing of business processes (activities constituting a service) by companies to other companies. This term is frequently associated with outsourcing of such activities (e.g. receiving and making calls on behalf of other companies popularly known as call centres), by foreign companies to Indian companies in the field of IT-enabled services.

Carrying Capacity : It is the measure of habitat to indefinitely sustain a population at a particular density. A more technical definition for carrying capacity is the largest size of a density-dependent population for which the population growth rate is zero. Hence, below carrying capacity, populations will tend to increase, while they will decrease above carrying capacity. Population size decreases above carrying capacity due to either reduced survivorship (e.g. due to insufficient space or food) or reproductive success (e.g. due to insufficient food, or behavioural interactions), or both. The carrying capacity of an environment will vary for different species in different habitats, and can change over time due to a variety factors, including trends in food availability, environmental conditions and space.

Cascading Effect : When tax imposition leads to a disproportionate rise in prices, i.e. by an extent more than the rise in the tax, it is known as cascading effect.

Cash Reserve Ratio (CRR) : A proportion of the total deposits and reserves of the commercial banks that is to be kept with the central bank (RBI) in liquid form. It is used as a measure of control of RBI over the commercial banks.

Casual Wage Labourer : A person who is casually engaged in others' farm or non-farm enterprises and, in return, receives wages according to the terms of the daily or periodic work contract.

Colonialism : The practice of acquiring colonies by conquest or other means and making them dependent. It also means extending power, control or rule by a country over the political and economic life of areas outside its borders. The main feature of colonialism is exploitation.

Commercialisation of Agriculture : It implies production of crops for the market rather than for self-consumption i.e. family consumption. During the British rule, the commercialisation of agriculture acquired a different meaning—it became basically commercialisation of crops. The British started offering higher price to farmers for producing cash crops rather than for food crops. They used these cash crops as raw materials for industries in Britain.

Communes : Known as people's communes, or *renmin gongshe* in China, were formerly the highest of three administrative levels in rural areas in the period from 1958 to 1982-85, when they were replaced by townships. Communes, the largest collective units, were divided in turn into production brigades and production teams. The communes had governmental, political, and economic functions.

Consumption Basket : Group of goods and services consumed by a household. In order to estimate the consumption pattern of people, statistical agencies identify such items. For instance NSSO has identied 19 groups of items in the consumption basket. Some of them are (i) cereals (ii) pulses

(iii) milk and milk products (iv) edible oil (v) vegetables (vi) fuel and light and (vii) clothing.

Default : Failure to make repayment of the principal and interest on a debt e.g. sovereign debt (loan obtained by the government) to the lenders, say, international financial institutions, on the scheduled date, causing loss of credibility as a debtor.

Deficit Financing : A situation where the expenditure of the government exceeds its revenue.

Demographic Transition : It is a concept developed by demographer Frank Notestein in 1945 to describe the typical pattern of falling death and birth rates in response to better living conditions associated with economic development. Notestein identified three phases of demographic transition, pre-industrial, developing and modern industrialised societies. Later another phase, post-industrial was also included.

Dereservation : Allowing an individual or group of enterprises to produce goods and services which were hitherto produced by a particular individual or group of enterprises. In India, it refers to allowing large-scale industries to produce goods and services which were produced only by the small-scale industries.

Devaluation : A fall in the exchange rate which reduces the value of a currency in terms of other currencies.

Disinvestment : A deliberate sale of a part of the capital stock of a company to raise resources and change the equity and/or management structure of a company.

Employers : Those self-employed workers who by and large, run their enterprises by hiring labourers.

Enterprise : An undertaking owned and operated by an individual or by group of individuals to produce and/or distribute goods and/or services mainly for the purpose of sale, whether fully or partly.

Equities : Shares in the paid up capital or stock of a company whose holders are considered as owners of the company with voting rights and dividends in the profit.

Establishment : An enterprise which has got at least one hired worker for major part of the period of operation in a year.

European Union : It is a union of twenty-five independent states founded to enhance political, economic and social cooperation within the European continent. The member countries of European Union are Austria, Belgium, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany,

Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Netherlands, Portugal, Spain, Sweden, United Kingdom, Malta, Poland, Slovakia and Slovenia.

Export Duties : Taxes imposed on goods exported from a country.

Export Promotion : A set of measures (including fiscal and commercial support measures and steps aimed at removal of trade barriers) taken by a government to promote the export of goods with a view to achieve higher economic growth and accumulation of foreign exchange earnings.

Export-Import Policy : The economic policies of the government relating to its exports and imports.

Family labour/worker : A member who works without receiving wages in cash or in kind in a farm, an industry, business or trade conducted by the members of the family.

Financial Institutions : Institutions that engage in mobilisation and allocation of savings. They include commercial banks, cooperative banks, developmental banks and investment institutions.

Fiscal Management : The use of taxation and government expenditure to regulate the economic activities.

Fiscal Policy : All the planned actions of a government in mobilising financial resources for meeting its expenditure and regulating the economic activities in a country.

Foreign Direct Investment : Investment of foreign assets into domestic structures, equipment and organisations. It does not include foreign investment into the stock markets. Foreign direct investment is thought to be more useful to a country than investments in the equity of its companies because equity investments are potentially 'hot money' which can leave at the first sign of trouble, whereas FDI is durable and generally useful whether things go well or badly.

Foreign Exchange : Currency or bonds of another country.

Foreign Exchange Markets : A market in which currencies are bought and sold at rates of exchange fixed now, for delivery at specified dates in the future.

Foreign Institutional Investment : Foreign investments which come in the form of stocks, bonds, or other financial assets. This form of investment does not entail active management or control over the firms or investors.

Foreign Institutional Investors (FIIs) : Banking and non-banking financial institutions of foreign origin e.g. commercial banks, investment banks, mutual funds, pension funds or other such institutional investors (as distinct from the domestic financial institutions investing) whose investment

in stocks and bonds in the country through stock markets have significant influence.

Formal Sector Establishments : All the public sector establishments and those private sector establishments which employ 10 or more hired workers.

G-20 : A forum of countries that intents to promote global economic stability and sustainable growth. This forum brings together finance ministers and central bank governors from 19 countries: Argentina, Australia, Brazil, Canada, China, France, Germany, India, Indonesia, Italy, Japan, The Republic of Korea, Mexico, Russia, Saudi Arabia, South Africa, Turkey, The United Kingdom, The United States of America. The European Union is also a member of G-20 and is represented by the President of the European Council and by Head of the European Central Bank.

G-8 : The Group of Eight (G-8) consists of Canada, France, Germany, Italy, Japan, the United Kingdom of Great Britain and Northern Ireland, the United States of America, and Russian Federation. The hallmark of the G-8 is an annual economic and political summit meeting of the heads of government with international officials, though there are numerous subsidiary meetings and policy research.The Presidency of the group rotates every year. For the year 2006 it was held by Russia.

Goods and Services Tax : It is a single indirect tax on the supply of goods and services. It was introduced in July 2017 by abolishing a variety of taxes, such as sales tax and excise that prevailed in India. Under GST, tax is imposed on the basis of value addition at each stage of the movement of goods and services. Different slabs of tax rates, such as 0 per cent, 5 per cent, 12 per cent, 18 per cent and 28 per cent, are imposed on almost all goods and services. This slab is same throughout the country.

Gratuity : An amount of money given by the employer to the employee at the time of retirement for services rendered by the employee.

Gross Domestic Product : The total value of final goods and services produced within a country's borders in a year, regardless of ownership. It is used as one of many indicators of the standard of living in a country, but there are limitations with this view.

Gross Value Added: The sum of a country's GDP and net of subsidies and taxes in the economy (GVA=GDP+Subsidies–indirect taxes).

Household : A group of persons normally living together and taking food from a common kitchen. The word 'normally' means that temporary visitors are excluded and those who temporarly staying away are included.

Import Licensing : Permission required from the government to import goods into a country.

Import Substitution : A policy of the state for development of economy in which import of goods is generally substituted by domestic production (through import controls, tariffs and other restrictions) with a view to encourage domestic industry on grounds of self-sufficiency and domestic employment.

Infant Mortality Rate : It is the number of deaths of infants before reaching the age of one, in a particular year, per 1,000 live births during that year.

Inflation : A sustained rise in the general price level.

Informal Sector Enterprises : Those private sector enterprises, which employ less than 10 workers on a regular basis.

Integration of Domestic Economy : A situation where the policies of government facilitate free trade and investment with other countries making the domestic economy work together with other economies in an efficient and mutually interdependent way.

Invisibles : Various items enter in the current account of the balance of payments, some of which are not visible goods. Invisibles are mainly services, like tourism, transport by shipping or by airways, and financial services such as insurance and banking. They also include gifts sent abroad or received from abroad and private transfer of funds, government grants and interests, profits and dividends.

Labour Laws : All the rules and regulations framed by the government to protect the interests of the workers.

Land/Revenue Settlement : With the British acquiring territorial rights in different parts of India, administration of territories was formulated on the basis of survey of land. It was decided in the interests of government in terms of revenues to be collected from each parcel of land in possession of either a *ryot* (means peasant) or a *mahal* (revenue village) or a *zamindar* (a proprietary land holder). Decision in each of these cases was meant for the rights of the latter over land for the purposes of either ownership of land or rights to cultivation. This system is known as land/revenue settlement. There were different land settlements formulated in India. They are (i) system of permanent settlement, which is also known as the zamindari system (ii) *ryotwari* system (a system of revenue settlement entered into by the government with individual tenants) (iii) *mahalwari* system (a system of revenue settlement entered into by the government with a *mahal*).

Life Expectancy at Birth (years) : The number of years a newborn infant would live if prevailing patterns of age-specific mortality rates at the time of birth were to stay the same throughout the child's life.

Maternal Mortality Rate : It is the relationship between the number of maternal deaths due to childbearing by the number of live births or by the sum of live births and foetal deaths in a given year.

Merchant Bankers : Banks or financial institutions, also known as investment bankers, that specialise in advising the companies and managing their equity and debt requirement (often referred to as portfolio management) through floatation and sale/purchase of stocks and bonds.

Morbidity : It is the propensity to fall ill. It affects a person's work by making him or her temporarily disabled. Prolonged morbidity may lead to mortality.

In our country, acute respiratory infections and diarrhoea are two major causes of morbidity.

Mortality rate : The word 'mortality' comes from 'mortal' which originates from the Latin word *mors* (meaning death). It is the annual number of deaths (from a disease or in general) per 1,000 people. It is distinct from morbidity rate, which refers to the number of people who have a disease compared to the total number of people in a population.

MRTP Act : An Act (Monopolies Restrictive Trade Practices Act) framed to prevent monopolistic practices and regulate the conduct or business practices of firms that are not in public interest.

Multilateral Trade Agreements : Trade agreements made by a country with more than two nations to exchange goods and services.

National Product/Income : Total value of goods and services produced in a country plus income from abroad.

Nationalisation : Transfer of ownership from private sector to public sector. This involves take over of companies owned by individuals or group of individuals by either state or central government. In some contexts, it also involves transfer of ownership from state government to central government.

New Economic Policy : A term used to describe the policies adopted in India since 1991.

Non-renewable Resources : Resources that cannot be renewed. They have a finite, even if large, stock. Some examples are fossils fuels such as oil and coal and mineral resources—iron, lead, aluminum, uranium.

Non-tariff Barriers : All the restrictions on imports by a government in the form other than taxes. They mainly include restrictions on quantity and quality of goods imported.

Opportunity Cost : It is defined with respect to a particular value or action and is equal to the value of the foregone alternative choice or action.

Pension : A monthly payment to a worker who has retired from work.

Per Capita Income : Total national income of a country divided by its population in a specific period.

Permit License Raj : A term used to denote the rules and regulations framed by the government to start, run and operate an enterprise for production of goods and services in India.

Planning Commission : An organisation set up by the Government of India. It is responsible for making assessment of all resources of the country, augmenting deficient resources, formulating plans for the most effective and balanced utilisation of resources and determining priorities.

Poverty Line : The per capita expenditure on certain minimum needs of a person including food intake of a daily average of 2,400 calories in rural areas and 2,100 calories in urban areas.

Private Sector Establishments : All those establishments, which are owned and operated by individuals or group of individuals.

Productivity : Output per unit of input employed. Increase in the efficiency on the part of capital or labour leads to increase in productivity. This term is generally used to refer to productivity increase in labour inputs.

Provident Fund : A savings fund in which both employer and employee contribute regularly in the interest of the employee. It is maintained by the government and given to the employee when he or she resigns or retires from work.

Public Sector Establishments : All those establishments which are owned and operated by the government. They may be run either by local government, state government or by central government independently or jointly.

Quantitative Restrictions : Restrictions in the form of total quantities or *q*uotas imposed on imports to reduce balance of payments (BOP) deficit and protect domestic industry.

Regular Salaried/Wage Employee : Persons who work in others' farm or non-farm enterprises and, in return, receive salary or wages on a regular basis (i.e. not on the basis of daily or periodic renewal of work contract). They include not only persons getting time wage but also persons receiving piece wage or salary and paid apprentices, both full time and part-time.

Renewable Resources : Resources that can be renewed through natural processes if they are used wisely. Forests, animals and fishes, if not over-exploited, get easily renewed. Water is also in that category.

South Asian Association for Regional Cooperation (SAARC) : It is an association of eight countries of South Asia — Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, Sri Lanka and Afghanistan. SAARC provides a platform for the peoples of South Asia to work together in a spirit of friendship, trust and understanding. It aims to accelerate the process of economic and social development in member countries.

Self-employed : Those who operate their own farm or non-farm enterprises or are engaged independently in a profession or trade with one or a few partners. They have freedom to decide how, where and when to produce and sell or carry out their operation. Their earning is determined wholly or mainly by sales or profits from their enterprises.

Social Security : A government or privately established system of measures, which ensures material security for the elderly, disabled, destitute, widows and children. It includes pension, gratuity, provident fund, maternal benefits, health care etc.

Special Economic Zone (SEZ) : It is a geographical region that has economic laws different from a country's typical economic laws. Usually the goal is to increase foreign investment. Special Economic Zones have been established in several countries, including the People's Republic of China, India, Jordan, Poland, Kazakhstan, the Philippines and Russia.

Stabilisation Measures : Fiscal and monetary measures adopted to control fluctuations in the balance of payments and high rate of inflation.

State Electricity Boards (SEBs) : These are part of the state administration that generate, transmit and distribute electricity in different states.

Statutory Liquidity Ratio (SLR) : A minimum proportion of the total deposits and reserves to be maintained by the banks in liquid form as per the regulations of the central bank (RBI). Maintenance of SLR, in addition to the Cash Reserve Ratio *(*CRR), is an obligation of the banks.

Stock Exchange : A market in which the securities of governments and public companies are traded. It provides the facilities for stock brokers to trade company stocks and other securities.

Stock Market : An institution where stocks and shares are traded.

Structural Reform Policies : Long-term measures like liberalisation, deregulation and privatisation aimed to improve the efficiency and competitiveness of the economy.

Tariff : A tax on imports, which can be levied either on physical units, e.g. per tonne (specific) or on value. Tariffs may be imposed for a variety of reasons including: to raise government revenue, to protect domestic industry from subsidised or low-wage imports, to boost domestic employment, or to ease a deficit on the balance of payments. Apart from the revenue that they raise tariffs achieve little good—they reduce the volume of trade and increase the price of the imported commodity to consumers.

Tariff Barriers : All the restrictions on imports by a government in the form of taxes.

Trade Union : An organisation of workers formed for the purpose of addressing its members' interests in respect of wages, benefits, and working conditions.

Unemployment : A situation in which all those who, owing to lack of work, are not working but either seek work through employment exchanges, intermediaries, friends or relatives or by making applications to prospective employers or express their willingness or availability for work under the prevailing condition of work and remunerations.

Urbanisation : Expansion of a metropolitan area, namely the proportion of total population or area in urban localities or areas (cities and towns), or the increase of this proportion over time. It can thus represent a level of urban population relative to total population of the area, or the rate at which the urban proportion is increasing. Both can be expressed in percentage terms, the rate of change expressed as a percentage per year, decade or period between censuses.

Worker-Population Ratio : Total number of workers divided by the population. It is expressed in percentage.

NOTES